Some documentation for

Minos (V2.5.2.1)

Contest Logging Software

(V2.5.2.0)

This document is based on,
and specifically relevant to,

V2.5.2

Preface

The **Minos Contest Logger** is supplied under Version 3 of the GPL or any later version.

The Development Team are Mike Goodey (G0GJV), David Balharrie (M0DGB/G8FKH), Neil Yorke (M0NKE), Dave Sergeant (G3YMC), Peter Burton (G3ZPB), and Ken Punshon (G4APJ)

The **Minos Contest Logger** is **not** a supported product; no commitment or liability is or will be accepted to address any problems that may be encountered in using it.

However, any bugs or deficiencies should be notified to assist in the continuous development and improvement. If you use Minos and want to discuss the program or have any problems with it you are encouraged to join the project discussion mailing list on Groups.io.

Either visit https://minos.groups.io/g/main or send a blank email to main+subscribe@minos.groups.io

Preface

This document is intended to provide information ranging from simple installation and usage through to detailed internals, so don't worry if some of it is of no use/interest to you. There have been extensive changes since Minos V1. With V2, the software is (usually) installed (see section 1.1), and the Applications (see section 4) are a major new facility which has continued to expand since V2.0.

As of V2.0.7 the user has the ability to configure the layout of the Logger main panel (see section 3.5). Unless otherwise stated, all descriptions and illustrations in this document use the default configuration. Some illustrations may have been captured from older versions of the software and thus differ slightly from the latest version.

Certain HF Contests are supported, but this document is written from the perspective of VHF/UHF. There *may* at some point be an 'HF Supplement' describing any relevant information if significant differences are identified. Minor differences will, in general, be noted herein.

This is a 'live' document that may be revised on the basis of any comments, feedback, or additional information received (via the project discussion mailing list). No commitment or liability is or will be accepted to address any problems that may be encountered in using it.

The software it describes is also under continuing development, so may not exactly match what is described here. As with software, expect the document to have bugs; some of these will be known, others will not. Please report any errors that you find (after checking that you have the latest version of the document).

There is an implicit assumption that readers and users of the software are 'PC literate' and know about windows, panels, scroll bars etc.

The Logger runs on Windows 7 or later, but the document is written from the perspective of running the programme on Windows 10 Home (various versions, latterly 22H2). At this time there are no known problems with running it on other versions, indeed it has been extensively tested on a number of other platforms, but no guarantees are given.

The Logger also builds and runs native on Linux. The intention is to also provide release binaries for Ubuntu (32 and 64 bit), and Raspberry Pi, although it is more common for users of these systems to build Minos. Further platforms (other varieties of Linux, Android, Mac) may eventually be added. It is yet to be decided what will happen with documentation for these other platforms.

The general usage should be the same, or very similar; any significant differences will be noted as and when discovered/reported to the author.

Please consider any trademarks, copyrights, etc. that have not been acknowledged as acknowledged.

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1 Introduction

The Minos Contest Logger is a logging program originally written by Mike Goodey G0GJV, now being developed by G0GJV and others. All RSGB and Region 1 VHF and UHF amateur radio contests are supported, along with 'simple' HF Contests[1] such as the 'RSGB 80CC'. The Logger supports real time or post event data entry and production of an entry log file for submission to the adjudicators.

Operation of the program is intended to reflect the normal practice of filling in a log book. The fields appear in that order on the screen, and are below a similarly laid out list of previous contacts - similar to a log book.
It is mainly intended for **on-line** log entry, but can be used to enter logs post event. The less done after the event the better; all relevant data should be captured during the contest.

A single operator can't keep both computer and paper up to date together. With only one operator, keep the computer log up to date - it is your duplicate check system as well as the log that will eventually be used to submit your contest entry.

Where possible a second operator/logger should use the computer, and the main operator should maintain a traditional paper log. This way, the main operator always has a piece of paper to scribble notes on, as well as a providing a backup log.

The user is ALWAYS right - even if insisting that a W9W/MM worked him from ZZ99ZZ square. Although errors may be reported, in the end the program can be made to log it like that anyway. However, a report of 6 and 0, or invalid dates and times will **not** be allowed.

Error messages should never get in the way of entering more data. So, except in unusual circumstances, the program will not display error messages that must be acknowledged.

For more on the history of Minos, and its predecessor GJVLog, see http://minos.sourceforge.net/history.html

[1] Others are quite possible, but difficult, due to things such as: no proper integrated CW keyer support; contests such as 'Club Calls' have no assistance on the text entered (such as "Member BARC", or "HQ BARC", or "No Club"); no multiplier support e.g. in IOTA

1.1 Installation

The Minos installation programme (MinosInstall_2_5_2.exe) is downloaded (and saved to some convenient location) from https://sourceforge.net/projects/minos/files/

This release of Minos V2.5 can be installed to the same location as used for the immediately previous version (V2.4) or the Beta vesions since then. It can also be installed to an alternate location and run simultaneously, but **don't try to share log files between the two versions**, active or not, – the results are indeterminate (but likely problematic) if V2.4 is used to open a V2.5 log file.

It is prudent to save a copy of the **\Configuration** folder prior to installing the new version in case you want to revert back.

If currently running a version earlier than V2.4 **remove it** and then install V2.5. ***However***, be aware that the format of the Rig and Rotator configuration files changed during development so earlier versions should be deleted and the details set up again. If you fail to do this an error message will inform you of the problem when you invoke the application(s). Also be aware that this version of Minos uses Hamlib 4.5 (the latest release) which in some respects is incompatible with the now obsolete Hamlib V3.x.

Administrator privileges are needed to invoke the installation programme. The usual 'changes to system' dialogue may be seen, depending on system version and configuration:

- Open File – Security Warning (click Run)
- User Account Control 'changes to device' (click Yes)

The standard installation dialogue then follows, as shown below.

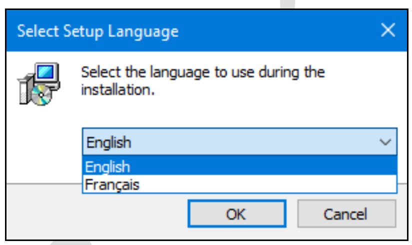

Figure 1 – Choice of language for installation

Accept the licence agreement:

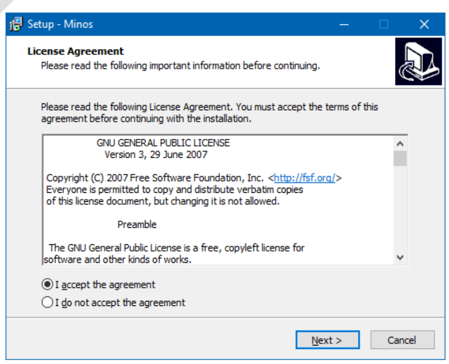

Figure 2 – Licence agreement

The default location for the installation is **“C:\Minos’**. This can be changed, but if this is done via the ‘Browse’ button be sure that it ends up where you want (by default a sub-folder is created). To avoid problems with access controls, Minos should **not** be installed into system folders such as **‘C:\Program Files’** or **‘C:\Program Files (x86)’**.

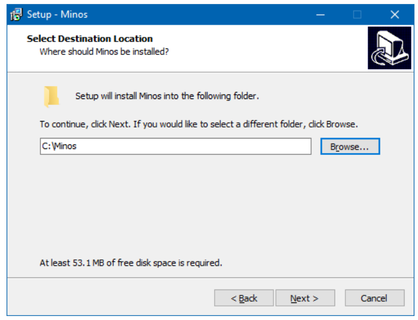

Figure 3 – Target location

There may of course be a message that the folder already exists, or needs to be created.

The log (.minos) files are by default placed in a sub-folder of the programme ‘root’ directory (see a description of the folders in Section 5.1).

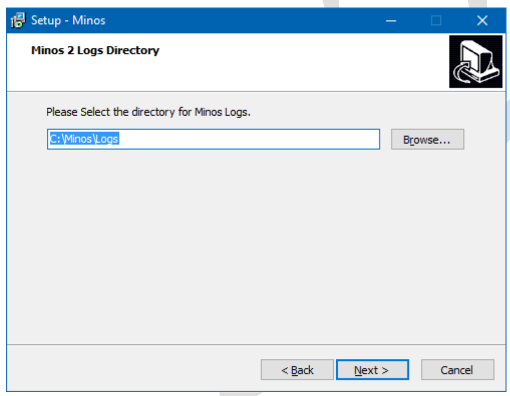

Figure 4 – Location of logs

By default a start menu entry is created, but this can be suppressed if so desired.

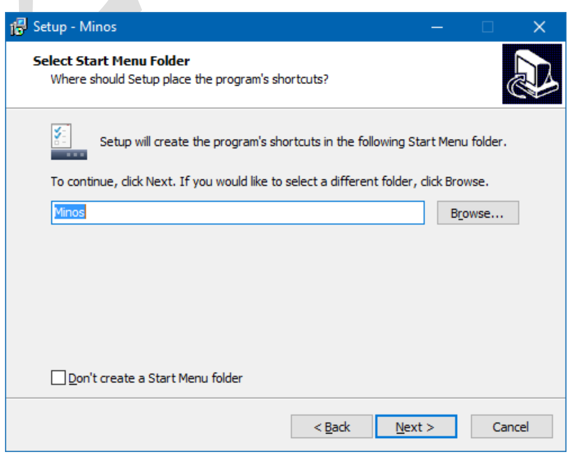

Figure 5 – Start menu

The ‘**Additional Tasks**’ panel (Figure 6) gives the option of creating a desktop shortcut for invoking the Logger in a convenient fashion.

There is also an option to associate the **'.minos'** extension with the Logger so that double-clicking a log file will open whichever version of the Logger was installed last.

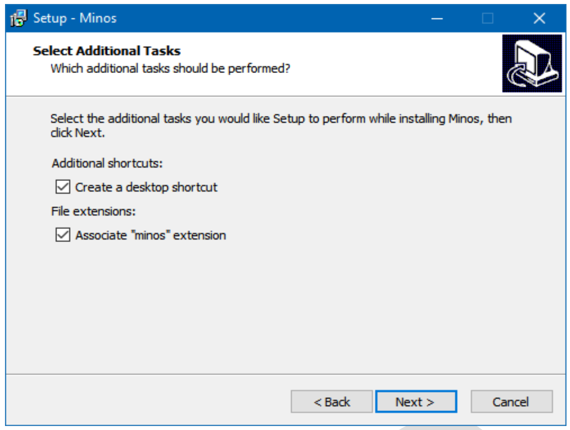

Figure 6 – Additional Tasks

Then there is a final check before installation starts:

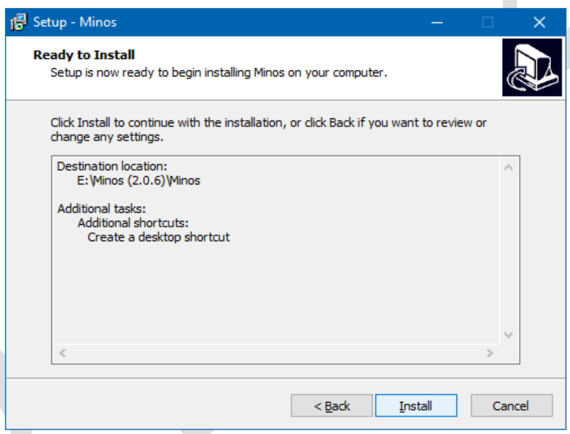

Figure 7 – Ready to install

After a short period of activity the completion notification (Figure 8) gives you the option to run the programme immediately. If this is not your first installation you may want to copy over configuration files before starting the programme.

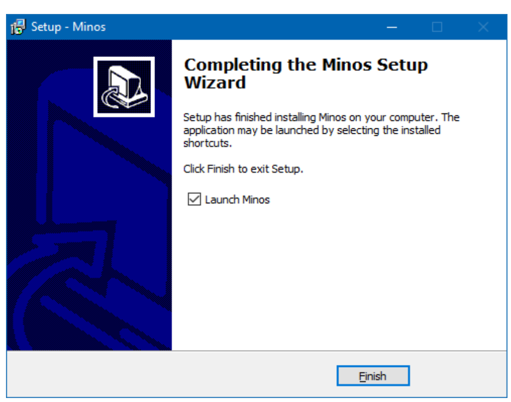

Figure 8 – Installation complete

1.1.1 Uninstallation

The Logger is uninstalled using the standard '**Control Panel/Programs and Features**' mechanism.

This removes all the binaries (**\Bin**), and the general Configuration information (that is not specific to the particular station installation), leaving (under the root folder):

- **\Configuration**: Station specific (The Display, Entry, ListPreload, LogsPreload, MinosConfig, MinosLogger, MinosRotatorConfig, QTH, and Station INI files
- All application configuration such as defined rigs and rotators is also saved, including screen layouts in ScreenConfigs.json.
- **\Lists**: (CSL files).
- **\Logs**: (Minos files).
- **\RigRecording** if present from use of RigRecorder
- **\TraceLog**: (diagnostics).
- Changes to fonts and panel layout for the Logger will be retained in the Registry[2].
- Any personal files placed in the **\Bin** folder will be retained (and hence the folder).
- Any other folders that the user has created under the main Minos folder

1.1.2 Options Panel

Options that were previously scattered around several pages are now all configured from **"Tools/Options"** (Figure 9).

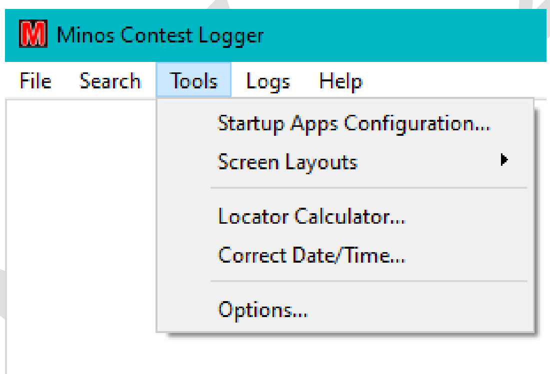

Figure 9 – Tools Menu

Other items on the menu are covered elsewhere in this document:

- **Startup Apps Configuration** (Section 4.1)
- **Screen Layouts** (Section 3.5)
- **Locator Calculator** (Section 3.3)
- **Correct Date/Time** (Section 3.4.4)

Options invokes a panel as shown in Figure 10 which has six tabs with functions described in the relevant (referenced) section(s) of the document:

- **Wanted Bands:** Selects the bands to be used by Minos.
- **Cluster/Bandmap:** Control of Cluster Display (Section 4.3.4.4) and Bandmap (Section 4.3.5.5).
- **General Options:** Age Protection (Section 2.1.1) and Default Directories.

[2] These can be reset using "**File/Exit Minos Contest Logger and Clear registry**".

- **Display Options:** Including:
  - Language (Section 1.1.3)
  - QSO detail options (Section 2.2)
  - List Spacing (Section 3.5.2)
  - Fonts (Section 3.5.3)
  - Auxiliary Headers (Section 3.2)
- **Log Radio Settings:** Radio general settings and presets (Section 4.2.1.1)
- **UDP Broadcast:** Interface to backend Loggers (Section 3.6.2)
- **WSJT-X:** Interface to frontend MGM apps (Section 3.6.1.1)

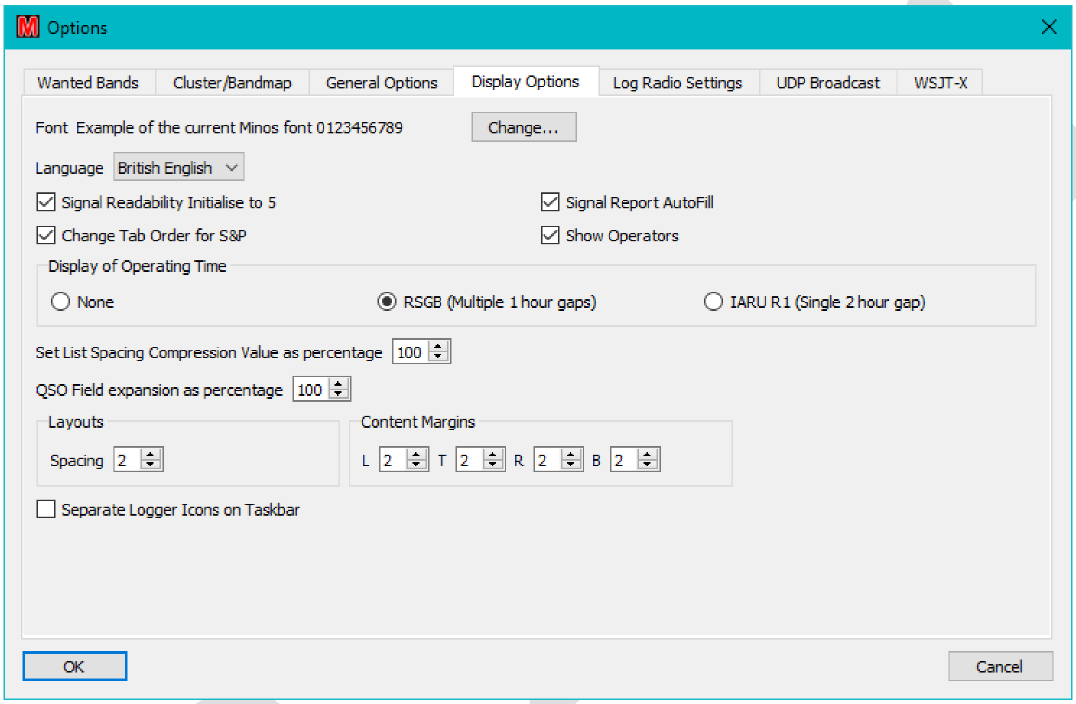

Figure 10 – Options Panel – Display Option tab

1.1.3 Choice of language

The language in use can be changed from the system/user default[3] within the **'Display Options'** tab of the **'Options Panel'** . If the language is changed then Minos restarts all applications.

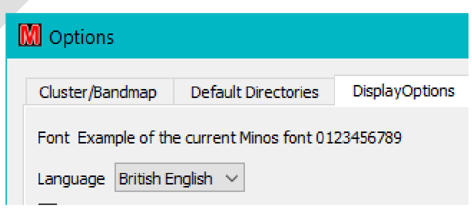

Figure 11 – Setting the Language

The choice is currently either English or French, but if anybody wants to volunteer to add other languages they will be welcomed[4].

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[3] If the language is chosen to be different from the default then some mixed language messages may result. Also any entities (e.g. Contest Sets) created, including Default, will remain in the original language.

There is no intent to make this, or any other, documentation available in anything other than English. Nor will the focus of the app be changed from UK contests (so NGR, PostCodes etc remain, and such things as Départments will **not** be added).

1.2 Basic Configuration

Ensure that the computer clock is reasonably accurate (within 1 or 2 seconds or better, particularly for MGM contests), and set to BST if applicable (as can be seen at the bottom right of Figure 16, Minos calculates and displays UTC[5] - in the Auxiliary panel as well).

In order to perform the initial set-up, invoke the program and the Splash Screen will be displayed.

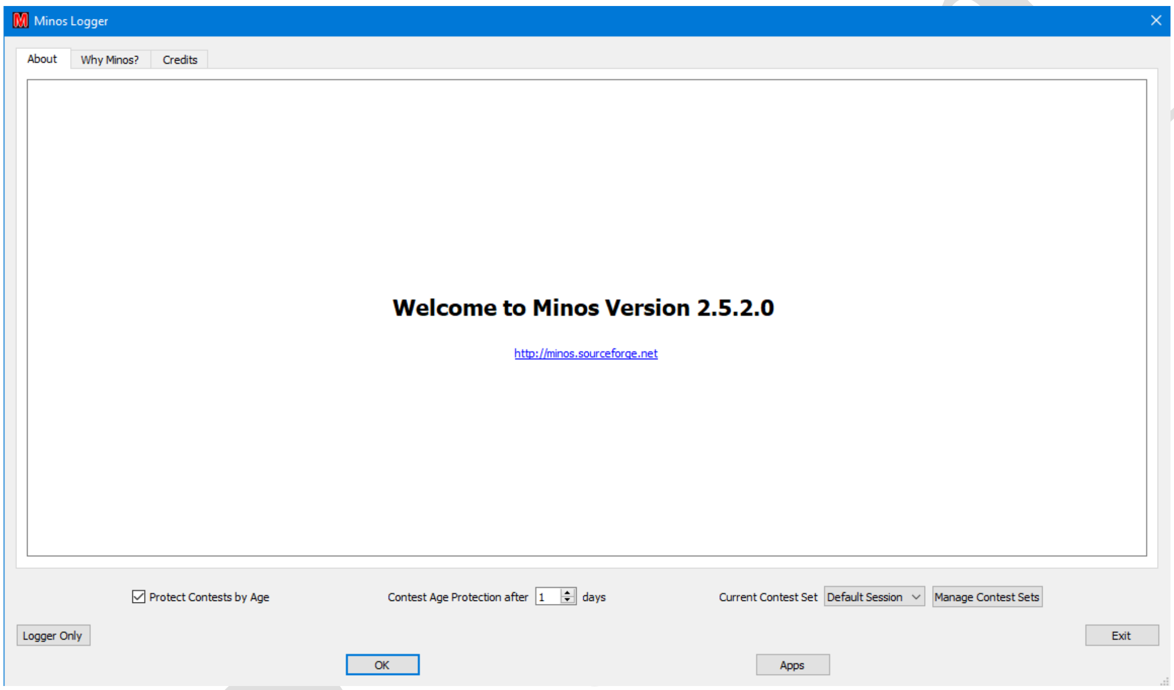

Figure 12 – Welcome screen

Click on **"OK"**[6] (or "**Logger Only**") to dismiss the splash screen. **Ignore the "Apps" button for now**. **Before trying to control a Rig or Rotator, or use any of the other integrated applications, read Section 4.1 and 4.2.**

For now, also ignore the **'Contest Age Protection after'** field which is described in 2.1.1 and the two buttons associated with **'Current Contest Set'** which are described in Section 3.4.2.

You will certainly want to expand the Logger main panel. Depending on your screen size and resolution you may want to expand to full screen, particularly if you have a wide screen.

Click on "**File/New VHF Contest**" to be presented with **'Details of Contest Entry'** (see Figure 13)[7].

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[4] The process is (technically) easy, but somewhat time consuming.
[5] See section 3.4.4 for information about time adjustments that can be made.
[6] You may be interested to read the "**Why Minos?**" tab for background on the origin of the name.
[7] **"New HF Contest"** will display the same panel except for there being a single **'HF Calendar'** button at the top.

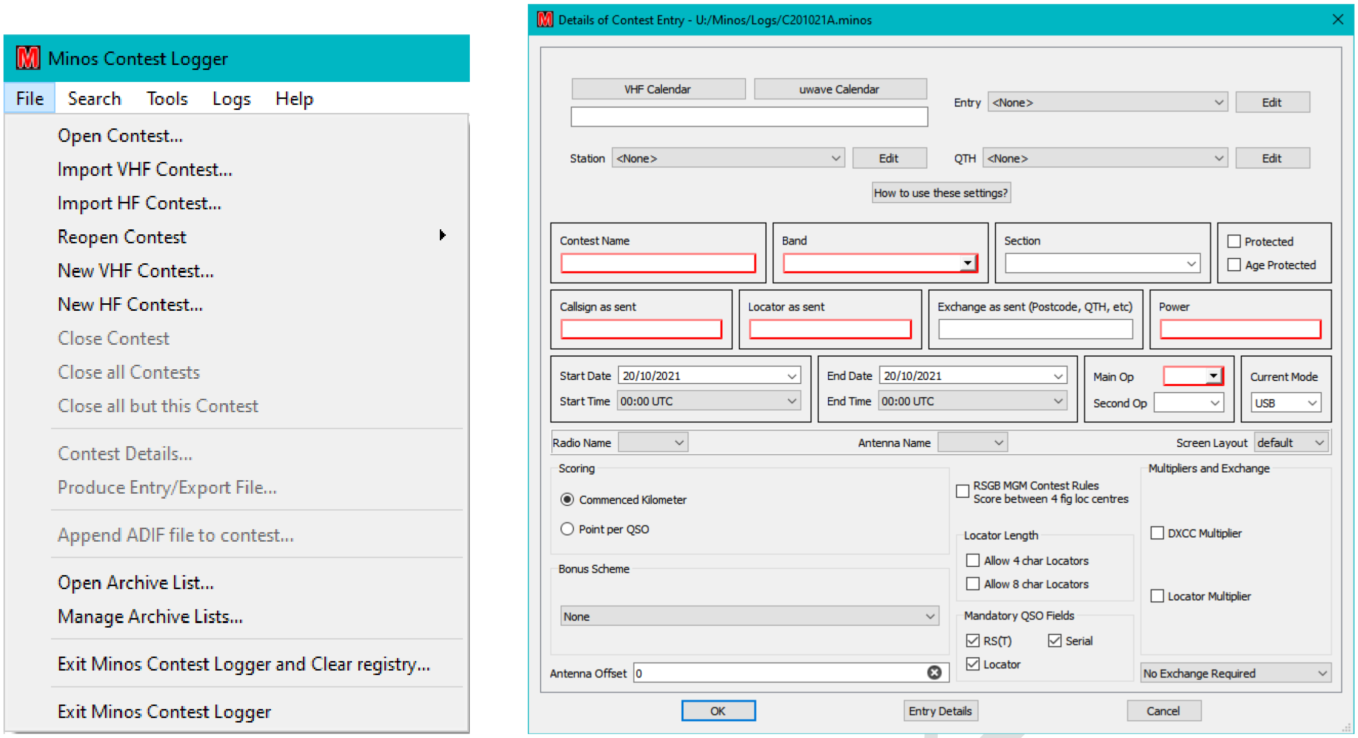

Figure 13 – File options and Contest Entry Details

This then allows the set-up of various items that will be used in future. The red outlines indicate the boxes that still require information before a contest can be validly created. When you are creating a contest the red edgings will disappear as the required information is supplied.

Click on VHF Calendar and then (because it can, and does, change) "**Download Latest Calendar**"[8]. At the time of writing, Minos ships with VHF calendar version 22.006 23/06/22[9]. Please check for updates before use, and certainly in mid December for the subsequent year.

Clearly a new download of the calendar will be needed on **at least** a yearly basis[10].

For now, close the VHF calendar in order to edit the grouped settings[11], which can be applied to a contest all in one go. There are four basic groups of settings:

- **Entry**: All the extra bits for a real entry (entrant callsign, group name, contact details) which are stored in Entry.ini
- **Station**: Details of the rig, power, and antenna (stored in Station.ini); you will typically require one of these for each band, if only due to antennas.
- **QTH**: Where the station is located for this contest, including Maidenhead Locator used. The data is stored in QTH.ini.

[8] Any message of the form "3 of 4 files downloaded. We don't expect to load them all." can be ignored.
[9] Microwave 22.001 23/01/22 and HF 22.011 03/08/22
[10] The intention is that this facility should be available on an indefinite basis, but there is no commitment to such activity.
[11] If you have created these entries for an earlier version of the program you can copy over the INI files (to the Configuration folder). It is also useful to copy over any CSL files you may already have (to the Lists folder).

- **Contest**: Typically derived from the contest calendar, specifying contest name, band(s), and scoring. If the calendar is not used then individual boxes can be configured (see Figure 15) or modified from the calendar derived values.

To define these settings, click on Edit, then select New Setting[12]. Assign a meaningful name and fill in the requisite data. A typical collection of grouped settings might for example be called:

- **Entry**: Self, Club, Contest Group
- **Station**: 50MHz, 70MHz, 144MHz, 432MHz, Portable-144MHz
- **QTH**: Home, VHF NFD, UKAC Portable

Click on the setting name on the left to select an existing setting and then its components are shown in the right hand pane, and can be edited individually. There is **no** validation of field content at this time, although errors may be highlighted later.

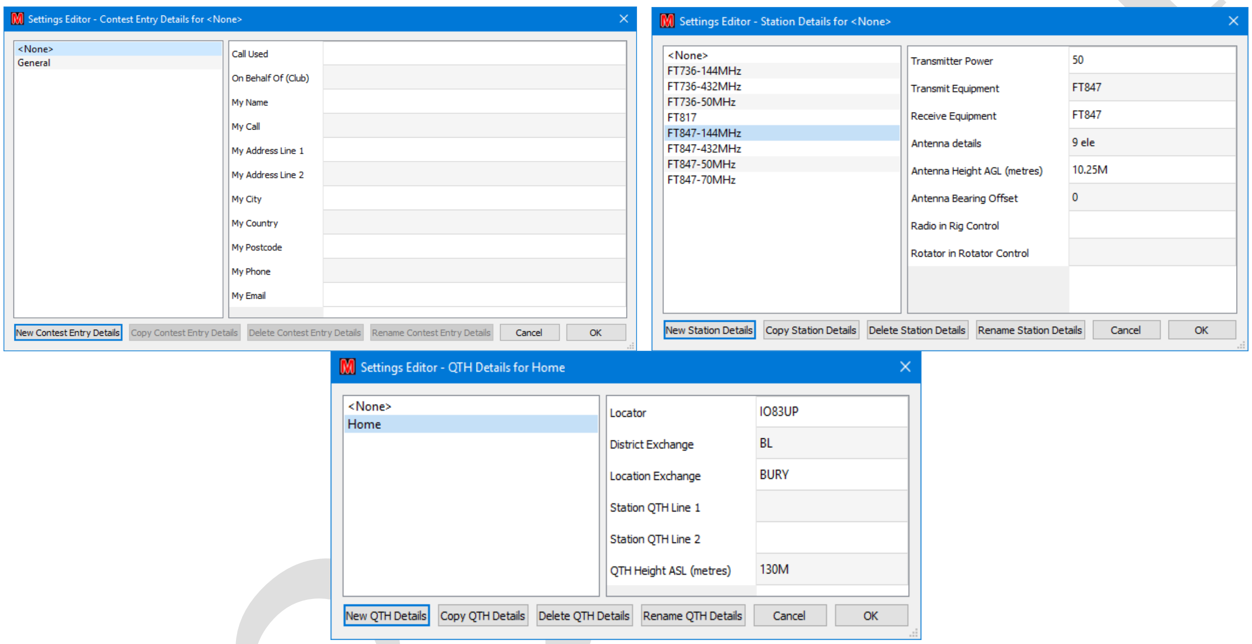

Figure 14 – Example Grouped Settings

Once the settings have been defined, use "**Entry Details**" to check that meaningful and complete data has been supplied.

During a contest "**File/ Contest Details**" can be used to check, and potentially change, these details (e.g. to add operators[13] or comments).

For now, "**Cancel**" the contest to finish set up; this will lose any other data entered into the contest (but you will do this well before the contest won't you?).

[12] Once you have created entries, an existing setting can be copied and minor changes made to cope with such things as the same rig being used on multiple bands.
[13] Operators can also be added via the pull down lists in the data entry panel.

2 Normal operation

Invoke the program from the shortcut on your desktop or from the start menu[14].

Any contest files that were left open last time will remain open[15] (see section 3.4.1 for discussion of multi-band contests that would typically use this facility). Similarly, any Archive (CSL) files (see section 3.4.3) will be re-opened; these are implicitly Read-Only unless you deliberately edit them.

For normal single contest operation close anything other than the current contest, or ensure that the non-current contests are marked as **'Protected'** to prevent you accidentally modifying them (see section 2.1.1).

If want to re-open a contest that you recently closed then "**File/Reopen Contest**" gives a list of the last 5 contests opened[16]. Opening one of the files on the list causes the entry to be removed, but it will be placed back at the head of the list when the log is closed again.

There are no known limitations (e.g. number of contacts, bands, field lengths….), but if you discover any please notify the support group (see Preface). Although it is possible to open at least 20 concurrent contests the 'tab space' at the top of the screen (on a wide, 1920x1080, screen) is full when 10 are open and it becomes complex to switch between them[17].

2.1 Creating a new contest

Use "**File/New Contest**" which will bring up the Contest Details panel as described in section 1.2.

Select the contest from VHF Calendar (normally for data entry during the contest; see section 2.2.1 for details of post-contest data entry) or manually configure as required (but be careful, it is easy to introduce errors).

Once the Applications have been defined (see section 4.1) the **'Radio name'** and **'Rotator name'** pull-down lists can be used to define the default radio/rotator to be used for the contest. The connection can also be established/changed through the QSO entry panel (see Figure 87 and Figure 111) – at which point these boxes will be filled.

The **'Screen Layout'** pull-down list allows selection of the layout to be applied from those defined (see Section 3.5).

Selecting an entry from **'Current Mode'** sets the initial mode for the contest. It can of course be changed later within the RigControl Panel.

Select the requisite bundled settings from the drop down lists for Entry, Station, and QTH. Any bundles set to **'<None>'** will be ignored (and you will have to specifically enter the relevant data before submitting the entry).

If the setting you want isn't there, use the "**Edit**" button to define a new setting or copy and then modify an existing setting in the manner described in section 1.2.

[14] The logger can also be invoked by directly clicking on the image (MqtLogger.exe in the \Bin folder) or by double clicking a log (.minos) file if the file association has been set up during installation.
[15] A warning message will be given if the file cannot be found. **Beware:** if you swap languages, the history of open files may be lost.
[16] If the file that you attempt to re-open has been deleted then no error message will be given, but the entry will be removed from the list.
[17] From the operator perspective 5 is probably more than enough!

The **'Contest Name'** can be entered and **'Band'** selected from the pull-down list if the VHF calendar is not used.

Note that the **'Section'** selected is no longer relevant to RSGB contests; it is determined by the Contest Entry Upload Robot. However, you may want to enter the requisite value here for completeness of the log[18] or for entering a non-RSGB contest.

'Callsign as sent'[19] will be derived from the **'Entry'** information and **'Locator as sent'** from the QTH information; both can be overwritten as required.

If the contest is not selected from the Calendar then use the relevant fields on the **'Details of Contest Entry'** panel (see Figure 15) to set the name, date/time, scoring, multipliers etc. Any bonus scheme in use must also be selected from the pulldown list at the bottom left.

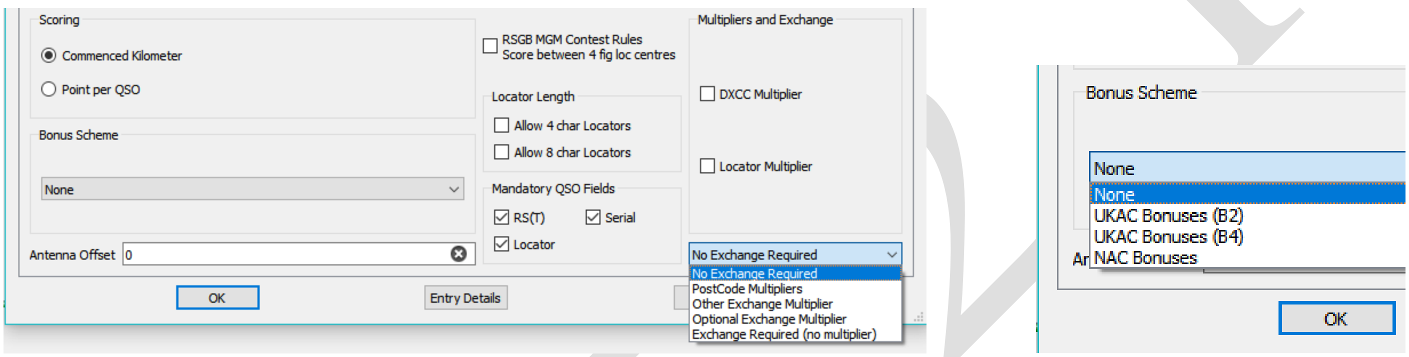

Figure 15 – Contest details and Bonus schemes

Use "**Entry Details**" to check that everything is complete and as expected. Any changes specific to the particular contest can be made via "**Entry Details**" at any point prior to exporting the log. If an error needs to be corrected it is possible to re-export the entry and re-submit (up until the closing date of the contest).

Click "**OK**" and the **'Save new contest as'** panel will generate a default name of the form **G4APJ_2017_09_02_144 MHz.minos** (the 'G' prefix will vary according to the country designated in the callsign) for the file located in the **'Logs'** folder[20]. In the event that the file already exists a warning message will be displayed and a new filename created with **'_1'** (etc.) appended. It is possible to specify some other name, or browse to another folder if you want, but this may introduce undue complexity if you are unsure of what you are doing.

If you have not provided sufficient information when you click "**OK**" the "**save as**" panel will not appear and the cursor will be moved to the first field identified as requiring input. All fields still required remain outlined in red.

Clicking "**Save**" to the **'Save new contest as'** panel causes the blank Data Entry panel to be displayed ready for entering details of QSOs (see Figure 16). The contest just created is highlighted (red text on tab)[21].

[18] This can be done from the pull-down list defined for a contest from the calendar, or by entering any arbitrary text (which is completely ignored by the Logger).
[19] The **'Main Op'** field is derived from this callsign, removing prefix country information and any suffix such as /P. Once derived any changes to the **'Main Op'** field as a result of changing callsigns must be done manually.
[20] Note that the Contest Name does not appear in the filename, regardless of how it was entered.
[21] Protected contests show green and those not selected show blue.

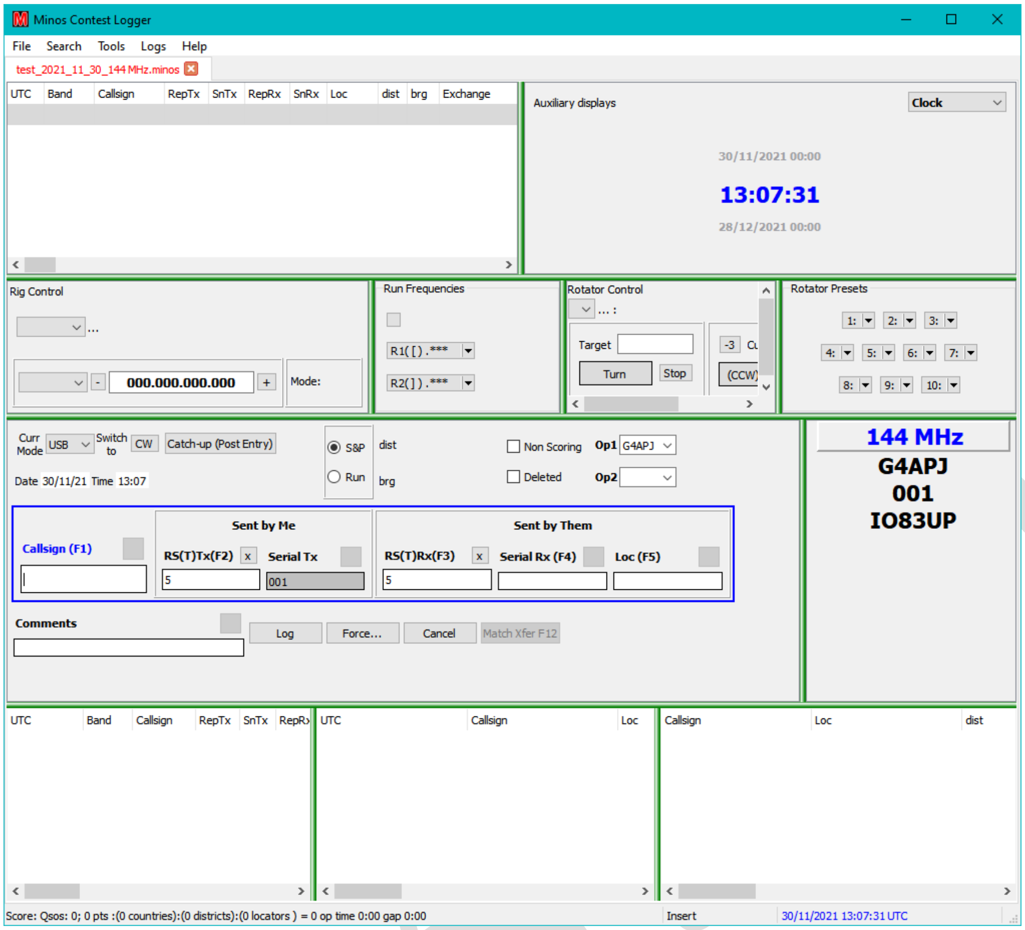

Figure 16 – Initial screen for entering QSO details

Figure 16 uses the default panel layout which includes the Rig and Rotator panels. See section 3.5 for information on how to change this default to remove these panels if they are not being used (or add others that are required).

The width of the sub-panels (with green edges) can be adjusted by sliding them across. The heights can be adjusted up/down in similar manner, even hiding panels to minimise screen usage on a temporary basis[22]. Column widths can also be adjusted. The worked/un-worked filters and the fonts/panel locations/column widths are saved across invocations (and re-installation). Defaults can be reset using the "**File/Exit Minos Contest Logger and Clear registry**" option which resets all Minos components.

2.1.1 Protected Contests

Protected contests are treated as 'Read-Only' (which prevents changes to e.g. screen layout).

For normal single contest operation close anything other than the current contest page(s), or ensure that the non-current contests are marked as **'Protected'** to prevent you accidentally modifying them.

Contests can be protected in two ways:

- Manually, by selecting **"File/Contest Details"** for the requisite contest and then ticking the **'Protected'** tick box on the right hand side just above the **'Power'** box, see Figure 18.

[22] Panels can also be 'detached' – see Section 3.5.1

- Automatically, after the number of days specified in the **'Contest Age Protection after'** field on the Welcome screen (see Figure 12) have passed since the end of the contest. If the contest has been age protected then both the **'Protected'** and **'Age Protected'** boxes (Figure 18) are set.

The protected status is preserved between invocations, and hence must be (temporarily) removed in order to change the content, including screen layout. This is done by clearing the **'Protected'** box, but note that it will revert to protected on the next invocation of the programme. If a contest has been age-protected then **both** boxes must be cleared to unprotect it!

If a contest has been protected then all the **'Contest Details'** except for the **'Protected'** field are greyed out and any connection to RigControl or Rotator will be disabled. The QSO panel is highlighted and a message displayed (see Figure 17).

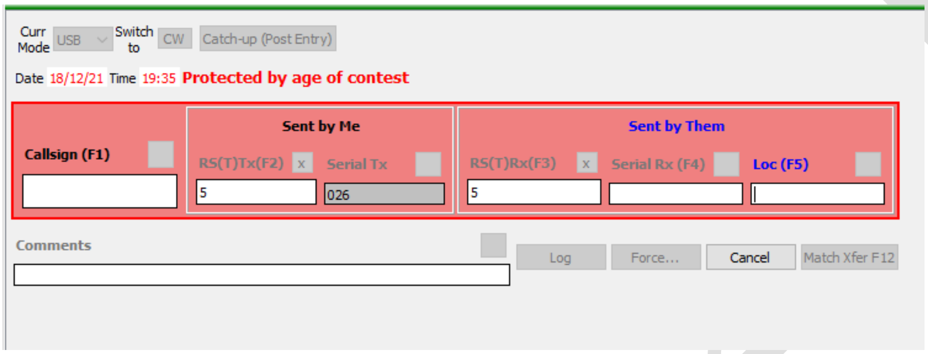

Figure 17 – QSO details in Protected Contest

The **'Callsign'** and **'Locator'** fields are still accessible, but validation is disabled, and the next QSO number is hidden from the Next QSO details panel.

2.1.2 MGM Contests

In order to support the MGM contests (beginning in April 2018) there are a variety of options to control QSO requirements (see Figure 18). These will be filled in automatically, and correctly, if the contest is selected from the VHF calendar (which will always be the preferred method).

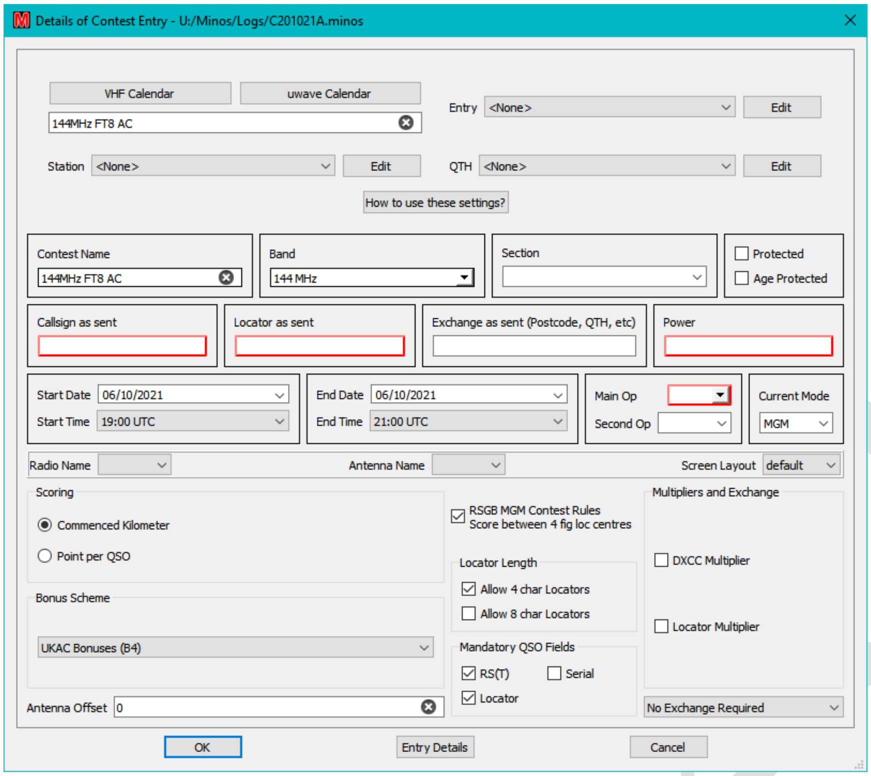

Figure 18 – MGM Contest Details

Once the **'RSGB MGM Contest Rules'** box is ticked (either automatically from the selected contest or manually) it sets things up for the scoring of the RSGB MGM contests[23].

This sets **'Locator Multiplier**', **'Allow 4 char Locators'** (scores are calculated as centre to centre of the 4 character Locators), and clears Serial in the Mandatory Fields (i.e. serial numbers are not only not required, but cannot be entered[24]).

Once you set a QSO mode to MGM (in any contest) the validation on the report is removed - it just requires it to be present – **so be careful what you type[25]**.

If the MGM software being used can create ADIF files (e.g. WSJT-X ADIF Export) the log may be imported into Minos. Create the contest then select **"File/Append ADIF file to contest"** (see Figure 13) and browse for the .adi file to be included[26].

Beware: the whole of the .adi file will be included so make sure the file only includes the data for the contest! Any records outside of the contest duration will be marked invalid but make cause problems with e.g. duplicate QSO marked within the contest. If the adi file is imported multiple times then many duplicate contacts will be created (as duplicates).

Concurrant logging is also now possible, and the preferred option, see section 3.6.1.

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[23] In V2.0 the options were locked to prevent changes; this restriction has been removed so be careful.
[24] Any of the **'Mandatory Fields'** not ticked is prohibited.
[25] If you switch back from MGM to e.g. USB in a mixed-mode contest you must re-enter the default report on the first contact.
[26] Defaults to the Minos log location.

If you have cause to use “**Insert Before**” or “**Insert After**” (Section 2.2.5) with an MGM contest don't worry about the SerialTx being blanked out, resulting in a serial of 000.

2.2 Entering QSO details

When the Data Entry panel is displayed the cursor will initially be in the Callsign field; to return to that field click on it with the mouse or hit F1. By default the Mode (top left of data entry panel) is set to USB, but it can be changed by means of the associated pull-down list, or will track changes made to the physical rig.

There is no required sequence of data entry, although the default order of the boxes (as selected by the **‘TAB’** or **‘RETURN’** key) reflects normal contest exchanges according to the setting of **‘S&P’/‘Run’**. This facility is enabled via **“Tools/Options/Display Options”** and ticking **“Change Tab Order for S&P”**.

‘RETURN’ is normally used to move between fields[27], by selecting the next empty/invalid field[28]. For the RS(T) fields, if Autofill has been set[29], the remainder of the field will be filled with 59 or 599 as appropriate to the mode (not for MGM); any data entered will not be overwritten.

Fields may be selected ‘out of order’ using the mouse or the designated Function key (F1/F2/F3/F4/F5 in Figure 19); a field may be partially filled (although of course must be completed before the QSO can be normally logged – see section 2.2.2 for details of logging incomplete QSOs).

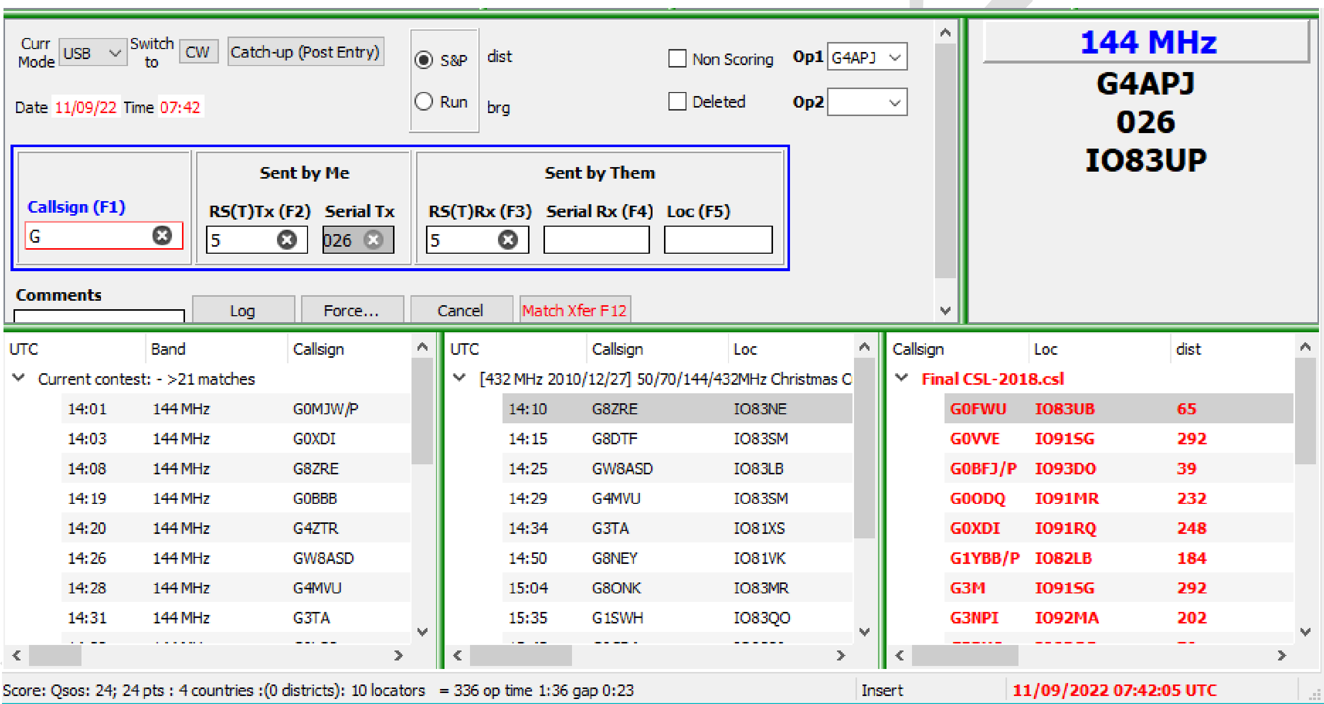


Figure 19 – Details of contact

The contents of individual boxes can be deleted by clicking the white ‘x’ on the black background at the end of each box. This replaces the experimental ‘x’ above those boxes in the QSO panel.
Additional buttons and functions become available when Bandmap is active – see Section 4.3.5.3 for details.

---

[27] Clicking on the “**Log**” button has the same effect.
[28] For a new contact this will be the same sequence as if TAB were used.
[29] This is set under “**Tools/Options/Display Options/Signal Report AutoFill**”.

Upper case is forced as data is entered in the Callsign, Loc, or Exchange[30] fields, and Minos displays possible matches (using complex rules documented in section 5.2) from the current contest[31], any other contests that are open, and any open CSL file(s) with a maximum of around 20 entries being displayed.

On RS(T) fields in non MGM contests, only numeric characters and a tone of A are valid; you can enter e.g. 59A for auroral contacts. In MGM contests no validation of RS(T) fields is done. On serial number fields only numeric characters are allowed, except in the received serial number; "-" shows no serial number is received (e.g. for NAC contests) .

As a Locator is entered the distance and bearing is calculated and displayed in the top right of the panel. Partial Locators give approximate figures, designated by brackets.

Note that data is not validated as it is entered, only when you attempt to log the entry. This allows for partial information to be entered (e.g. during QSB) and then be enhanced/corrected later in the contact.

If a suitable match is highlighted (with the grey bar, see Figure 19) then the contents of the selected entry can be transferred to the data entry boxes by clicking on the "Match Xfer" button or hitting the "F12" key[32] (or by double clicking the requisite entry). This *may* be faster than retyping the data, and should reduce typos (but beware, /P stations are not the only ones that may change location/locator details).

The current QSO may be abandoned, and all entry fields reset to default, by hitting the **'ESC'** key or clicking on the **"Cancel"** button. Hitting **'ESC'** or **"Cancel"** again recovers the fields, provided nothing has been typed since the original cancellation.

This allows the re-use of the current contact number; if the QSO has progressed far enough to make this undesirable tick the **"Non-scoring"** box before **"Forcing"** the entry to be logged.

There are a number of circumstances that might require **"Non Scoring"** to be used without **"Force"**:

- All the details have been entered, but you know that the other end missed them.
- To mark the FIRST contact with a station as a duplicate, rather than a subsequent one.
- The contact is known to be invalid – maybe with a station who will be one of your operators on the cover sheet.

Once all fields are filled with valid data, **'RETURN'** will log the contact[33]. This happens when the program sees that there is valid data in each required field, which under some circumstances may be before you think the contact data is complete. In this situation use cursor up to select what is now the previous contact, and edit in the required additions.

If the QSO data is incomplete[34] and you cannot log normally, then click on **"Force"** which will result in the dialogue described in section 2.2.2.

If your computer logging falls behind, and you miss one or more contacts, it is possible to override **'Serial TX'** by double clicking the field and entering the required next serial number. Minos will then check that you intend to do this and that you want to fill in the contacts later.

---

[30] If Postcode multipliers or similar are in effect.
[31] Duplicates are always displayed first.
[32] Watch out: the callsign may also get amended from that entered (e.g. /P added).
[33] The 'running totals' at the bottom of the Logger will be updated (see Figure 16 and Figure 21).
[34] For instance, some stations may not know their locator; you can work it out later from the geographical location they give you and then edit the QSO details.

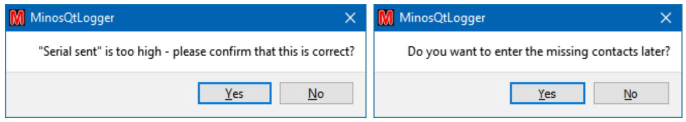


Figure 20 – Skipping contacts

Dummy entries will be created for the missing serial numbers. You can then fill in these missing contacts later (see section 2.2.4).

If you send some serial numbers twice, then double click and overtype the serial sent field to make the log reflect what you actually did.

RSGB 6 hour contest rules state: "Entrants may choose between any continuous 6 hour period in which to operate (e.g. 1500-2100, or 1917-0117) or alternatively split the 6 hour period into any number of segments as long as they are separated by at least 1 hour off time."

The bottom line of the Logger panel (see Figure 21) provides information to facilitate such operation:

"op time" is the total operating time the adjudicator would see if the log were entered as it stands, so should not go over 6 hours.

"gap" is the gap since the last QSO. If taking a break then this should be 60 minutes or more for the break to be included. **Beware:** if op time is 5hr 40 mins, and gap is anything over 20 mins, don't (re)start operating until the gap is up to 60 mins, or any following QSOs will not count!

2.2.1 Colour Coding

During normal operation various colours are displayed on the screen to indicate status or progress. Figure 21 shows an example part way through contest[35].

[35] The screen layout has been modified from the default to remove the RigControl and Rotator application panels.

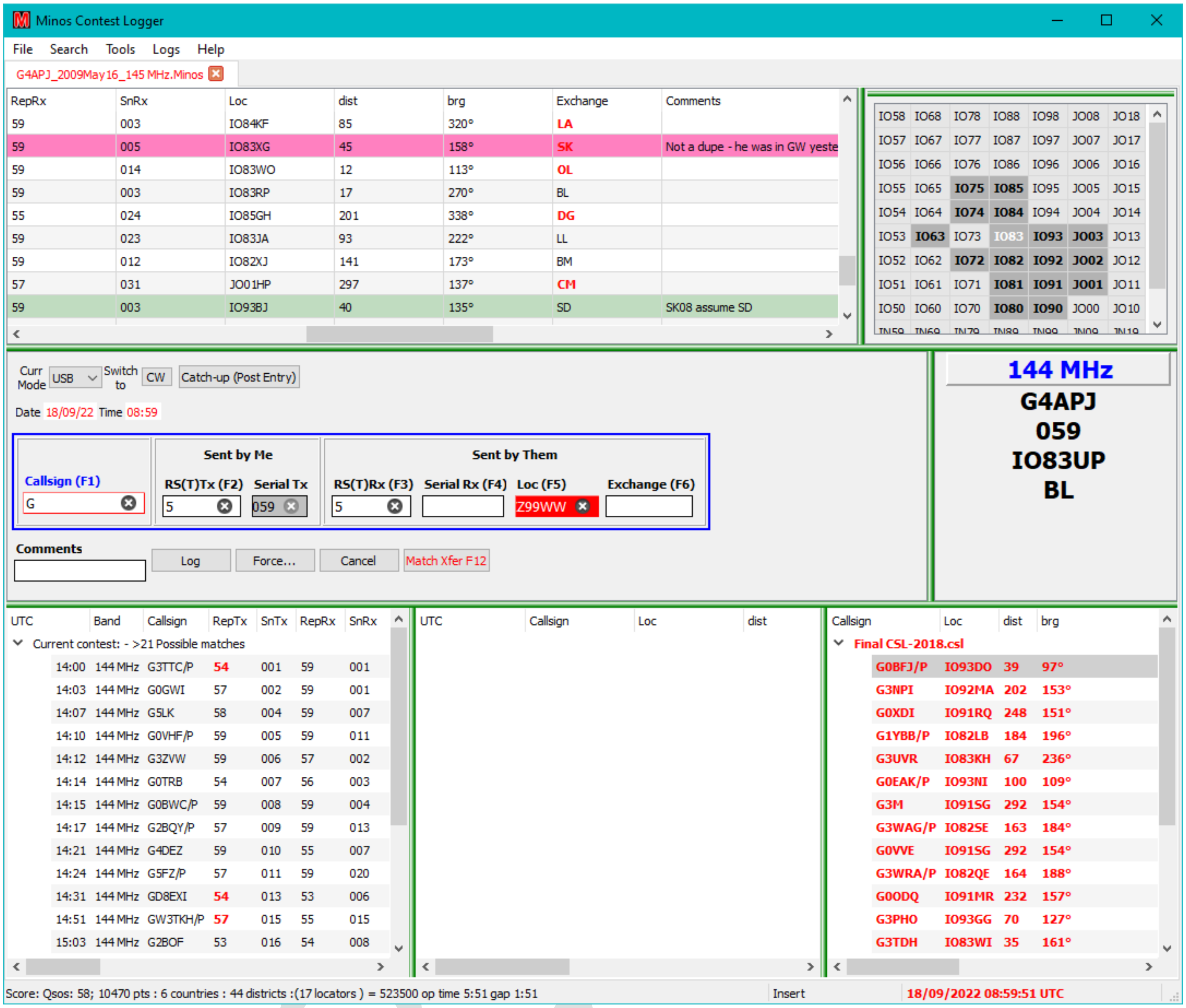


Figure 21 – Colour coding

On the main display panel:

- **Red Callsign**: New DXCC multiplier (where DXCC multiplier scoring active)
- **Red Loc**: New Maidenhead Locator (with Locator multipliers active)
- **Red Exchange**: New Postcode (or other Exchange, when multipliers active)
- **'Greenish' horizontal bar**: QSO where the details have been edited
- **'Mauve' horizontal bar**: record had to be forced; turns green once missing data entered
- **Colours in Locator stats:** Depending on the bonus scheme in use, various colours may be displayed[36]; grey indicates no bonus scheme in use. Pale indicates unworked, bold indicates worked.

Within the data entry area:

- **Red Date or Time**: current value is invalid for the designated contest duration (outside the contest time limits).

[36] Which also show up in the other panels displaying Locator information.

- **Blue text** above the data entry boxes: indicates the currently selected field
- **Red box** around callsign or locator indicates incomplete data.
- **Solid red fill** shows possible invalid data (including duplicate callsign)

Inside a match box:

- **Blue horizontal bar**: Currently selected match that will be used in a transfer operation (using F12).
- **Red Callsign/Loc/Exchange**: As per main panel

The content of the bottom right panel ('CSL match') is in red to show that panel is selected. See Figure 19 for an example of another contest panel selected.

2.2.2 Forcing a contact

If the data for a QSO is incomplete and cannot be normally logged then the "**Force**" button (or **"Alt/F"**) can be used to store the current details pending further communication with the station or later calculation (e.g. when the other station does not know its Locator). This results in a panel as shown in Figure 22.

The current errors in the QSO are displayed, along with various options that may be selected:

- **Unfilled**: Data is missing
- **Backpacker**: If a back packer, or other station, has moved since you worked them earlier this allows the second contact to be marked as valid rather than a duplicate.
- **Insert manual score now**: (if you really want to).
- **Non Scoring**: (the default)
- **Don't Print/Export Entry**: to logically remove a contact
- **Force Country for Multiplier**: To force the use of a particular country for country multipliers.
- **No District Available**: ('auto' for non GB stations)
- **Cross band**: half score

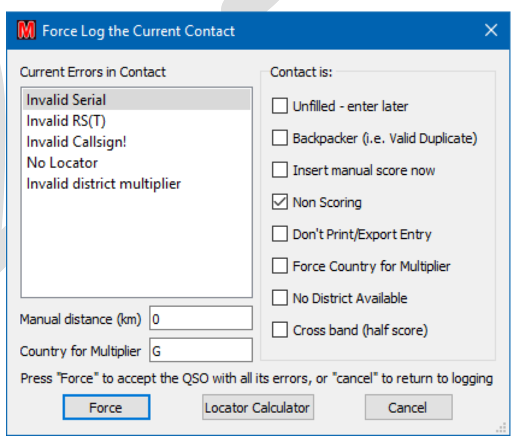


Figure 22 – Force log dialogue

The **'Current Errors in Contact'** may also indicate **'Locator probably not within country'**.

Clicking **"Force"** (check focussed, so also **'RETURN'**) actually writes the record.

2.2.3 Modifying a previous contact

As and when more data is forthcoming for a Forced log entry the record may be edited to enhance or correct the QSO entry. Double click the requisite entry, which results in a display of the form shown in Figure 23.

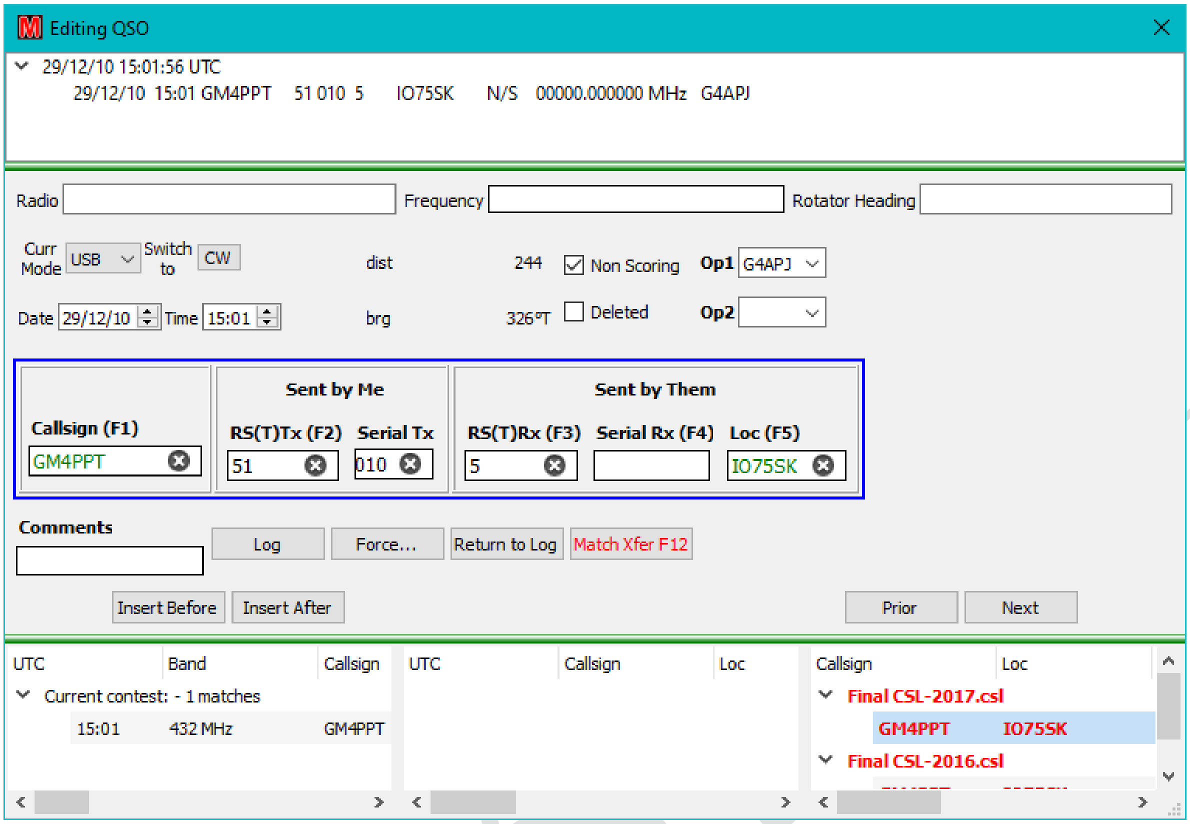


Figure 23 – Editing the details of a QSO

Fields can be modified in standard windows fashion by clicking into, or double clicking to select the whole of, the existing field and then making modifications. When date or time fields are thus selected there is the additional option to click the up-down controls that appear.

Once the data has been completed, ensure **'Non Scoring'** has been cleared (just above the Locator field) before logging the record by clicking **"Log"**[37]. If the data is still not complete then **"Force"** will invoke the force panel dialogue.

The QSO may also be deleted using the tick box **'Deleted'** just to the left of Op2 and clicking **'Log'**.

The modified entry in the main log display turns green (see section 2.2.1 for information on colour coding of records) if all information has now been supplied.

A message box (Figure 24) allows confirmation that the changes should be applied. For valid entries the distance and bearing are then calculated and the record is included in the score.

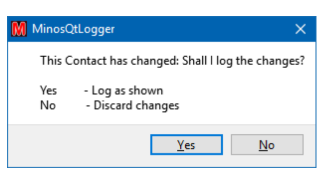


Figure 24 – Confirmation of changes

[37] Note that if the serial number sent is changed it is assumed that it is a correction and no intermediate serial numbers will be offered to be created.

“Prior”, and **“Next”** may also be used to navigate between entries that still require amendment. If the current selected entry has been modified when one of these buttons is selected then the confirmation of changes dialogue will appear before the action is taken. If **“No”** is selected then the original record will be restored and remain selected.
“Return to Log” saves any changes made and finishes the QSO editing activity.

Use **“X”** to **discard** any changes that you have made and return to the log.

2.2.4 Completing unfilled contacts

If unfilled contacts have been created then the **“First Unfilled QSO”** button will appear within the QSO data entry panel (see Figure 25).

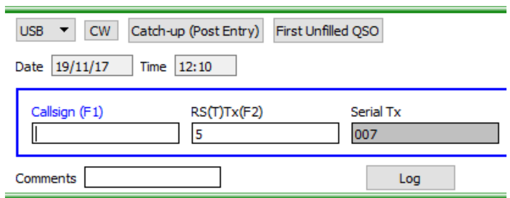


Figure 25 – First Unfilled QSO button

Click on this button to locate the unfilled QSOs. If any are found a similar dialogue to the edit QSO will appear (see Figure 27). Correct the problems, if only by marking the QSO as non-scoring, then use “Next” to move on until all have been processed.

There shouldn’t be any left in the log with **‘UNFILLED CONTACT’** in the comments, nor any incomplete information marked in ‘mauve’!

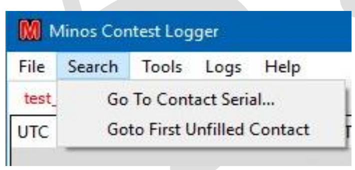


Figure 26 – Search pulldown menu

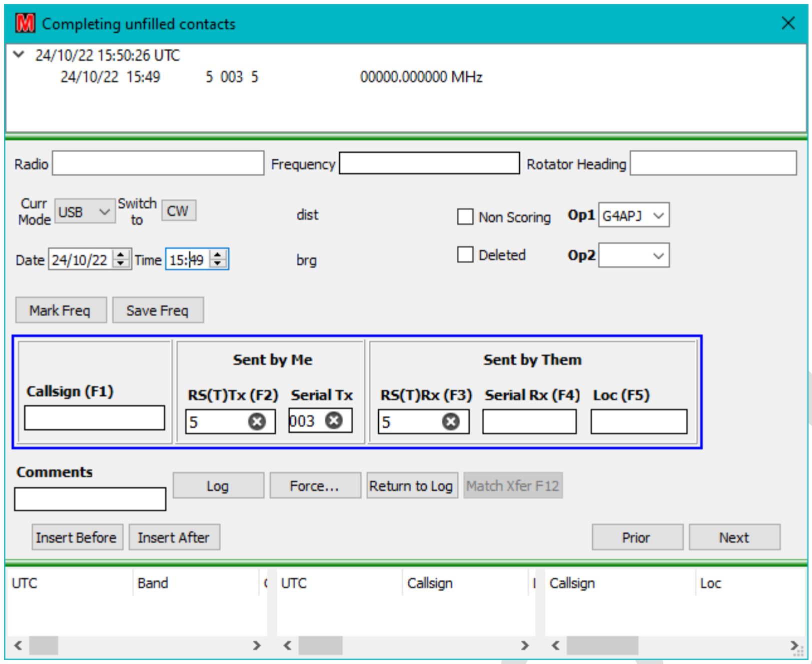


Figure 27 – Completing unfilled contacts

"**Return to Log**" **saves** any changes made and finishes the QSO editing activity.

Use **"X"** to **discard** any changes that you have made and return to the log.

2.2.5 Insert Before/After

When editing a QSO or filling any unfilled QSOs the **"Insert Before"** and **"Insert after"** buttons allow the creation of new QSO records in the indicated position relative to the currently selected QSO using the "Edit QSO" dialogue.

These buttons only become active after the first two records have been written, and do not allow additions before the first record or after the last.

2.3 Submitting an entry

Select **"File, Produce Entry/Export File"** and check that the 'administrative' details displayed are complete and correct (see Figure 28)[38]. Any last minute changes made at this point, e.g. overall comments, are written to the EDI file so that they get seen by the adjudicator; they do not necessarily appear on the claimed score page.

[38] If changes have been made to Entry.ini or the other files since the contest was created then the requisite entry will have to be re-selected to get the data into the Entry data or use **"Entry Details"** at the bottom of the Contest Details form (see Figure 13).

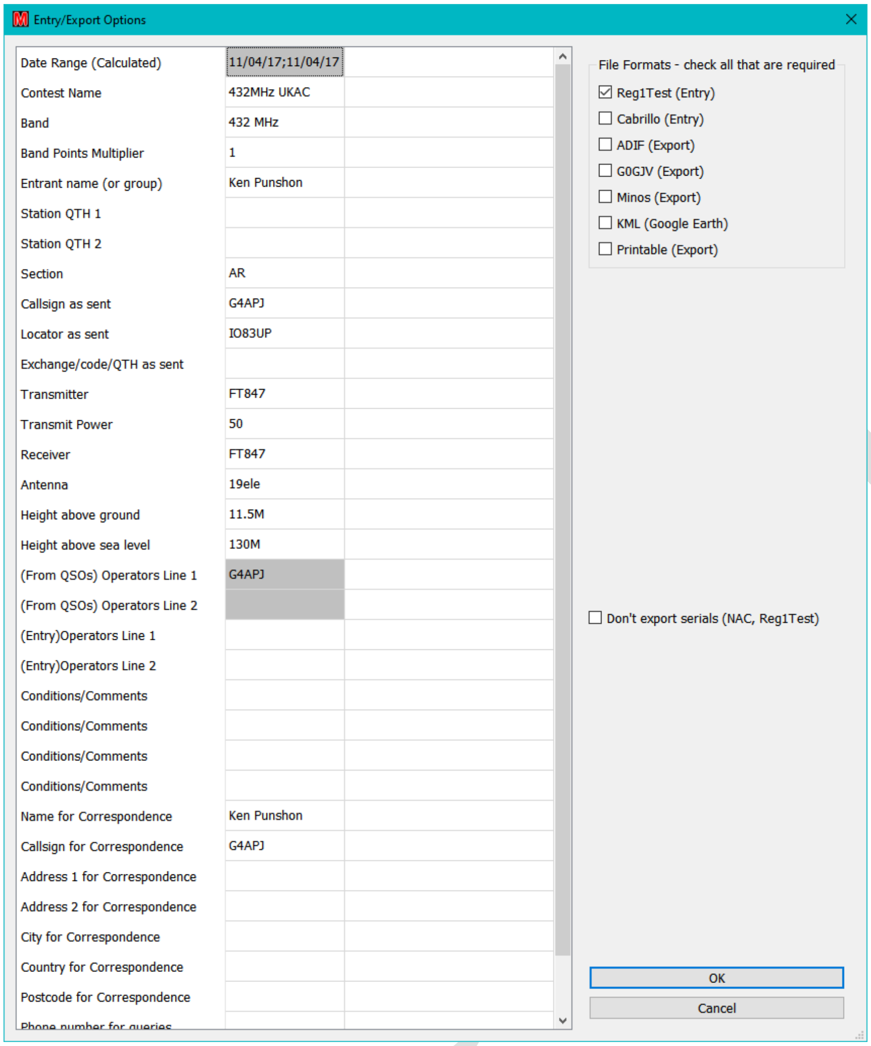


Figure 28 – Checking the Entry data

Select the required format[39] (for non-HF contest submission this is unlikely to be anything other than **Reg1Test** that is accepted by the VHFCC up-loader).

The other formats that may be used are:

- **Cabrillo:** the default, and preferred, format for HF contest submissions worldwide
- **ADIF**: to transfer data to other software packages
- **G0GJV**: (probably now redundant) to transfer to the old DOS Logger
- **Minos**: produces a ‘cleaned up’ log with all edits log cleaned out. It may also be used to produce a shorted abstract, e.g. for a Trophy Contest from a full IARU 24 hour event.
- **KML**(GoogleEarth): If you have GoogleEarth installed, generate one, and double click on it. It gives you all the contacts as pins on GoogleEarth.
- **Printable:** produces a plain text file for checking or printing

As can be seen in Figure 28 there is an option to suppress the export of serial numbers (NAC, Reg1Test).

Click **“OK”** to be presented with the “**Save as**” dialogue.

[39] Multiple formats may be selected; each will produce its own format output in turn.

Select requisite folder and filename (.EDI). You may want to have your own naming scheme if intending to save these files. The default location for the EDI file is the same as the log files.

Click "**Save**" to write the file.

Check the content of the resultant file. Note that in an EDI file any per-QSO comments will have been abstracted and collected under the [Remarks] header in order to comply with the EDI field definitions; this can be confusing at first sight.

For RSGB contests, go to the entry robot (http://www.rsgbcc.org/cgi-bin/vhfenter.pl) to submit your entry (you will get a copy back from the robot by email). If you wish to practise the submission process then take a look at the VHF Log Test Facility on:

https://www.rsgbcc.org/hf/rules/2020/rvhflogtest.shtml and
http://www.rsgbcc.org/cgi-bin/vhfenter.pl?Contest=Log%20Upload%20Tester

To exit the program, use "**File/Exit Minos**" or "**X**". If a contest is truly completed then it is advisable to close it first[40], possibly after marking it as **'Protected'**.

[40] The abstracted content of a contest may then be added to the archive list using one of the CSL utilities, or manually.

3 Additional Features

3.1 Post-event data entry

This facility is intended to allow transfer of data from a paper log. Do not use it to tidy up after the end of a contest.

Create the contest in normal fashion then click on **“Catch-up (Post Entry)”** in the Details of Contact panel (see Figure 29). The date will change to the start date of the contest (watch out in 24 hour, or longer, contest) and the cursor will move to the Time field which is set to 00:00 initially.

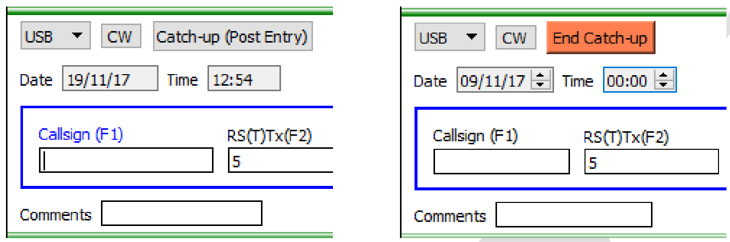


Figure 29 – Catch-up (Post Entry)

Enter the time (and date if needed) of the first contact and then enter QSO data in the normal fashion. The duplicate checking and CSL panels will operate in normal manner.

After each QSO is entered the cursor is set on the last character of the time field to allow amendment of the time to that of the contact. When finished click **“End catchup”** to return to normal operation.

Once all data entry is complete, submit the entry in normal fashion (see section 2.3).

3.2 Auxiliary displays

By default the top right of the Logger display is used to show auxiliary data in a single panel. “**Tools/Screen Layouts**” (see section 3.5) is used to control the number and position of such Auxiliary Displays and the default content.

A pull down list at the top of each panel controls what content is displayed. The panels selected are held within the contest context and are preserved when a contest is closed.

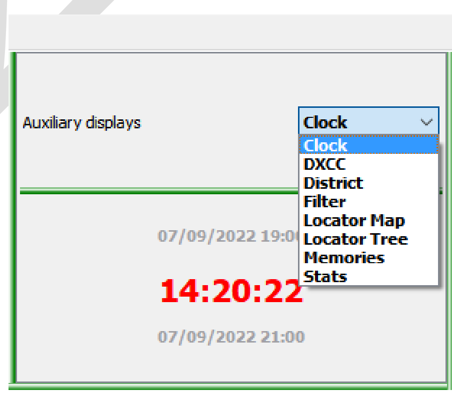


Figure 30 – Auxiliary display options

Different options may be selected to be displayed for each open contest, although **any filters configured apply to all (relevant) contests**. The option selected, and the filters applied, are tied to the contest and stored in the log (Minos) file.

3.2.1 Clock.

The default is to display the current time in UTC. Blue means (the currently selected) contest is in progress, red means it has either finished or not yet started. The start and finish dates/times are displayed in black above and below the current time.

Figure 31 – Clock panel (with header hidden)

3.2.2 Statistics

There are a number of statistics displays that can be selected from the pull-down list:

- **DXCC -** displays the DXCC entities, under control of the selected filters. Most of the time only the European entities will be meaningful.

Auxiliary displays — DXCC

Call	Wkd	Locator	brg	Country	Other calls
EI	1	IO53XD	264°	Ireland	EI EJ
G	48	IO92GS	150°	England	2E G M
GD	1	IO74RE	293°	Isle of Man	2D GD GT MD ...
GI	1	IO64PR	295°	Northern Ireland	2I GI GN MI MN
GM	3	IO76VT	342°	Scotland	2A 2M GA GM ...
GW	4	IO82DG	212°	Wales	2W GC GW MC...

Figure 32 – DXCC Statistics

- **District** – In Postcode exchange contests can be used to display the Districts, under control of the selected filters.

Auxiliary displays — District

Code	Wkd	Locator	brg	District
AL(G)	1	IO91TR	148°	St Albans
BA(G)	1	IO81UF	180°	Bath(Frome)
BD(G)	1	IO83XW	27°	Bradford(Skipto...
BL(G)	2	IO83UO	180°	Bolton(Bury)
BM(G)	2	IO92BL	168°	Birmingham
BN(G)	1	IO90WT	154°	Brighton

Figure 33 – District Statistics

- **Filter** – controls the display of the DXCC, and Districts content **only**. Selection of the geographical regions is meaningless for the Districts display, only worked/unworked are applied. These filtering controls get saved in the minos file for each contest. The panel select works per-contest

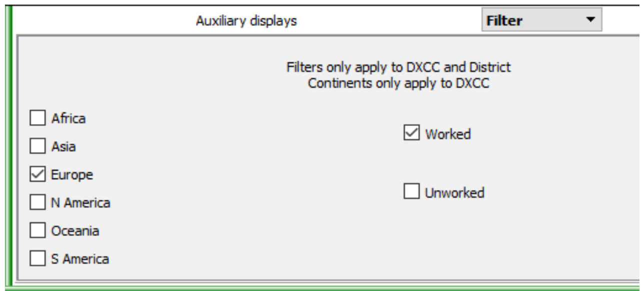


Figure 34 – Statistics Filter

- **Locator** – displays Locators using a colour coding scheme to indicate the relative bonuses[41] (if bonuses are in use). The initial display of squares will be extended, and 'sliders' added, if a square outside of this range is worked.

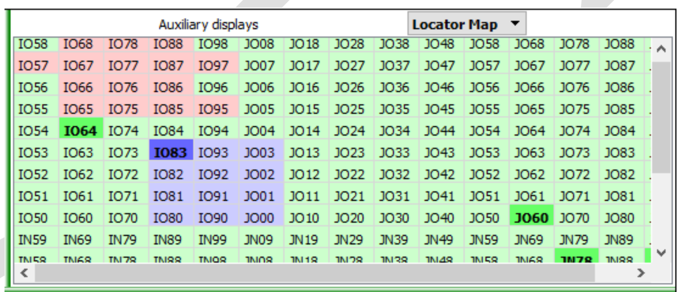


Figure 35 – Locator Statistics (Grid)

Clicking on a square with Locator Map loads the correct bearing to the Rotator panel; it does NOT start rotation.

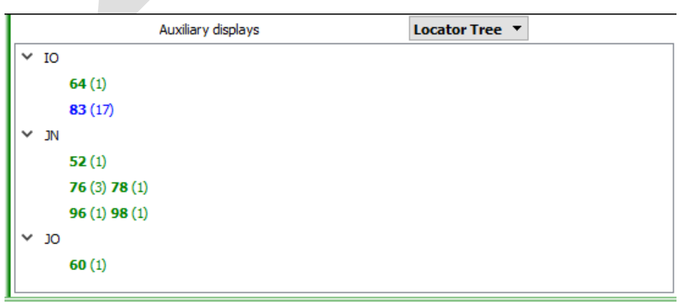


Figure 36 – Locator Statistics (Tree)

[41] Since in the UKAC all squares now have the same bonus this is no longer really relevant. Grey indicates no bonus scheme in use. To get the colours shown, the B2 scheme would have to be in use.

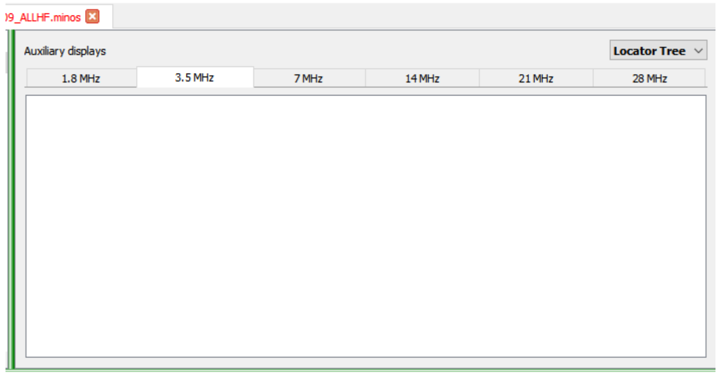


Figure 37 – Locator Statistics with multi-band HF contests

Both Locator displays have a tab for each band in multi-band HF contests (Figure 37).

- **Stats** – displays information on best contact and scoring rates. The intervals for which the DX statistics are displayed can be modified by use of the P1 and P2 boxes at the bottom of the panel.

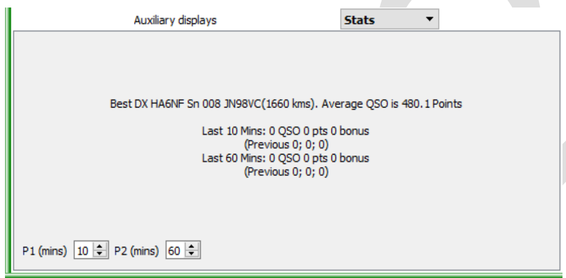


Figure 38 – Summary QSO Statistics

3.2.3 Memories

This auxiliary panel can be used standalone as a ‘scratchpad’ or in conjunction with RigControl (see section 4.2.1) to provide the extended function described here. When first invoked there will be no memories present, they have to be created. The created entries are stored within the context of the currently open contest[42] and are preserved when a contest is closed.

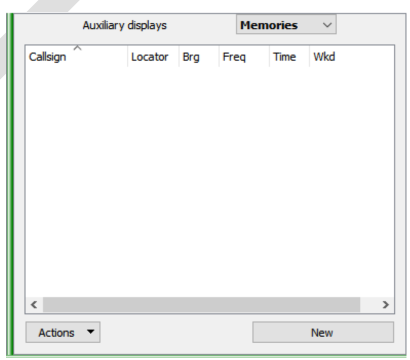


Figure 39 – Initial memories panel display

[42] Hence are stored on the local machine, not any remote machine being used as a connection to the rig.

The order and size of the columns shown in Figure 39 can be changed using standard drag and drop mechanisms, and the display order can be controlled for each (ascending or descending). The column containing the callsign on the left hand side cannot be moved.

If **"New"** is clicked when the QSO entry panel has a Callsign and/or Locator present that data will be pre-loaded into the new memory (it can of course be modified), along with the current time[43] (and bearing if available). If no data is present then an empty memory is displayed (although the time is filled in). See Figure 40 for examples. Clicking "**OK**" then loads the memory into the panel.

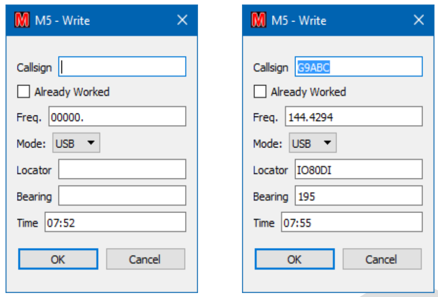


Figure 40 – Example memory content (blank and from QSO entry with RigControl active)

In order to save the memory some text is required in the callsign field; if nothing is supplied then "??" will be used, but beware when loading the contents back into the QSO entry panel.

The **"Already Worked"** box can be ticked if required which will then mark the memory entry in brackets in the same manner as any callsigns found in the log (see Figure 41)[44].

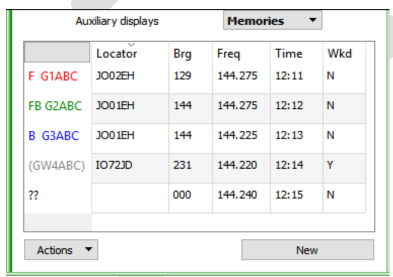


Figure 41 – Populated memories panel

If memories continue to be added then a scroll bar will appear once the available space is filled up. There is no practical limit to the number of entries, but they may be difficult to handle once scrolling starts.

The colour coding of the callsigns, and the preceding letters, indicate memories that 'match'[45] the current rig frequency and the beam heading:

- **Red F** frequency within +/- 2 kHz.
- **Blue B** bearing within +/- 15 degrees.
- **Green FB** both frequency and bearing within the limits.

[43] The time information in a memory is lost if you close the contest or exit the logger.

[44] This might for instance be used to mark a station that you want to work on another band.

[45] The match limits are currently hard coded.

Worked entries (manually designated or detected in the log) are enclosed in brackets. The '??' indicates a memory loaded with a blank (probably unknown) callsign in the QSO panel.

The **"Actions"** button provides a pull-down list of options (as shown in Figure 42) to Read, Write, Edit or Clear the selected memory or clear memories:

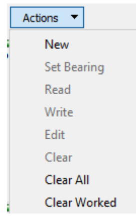


Figure 42 – Memory actions menu

Right-clicking an entry will provide the same list:

- **"New"** is an alternate way of creating a new memory.
- **"Set Bearing"** loads the rotator bearing from the memory into "Target" field of the Rotator panel.
- **"Read"** will load the callsign and locator information into the QSO entry panel, as will a single click on the callsign field of a memory.
- **"Write"** prepares to load the memory from the current QSO fields. It can be manually changed before confirming with "OK"; the current time will always be transferred.
- **"Edit"** is very similar to "Write". It opens the selected memory to allow the contents to be modified (without copying the QSO data).
- **"Clear"** deletes all contents of the selected memory (without confirmation).
- **"Clear All"** deletes all the stored memories (requires confirmation).
- **"Clear Worked"** deletes (without confirmation) all the memories that have been marked as worked as a result of being found in the log. Any entries that have been manually marked as worked are retained).

Memories may also be loaded by right-clicking on a match panel entry (see Figure 43).

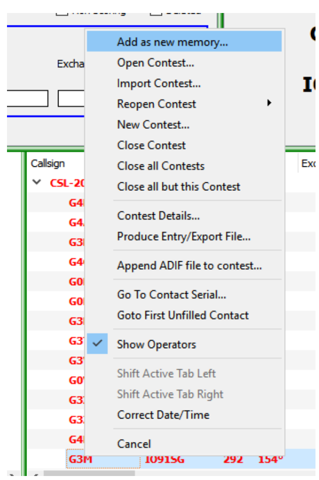


Figure 43 – Loading an entry from a match panel

A click (single or double) in the column to the left of the callsign field is equivalent to a read; a double click on any other part of the entry is equivalent to edit.

Be careful not to (double) click the callsign if editing (or clearing) an entry (e.g. if a station's frequency has changed) because the information will load to the QSO panel and the rig will QSY to the old frequency if you are using Rigcontrol.

If a locator is present in the QSO box then the calculated bearing will be added to a new memory. If there is no locator then a manual bearing may be entered when a memory is created. If a bearing is manually entered and there is a locator present then the calculated bearing overrides the manual data.

3.3 Locator calculator

The Locator Calculator can be invoked from the main screen using **"Tools/Locator Calculator"** or via a button on the **'Force'** panel. If a contest is open the current station locator will be pre-loaded as Station 1.

Depending on the option selected, you can enter a Locator, a Latitude/Longitude, or a (UK) National Grid Reference[46] into the **"Input grid ref"** box (see Figure 44) and convert between the various displays. Selecting **"Calc"** (or hitting **'Return'**) will then calculate and display the other two formats. If both Station details are given it also calculates the distance and bearing from Station 1 to Station 2.

Use **"Exit"**, '**X**', or "**Cancel**" to exit

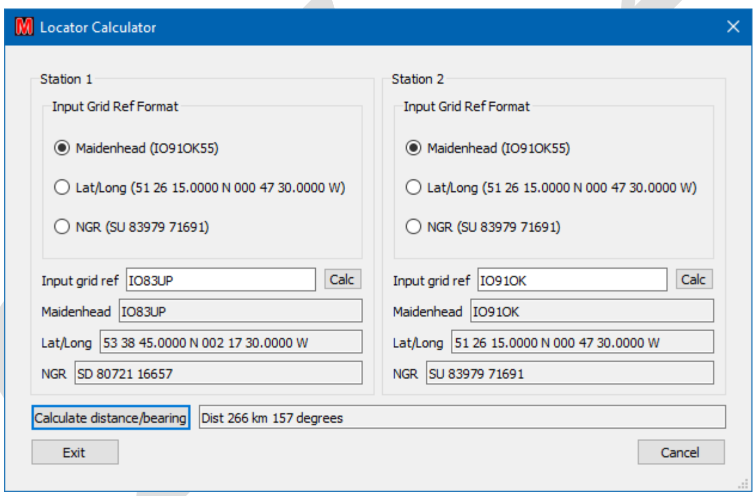


Figure 44 – Locator calculator

If the contest is set to **'allow 4 figure locators'** then the locator calculator will work out the closest point for a 4 figure locator e.g. JO88.

The distance is **always** calculated by the standard algorithm for the distance between the locator square centres.

Microwave locators using 8 figures such as IO80ST37 are supported. If the current contest/list has an 8 figure locator then the distance and bearing will be calculated to the accuracy of the 8 figure locators. During conversions, 8 characters always generated.

[46] Apologies to the Irish: the Irish National Grid requires a different centre meridian to allow for being somewhat west of the rest of mainland Britain, and hence is not calculated.

The precision used may appear excessive, but it is done in this way to enable conversion from NGR to Lat/Long and back again, and have some hope of ending up with the same result. Of course Locators are considerably less accurate.

But please note that precision doesn't necessarily imply accuracy; but those locations that can be checked accurately give a similar result.

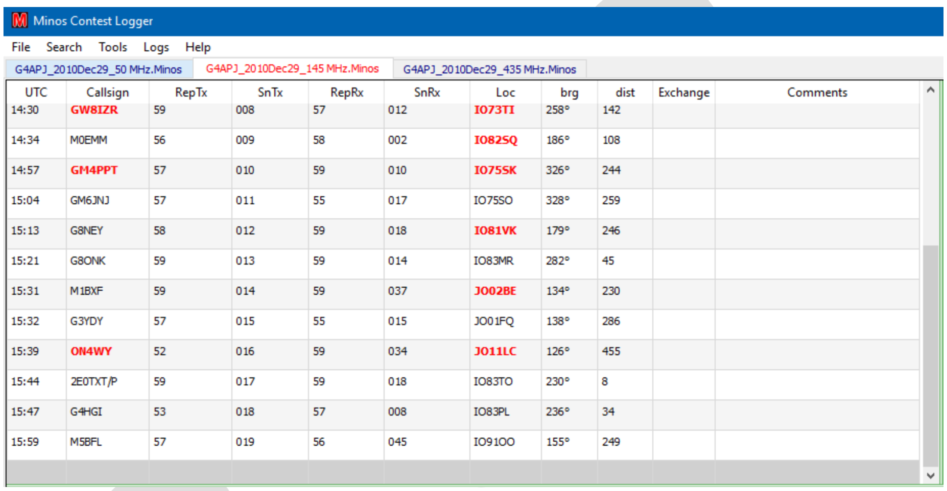
If the entered location is too far outside the UK then no NGR is displayed.

3.4 Advanced operation

3.4.1 Multiple Contests

It is possible to have multiple contests open simultaneously[47]. Each is displayed as a separate tab in the main logging screen as shown in Figure 45. The active contest can be selected by clicking on a specific tab or by using CTRL-TAB to move to the next tab in round-robin fashion.

It is possible to select/open multiple CSL (&LOG) files at once, but each will require acknowledgement. Contest sets (section 3.4.2) simplify the handling of multiple contests.



UTC	Callsign	RepTx	SnTx	RepRx	SnRx	Loc	brg	dist	Exchange	Comments
14:30	GW8IZR	59	008	57	012	IO73TI	258°	142		
14:34	M0EMM	56	009	58	002	IO82SQ	186°	108		
14:57	GM4PPT	57	010	59	010	IO75SK	326°	244		
15:04	GM6JNJ	57	011	55	017	IO75SO	328°	259		
15:13	G8NEY	58	012	59	018	IO81VK	179°	246		
15:21	G8ONK	59	013	59	014	IO83MR	282°	45		
15:31	M1BXF	59	014	59	037	JO02BE	134°	230		
15:32	G3YDY	57	015	55	015	JO01FQ	138°	286		
15:39	ON4WY	52	016	59	034	JO11LC	126°	455		
15:44	2E0TXT/P	59	017	59	018	IO83TO	230°	8		
15:47	G4HGI	53	018	57	008	IO83PL	236°	34		
15:59	M5BFL	57	019	56	045	IO91OO	155°	249		

Figure 45 – Multiband contest

This was originally implemented to allow a single operator to handle a multi-band contest, e.g. the Christmas Cumulatives or all the microwave bands in a contest in the 432MHz-248GHz contest using a single copy of the program[48].

Separate scores, multiplier lists, etc. are maintained for each contest, but the current list of operators is shared.

3.4.1.1 Multi-band (HF) Contests

For multi-band (HF) contests the screen layout will have to be modified (see Section 3.5) to display the **'HF Band Switching'** panel (Figure 46). The orange button shows the currently active band.

[47] Whilst it is possible to have multiple contests open to one band (or one rotator) this is **not recommended** and the results are indeterminate.

[48] Be careful! It is very easy to get confused and start entering data in the wrong contest log. This can be mitigated by use of RigControl (see "**Band Select**" description in section 4.2.1).

Clicking on the buttons will cause the rig to change to the selected band (last used frequency or band default selected through the Options panel – see Figure 85) and any new contacts will be logged in that band.

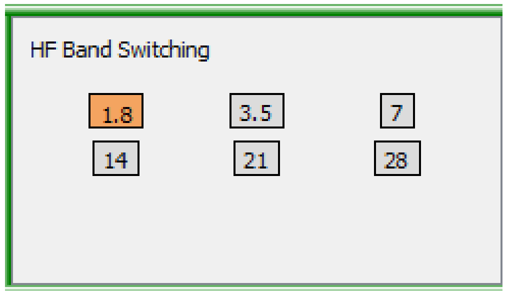


Figure 46 – HF Band Switching

3.4.2 Contest Sets

The concept of Multiple Contests (3.4.1) has now been extended to **'Contest Sets'**. These are managed via "**Logs/Contest Sets/Manage Contest Sets**" (see Figure 47) or the **"Manage Contest Sets"** button on the Welcome screen (see Figure 12).

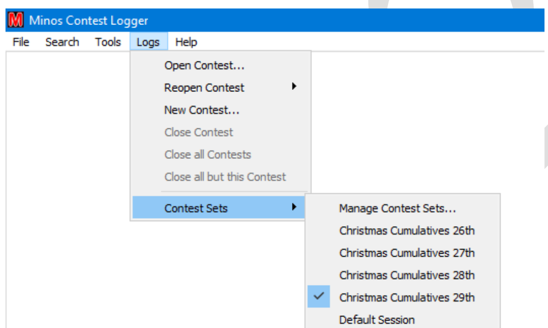
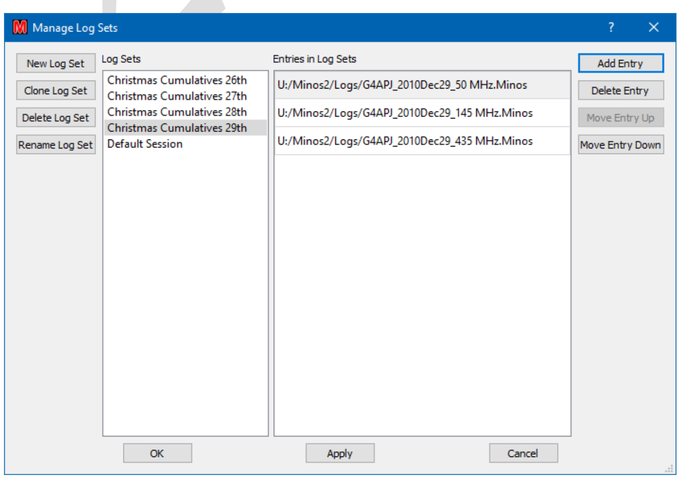


Figure 47 – Opening a Contest Set

Sets can then be created, logs added etc. When the programme is started all members of the highlighted set are opened[49]. If this is for the first time there is no requirement to acknowledge each and every **'Contest Details'** panel as there would be if each was opened in turn (or as a block).



[49] So be sure to leave **'Default Session'** selected if you do not intend to be working with a set.

Figure 48 – Contest Set Management

Contest Sets can be considered as a single entity when opening/closing them. In the example above, selecting **'Christmas Cumulatives 28th '** would close the three logs from the 29th in one go and open all three logs for the 28th.

3.4.3 Archive Lists (CSL)

Warning: however good and accurate you think your list is, DO NOT TRUST IT and don't copy details from it that you have not heard on the air. Remember: people, and especially portable contest stations, move from time to time, or sometimes just work out their locator more (or even less) accurately. Not only that, but if you 'invent' details that haven't been copied off air you are cheating, and if caught, you may well be disqualified from both that and other contests.

The format of a CSL file is described in section 5.1.3. It can be produced manually or the utilities detailed in Section 6 can used to create the file and maintain the content.

CSL files are opened using **"File/Open Archive list"**. Once a list is loaded the file is closed, so that any changes to the file will only be picked up the next time it is loaded. Multiple files may be used which are controlled via **"File/Manage Archive Lists"** [50] (see Figure 49), that also allows additional files to be added via the **"Open List"** button. In the event that a CSL needs to be removed this is also done via **"File/Manage Archive Lists"**.

As of V2.3.1 the **"Manage Archive Lists"** facility has been extended to allow ordering of the lists. Use the **"Move Up"** / **"Move Down"** buttons to move the position of the selected list. 'Other' contests will be searched in the order of the open contest tabs which can be adjusted if required.

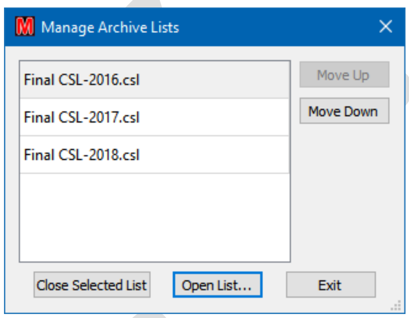


Figure 49 – Manage Archive Lists

All searching takes place in a low priority thread, so it doesn't interfere with other activity. Archive lists apply to all open contests.

3.4.4 Time adjustments

Adjustments to the date and time may be made by selecting "**Correct Date /Time**" under the Tools menu option (see Figure 50).

[50] At least one CSL file must have been opened for this option to be active.

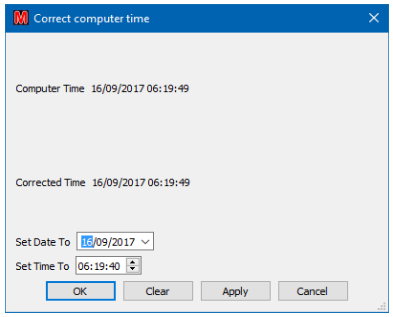


Figure 50 – Clock adjustment

3.4.5 Maintenance

The Contest Calendar should be updated at the very minimum on a yearly basis, preferably more regularly – especially in the first quarter when changes may still be in progress. This is done by clicking "**File/New VHF Contest**", "**VHF Calendar**" or **"uwave Calendar"**, "**Download latest Calendar**".

3.4.6 Additional functions on screen

Right clicking anywhere outside the match panels on the contest panel[51] allows a number of functions (see Figure 51). The first two groups of these are identical to the functions available from the File menu.

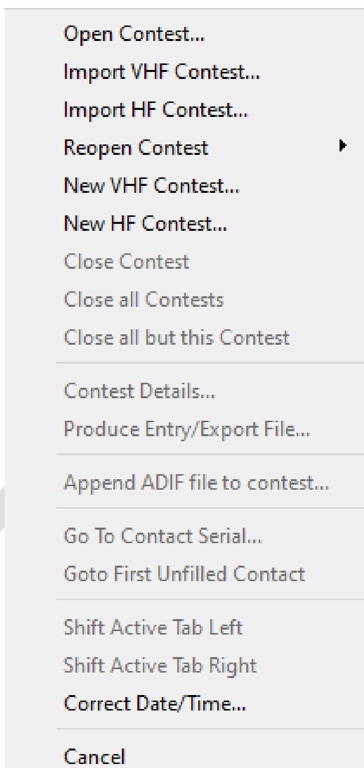


Figure 51 – Right click menu options on main screen

- **Append ADIF file to contest:** allows ADIF from another logger to be imported, as described in section 2.1.2.
- **Go To Contact Serial:** invokes the QSO edit dialogue for the selected contact number.

[51] Right clicking inside a match panel adds the option to add a memory – see Figure 43.

- **Goto First Unfilled Contact:** opens the QSO edit for the first located contact with missing information (section 2.2.4).
- **Shift Active Tab Left/Right:** Move the currently selected tab in the specified direction. The tabs can also be dragged.
- **Correct Date/Time:** Correct the PC date/time (see Section 3.4.4)
- **Cancel:** exits the option panel with no action

3.5 Screen layout configuration

Clicking on "**Tools/Screen Layouts**" allows selection of the screen format to use for the active contest or, by clicking on "**Configure Screen Layouts**" option, the definition of the formats to be contained in the list (Figure 52).

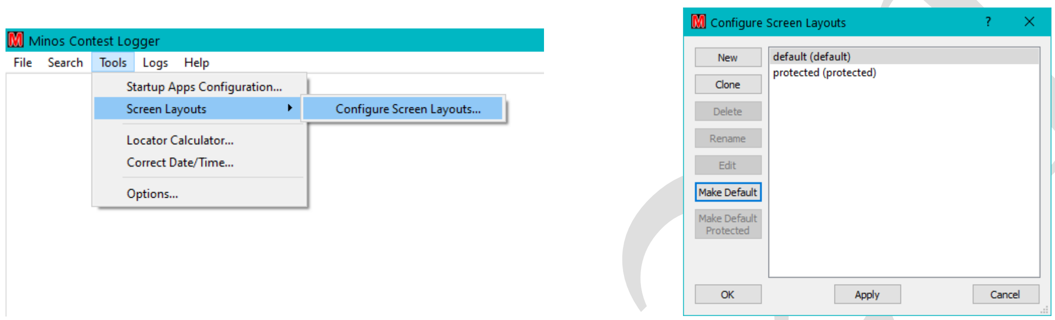


Figure 52 – Invoking the screen layout editor

A new layout is typically created by cloning the 'default' layout and giving it a name (Figure 53). A specific layout can be selected as the default for future contests by clicking **"Make Default"** and/or the default for protected contests by clicking **"Make Default Protected"**[52]. The layout used for an individual contest[53] can also be changed using the **"Screen Layout"** pull-down list on the mid-right of the contest details panel (see Figure 13).

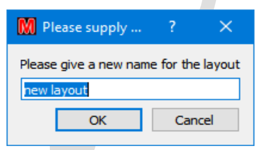


Figure 53 – Creating a newscreen layout

Clicking **"OK"** to a "**Clone**" operation or "**Edit**" to an existing (user defined) layout invokes a panel of the form shown in Figure 54 (the default layout) which allows the user to configure the content[54] and placement of the Logger panels.

[52] Beware:if there is a different layout for protected contests it can become confusing when a contest is (automatically) protected due to its age.
[53] The layout selected for a given contest is retained within the Minos log file.
[54] Possible panel displays are shown in Figure 55; note that only one Rig Control or Rotator Control panel may be selected ithin a given configuration. No error message will be given!

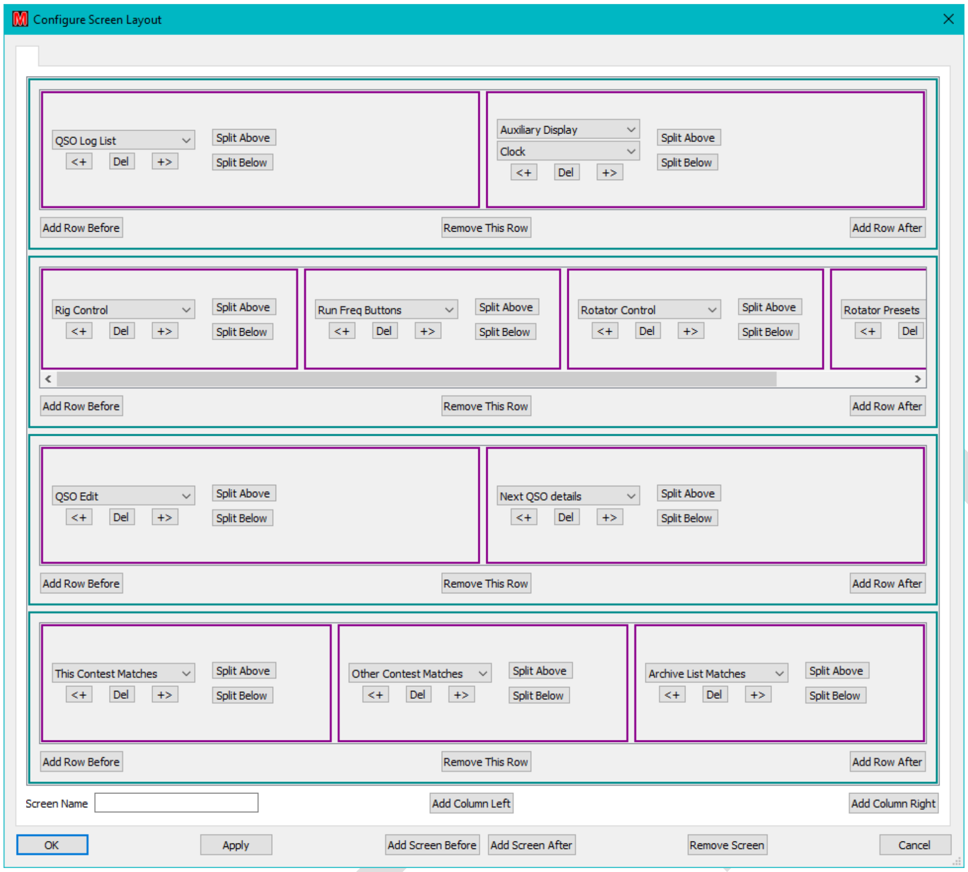


Figure 54 – Default screen layout configuration

Most of the buttons are self-explanatory, but for the sake of completeness a brief description of each is given here:

- **“<+” and “+>”:** Add a box to the left or right respectively of the current box. The content of that box is then selected from the pull-down list:

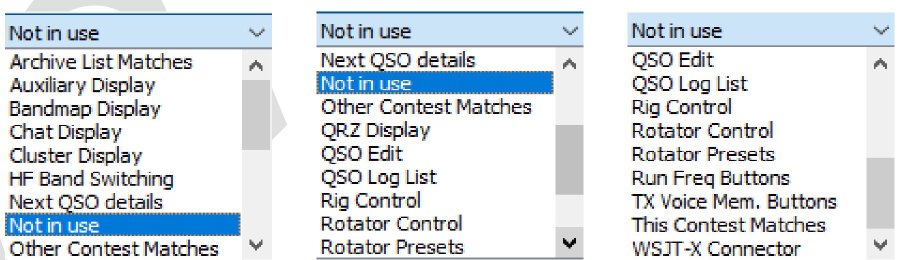


Figure 55 – Panel display options

- **“Del”:** Removes the panel from the display.
- **“Split Above”:** Divides the selected panel into two, inserting the default (Auxiliary) panel above the selected panel
- **“Split Below”:** As **“Split Above”** except the Auxiliary panel is inserted below.
- **“Add Row Before”/After:** Inserts a single panel before/after the row in which it is clicked. Additional boxes can then be added to that row using “<+” and “+>”.
- **“Remove This Row”:** Deletes the whole row (multiple boxes if present).

- **“Add Column Left/Right”** is typically used for the Bandmap Display[55] (Figure 56 right), although other panels may be configured. The extent of the column is marked using the CTRL key (Figure 56 left) and clicking to select the top and bottom rows, then clicking the requisite button[56].
- **“OK”, “Apply”, “Cancel”** options are available
- **Add/Remove Screen:** See section 3.5.1

Note: In Figure 55 the “HF Band Switching” entry in the pull-down list provides the facility to move between bands in multi-band HF contests (see Section 3.4.1.1). Band switching within a single contest is only valid in HF contests.

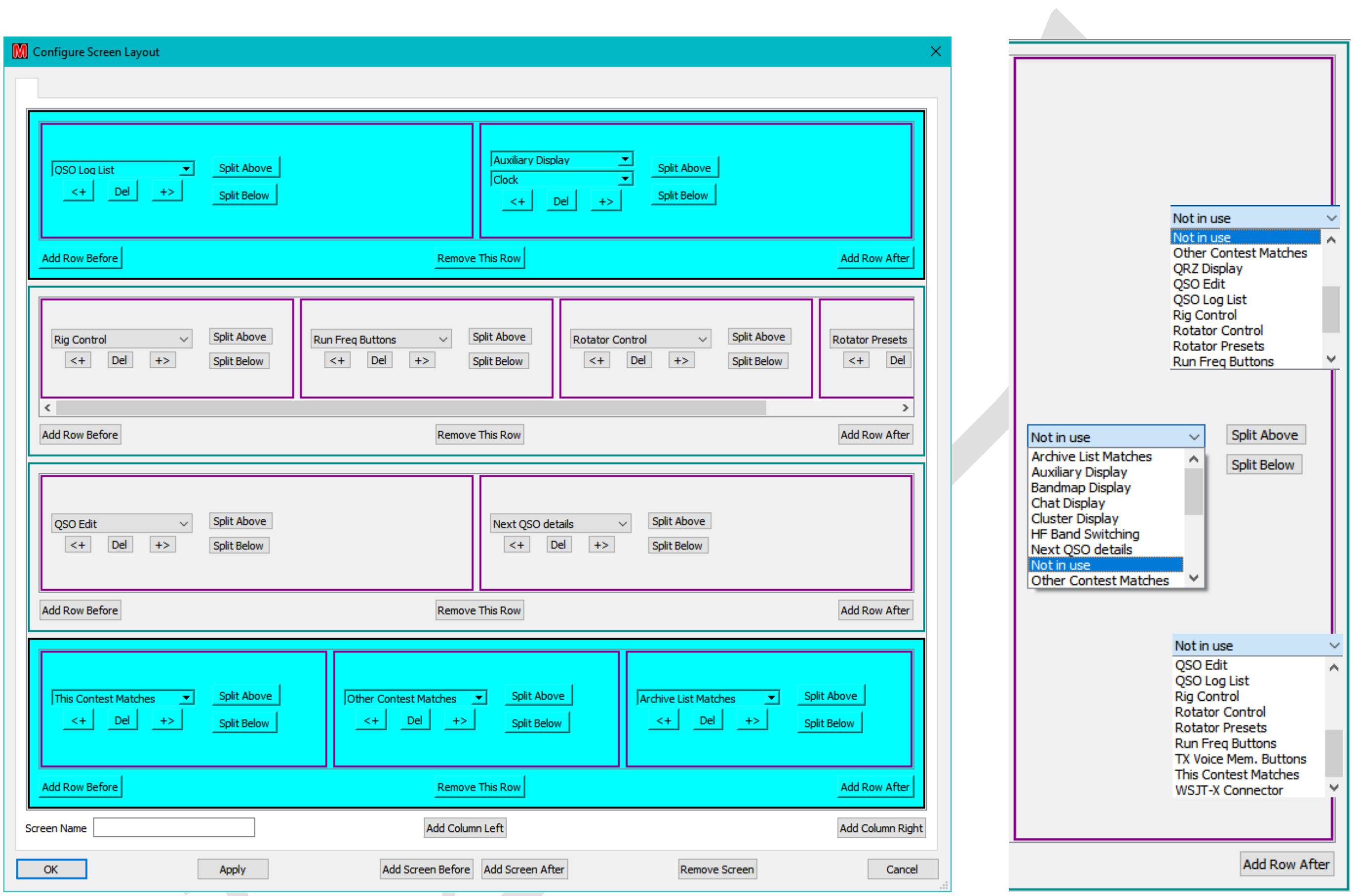


Figure 56 – Add column to left/right

In the event that all rows have been deleted[57] an additional option **“Add Row”** will appear (see Figure 57) to allow rows to be recreated (or click cancel and start again).

[55] Activating the Bandmap display enables extra functionality – see Section 4.3.5.3
[56] It is possible to create very complex (probably meaningless) configurations with multiple columns, but these are not recommended. To unscramble bad configurations create a new clone of the default and start again!
[57] Or none have yet been created in a **‘New’** layout.

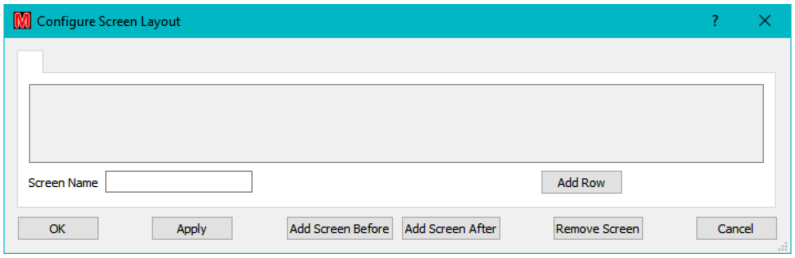


Figure 57 – Blank layout options

The Screen Layout configuration determines which panels are displayed where. To adjust the size of those panels slide the green bars (which change to red during adjustment) in the main Logger panel.

Note that (currently[58]) panels do not appear/disappear as applications are configured, hence reconfiguration is desirable (if no RigControl/Rotator then you probably want to remove that row). The content of visible panels may be affected by the active applications chosen.

There are multiple ways of achieving the same effect; be prepared to experiment/practise **before** using the facility live!

3.5.1 Floating Panels/Windows

The division of the screen panels into multiple 'floating' windows is configured using the buttons at the bottom of the Configure Screen Layout panel:

- **"Screen Name":** Allows the individual screens to be named (in the tabs in the Layout panel, on the floating window header, and when the cursor hovers over the icon).
- **"Add Screen Before":** Creates a new screen immediately before the screen currently selected. The new screen is selected to allow configuration of its content.
- **"Add Screen After":** Places the new screen immediately after the current one and selects it for configuration.
- **"Remove Screen":** Deletes the currently selected screen configuration.

Each screen created results in another icon on the taskbar. By default these are grouped together, but may be separated by ticking **"Separate Logger Icons on Taskbar"** within **"Tools/Options/Display Options"** (Figure 58 left).

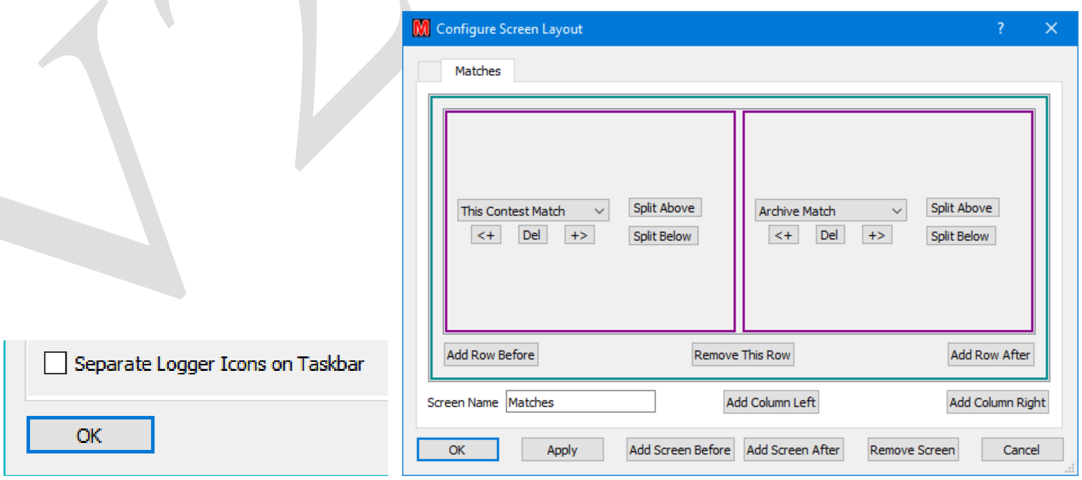


Figure 58 – Floating Panels configured

58 This *may* change at some point in the future; meanwhile use the **"Make Default"** facility.

Only one copy of any 'single use' panel (such as QSO Edit or Bandmap) is allowed, regardless of the number of screens created. If the selected panel is already configured elsewhere in the set of panels then **'None'** will be displayed when an attempt is made to select the existing panel.

Floating panels are of particular use on systems with dual (multiple) monitors.

3.5.2 Set List Spacing Compression

To allow more entries to be displayed within the various frames on the screen, adjust the height of table entries within the main Logger window by selecting the **"Set List Spacing Compression Value as percentage"** option from the **'Display Options'** tab of the **"Tools/Options"** panel (Figure 60)[59].

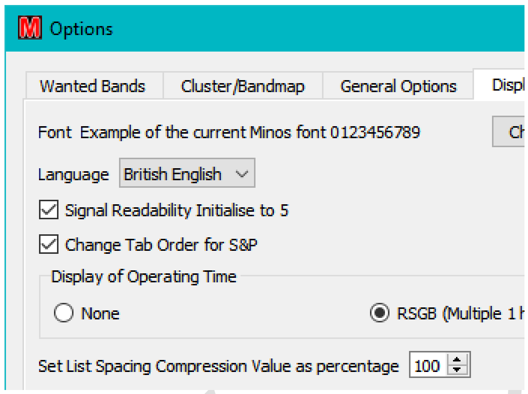


Figure 59 – Default list compression

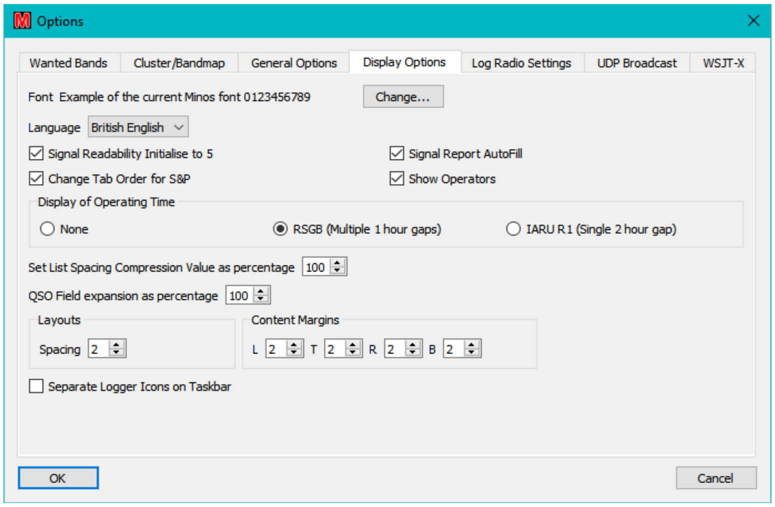
The minimum value without causing legibility problems is likely to be around 75, but it varies with operating system and screen settings. Be careful, if it's too small then Q can appear to be O or 0, and J can look like I.

Be prepared to experiment!

3.5.3 Changing Fonts

The display font in use[60] can be changed by clicking on **"Change…"** in the **'Display Options'** tab of the **'Tools/Options'** panel Figure 60 (left), which invokes a window of the form shown in Figure 60 (right). The default font (shown in that image) is also displayed at the top of the **'Display Options'** tab.

[59] This can be combined with reducing the font size, but it is possible to make entries illegible!
[60] Within the main Logger window(s) only.



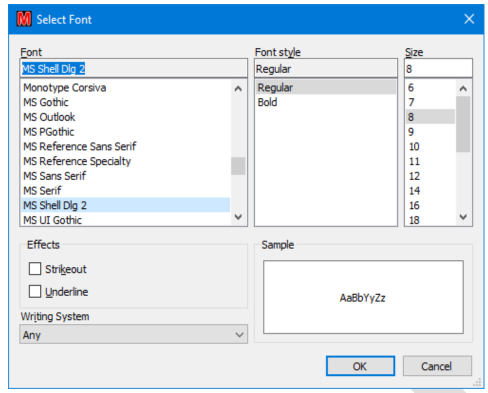


Figure 60 – Font selection

Beware, panel sizes may change and some fonts (notably ‘Wingdings’) can cause complete gibberish. To get out of such a problem, note that:

- **“File”** is the leftmost entry on the top banner of the main Minos window.
- **“Exit MinosQT and Clear registry”** is the last but one entry on the pull down list
- **“OK”** is the left (default) button in the confirmation.

3.5.4 Band marker colour

In the **‘Next QSO details’** panel the band in use is highlighted in colour (Figure 61).

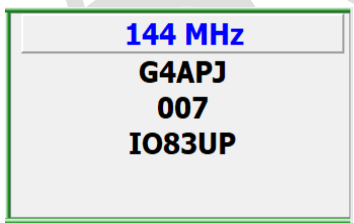


Figure 61 – Next QSO details

If you wish to change the colour used then edit **“\Configuration\bandlist.xml”** to change the colour at the end of the requisite line:

<Band type='VHF' flow='144' fhigh='146' unit='M' wlen='2m' UK='144 MHz' Reg1Test='145 MHz' ADIF='2m' Cabrillo='144' **Colour='blue'**/>

Available colours can be found by inspection of the other entries in the file.

Note that any changes made are lost on re-install!

3.6 External applications

The interoperation with other amateur radio programs has been significantly extended to allow (concurrent) interoperation with multiple frontend communications packages and multiple backend loggers.

3.6.1 Frontend communications

In order to simplify the logging of MGM contests Minos can not only interoperate with WSJT-X but also (at the same time) other clones such as MSHV. RigControl does not have to be active (solving the problem with MSHV port sharing) but this precludes meaningful of the Bandmap feature.

3.6.1.1 WSJT-X

If WSJT-X (or clone) is running on the same machine as Minos (currently the only option), then it should ‘just work’[61], subject to the **“Full WSJT-X link”** (see Figure 62) having been enabled in the options.

Inbound links are configured via **“Tools/Options”** and selecting the **‘WSJT-X’** tab as shown in Figure 62. Although Minos package labelling, and this documentation, assume that WSJT-X is the primary frontend, that is not a requirement and MSHV can certainly be used. However, the functionality may vary from what is described here.

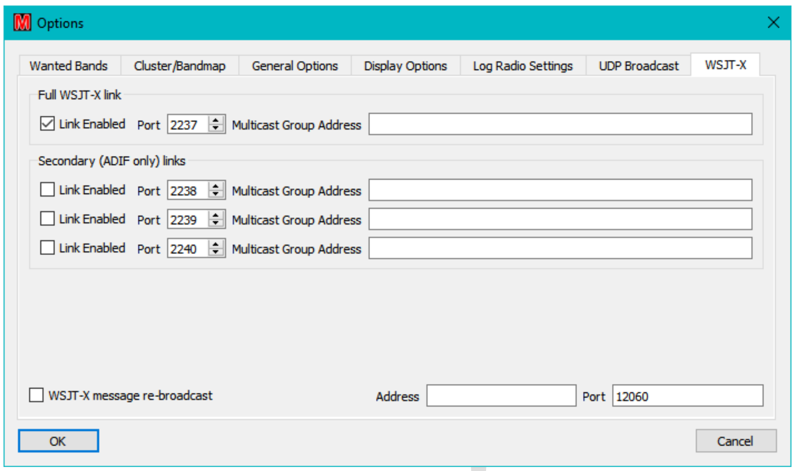


Figure 62 – WSJT-X link configuration

The primary **“Full WSJT-X link”** receives all the messages and must be enabled (**“Link Enabled”**), else the secondaries will not work.

Up to three **“Secondary (ADIF only) links”** may be **“Enabled”**, each having a separate **“Port”** selected. These secondary links only receive ADIF encoded logging of completed QSOs.

Although **“Multicast Group Address”** has been implemented to allow a port to be shared with other program(s) receiving the data, only WSJT-X is known to support such a facility and the field should probably be left empty for the time being.

“WSJT-X message re-broadcast” may be enabled to pass messages on to another program via the defined **“Port”** on the local machine – see Section 3.6.2.2. The use of the **“Address”** field is currently not supported.

In order to view all the WSJT-X messages[62] within Minos the screen configuration must be changed (see section 3.5) to include a “**WSJT-X Connector**” panel[63] which then displays both received and transmitted messages.

---

[61]Remote access is extremely unlikely to be supported in the future for reasons outlined in section 4.2.3
[62] Only possible on the Primary port since this is the only one that receives all the data generated.
[63] Beware: Screen layout only allows one WSJTX panel, but doesn't give an error message – it just defaults to “None” on the additional panel(s).

When WSJT-X (or clone) logs a contact the logging details will be replicated into Minos[64]. All contests listen for such messages and the QSO will be logged by **any** contests that:

- are not protected
- the contest period includes the date/time in the ADIF
- the band includes the frequency in the ADIF

This is regardless of whichever contest happens to be focussed at the time! If the current contest band does not match the frequency information of a packet received on the primary port then a warning will be issued (Figure 63)[65]. This message may appear for a short while on band change, then quickly disappear. If you don't have the WSJT-X window open you don't of course get band warning, but logged QSOs should always end in the correct contest tab(s).

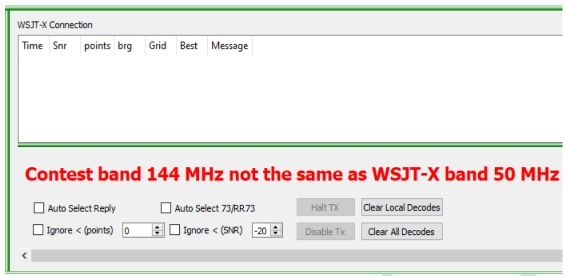


Figure 63 –WSJT-X Band warning.

The layout of the basic connector panel is shown in Figure 64. The colours in the panel are derived from the message aggregator source in WSJT-X and are dependant on position:

- Time: A cycle of four colours to help keep track of the 15 second periods.
- Message: Red if message contains own callsign, green for a CQ call.

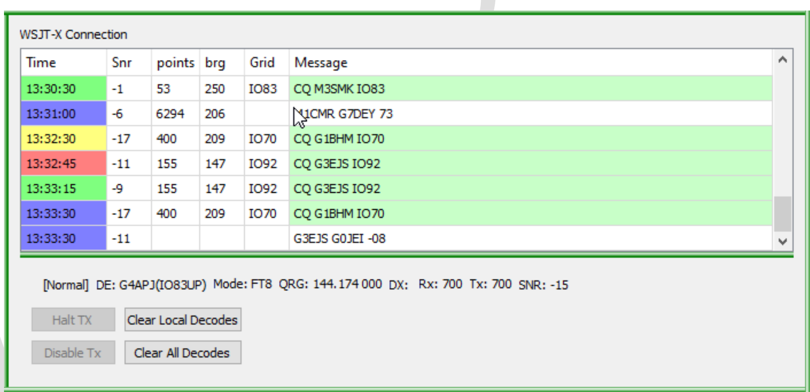


Figure 64 – Basic WSJT-X panel in operation

It is also possible to control some aspects of WSJT-X from Minos:

- Double click a CQ to call the station
- **"Clear Local Decodes":** Clears the MINOS panel content of the WSJT-X Connection display panel.

[64] Rotator Heading will not be available (WSJT-X has no concept of rotators). The bearing will be included, calculated from the Locator.

[65] Actually, the warning is issued as soon as Minos detects the band mis-match which will typically, but not necessarily, be before the QSO is completed and the contact is logged.

- **"Clear All Decodes":** Clears both the display panel **and** the both panels within WSJT-X. (WSJT-X 'clear all' also clears the Minos WSJT-X Connection panel).
- **"Halt Tx":** Immediately stops, and disables, further transmission.
- **"Disable Tx":** Equivalent to disabling the 'Enable Tx' button in Minos, i.e. transmission is disabled at the end of the current period.

The **'Best'** column appears in the display to indicate the highest scoring signal decoded in a CQ or initial response (e.g. G4APJ G0GJV IO91) to your CQ. The selection may be filtered on the basis of:

- **"Ignore < (points)":** Preference will be given to higher scoring QSOs

- **"Ignore < (SNR)":** Marginal signals that may be more difficult to work will be ignored**.**

If there are multiple entries then the best is always the latest so marked.

A number of other additional controls also appear in the display:

- **"Auto Select Reply":** Activates the automatic reply to the 'best' candidate signal. When it starts the reply, it disables itself again, so complying with the WSJT team's wish for every QSO to require a positive operator action.

- **"Auto Rearm":** If the station automatically selected and called goes back to somebody else then restart the selection process.

- **"Config CQ":** Initiates the 'Configure CQ' dialogue (Figure 65) to define which CQ calls will result in an auto-response in both (Con)test and non-(Con)test modes so that in a contest there is no automatic response to non-contest calls and vice versa.

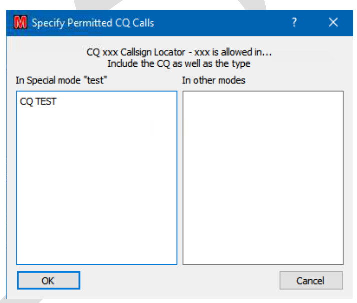


Figure 65 – Configure CQ auto responses

The "CQ" part has to be included as well, e.g. "CQ HQP", or plain "CQ" if you want to respond to ordinary CQ calls as well as the others selected. WSJT-X puts TEST in the CQ call when in EU (or NA) VHF contest modes (as per the MGMAC).

Scans continue until a match is found for a string defined for the current mode which causes a response. Everything processed is displayed.

3.6.1.2 MSHV

MSHV can be used either as a primary or secondary frontend. If operating as a primary then the functionality is very similar to that described above for WSJT-X. As a secondary, only completed QSOs are logged.

Since MSHV does not support sharing of the CAT control via RigCtlD, the ‘PTT via CAT’ functionality is prevented from operating if Minos has control of the port. If the rig supports VOX that should work; using DTR/DTS on a second port might work, but no such port is presently available to test it.

If Minos is going to control the rig then it should be invoked first so that MSHV is ‘locked out’ from the port.

To receive via MSHV you need to invoke **“Radio And Network Configuration”** then tick **“Enable Logged QSO ADIF Broadcast”** and **“Enable Decoded Text”** to activate the logging[66] (Figure 66). The port selected should match the primary/secondary port defined in Minos (see Figure 62).

Minos will ignore any data seen as a result of ticking **“Enable Logged QSO Broadcast”** (there appears to be a bug in it, but it has not been investigated).

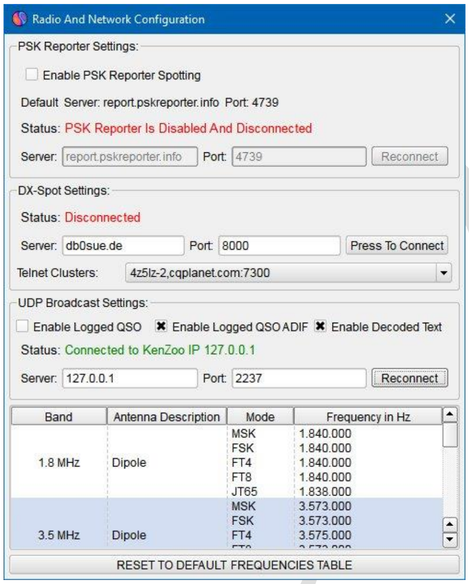


Figure 66 – MSHV configuration

3.6.2 Backend Loggers

Outbound links are configured via the **‘UDP broadcast’** tab of the **‘Tools/Options’** panel as shown in Figure 67 and the WSJT-X re-broadcast link (bottom line of **‘WSJT-X’** tab, see Figure 62).

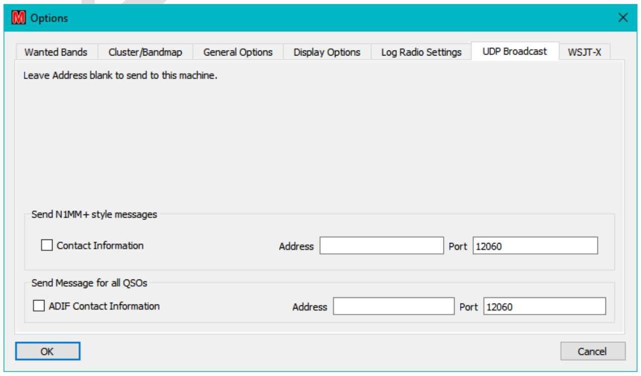


Figure 67 – Output Link configuration

[66] This is with version 2.29 (the latest at time of writing), but may depend on version.

When a contact is logged within Minos the details are also directed to any output ports that are enabled. Non-exhaustive testing has been done with two Loggers (individually and concurrently):

3.6.2.1 Log4OM

Invoke **“Settings/Program Configuration”** then select the **‘Connections’** option. As can be seen in Figure 68, either of the outputs (N1MM+ or ADIF format) from Minos can be used, although slightly different information will be transferred (most notably "N1MM_MESSAGE" includes the "Contest Id").

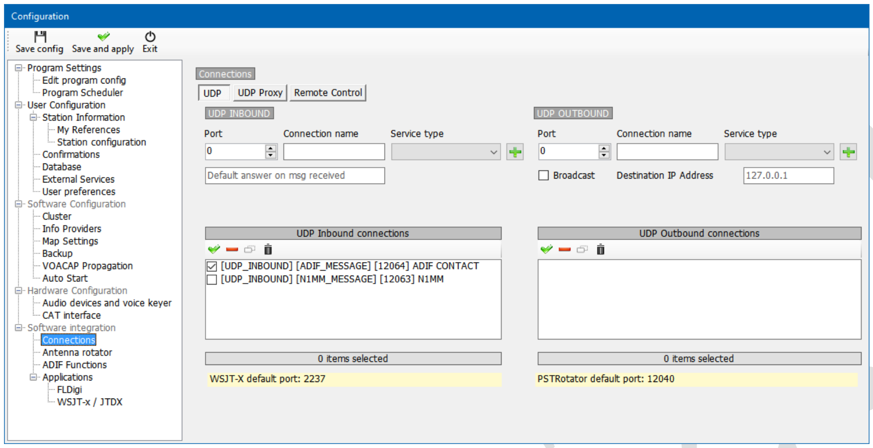


Figure 68 – Log4OM configuration

Contacts logged directly into Minos (as opposed to those logged within WSJT-X) are also transferred to Log4OM. **Beware**: auto upload from Log4OM to QRZ etc should be disabled else the QSO information would be available to the other parties straight away (thus invalidating the contact)!

3.6.2.2 N1MM+

N1MM+ is not really intended to be used in conjunction with Minos, but its use demonstrates the facility to ‘daisy-chain’ WSJT-X messages.

Within N1mm+ click on **“Config /Configure Ports, Mode Control, WinKey, etc...”** and select the **“WSJT/JTDX Setup”** tab (Figure 69). Make sure that the WSJT-X function is enabled, and using the same port as **“WSJT-X message re-broadcast”** (see Figure 62)

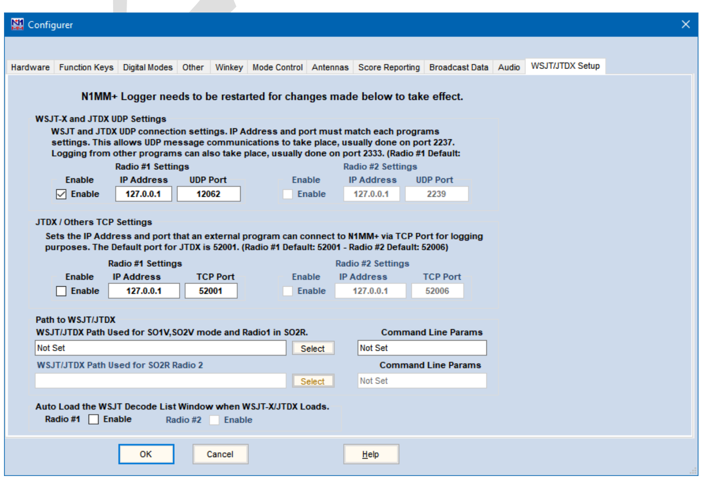


Figure 69 – N1MM+ configuration

Contacts logged into Minos are not transferred to N1MM+

4 Integrated Applications

Two applications form the infrastructure which the functional applications use:

- **MqtServer** is fundamental to the operation of the applications. It is a message broker which runs on all machines where you want the applications. It handles all the message-passing between Logger, apps, and remote servers.

- **MqtAppStarter** is the Application Starter. It is used *instead* of the Logger when you want applications but not the Logger. This will generally only be used in complex multi-machine environments (see section 4.4.3.1).

Only one copy of these infrastructure applications should be run on any given machine at any given time.

4.1 Configuration

In order to operate correctly the application configuration must be defined. The following description is for the use of the Logger to set up the applications. Usage of the Application Starter is covered in section 4.4.3.1.

The simplest was to configure the applications is by clicking the **"Apps"** button on the Welcome Screen which goes straight into the **'Minos App Configuration'** screen described on the following page.

If multiple application configurations are required, e.g. to cope with such things as significantly different configurations between band sets (HF/VHF/uWave) then the **'Manage Application Configurations'** screen (Figure 70) can be invoked either from **"Tools/Startup Apps Configuration"** [67] [68] or by holding the **'Shift'** key while clicking the **"Apps"** button during startup. Multiple application configurations can then be managed in a similar maner to the screen layout profiles described in Section 3.5.

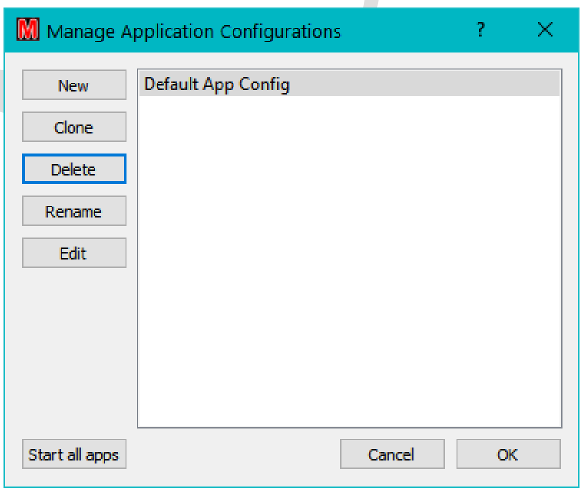


Figure 70 – Application Set configuration

[67] Reconfiguration is only allowed if the applications are inactive – use the **"Stop Apps"** button.
[68] In complex environments it can also be done using the MqtAppStarter application on a machine that is just acting as a server and thus not running the Logger.

The small **'Minos App Configuration'** screen should be expanded from its default size before configuring anything (see Figure 71). To add an entry click "**New Entry**" and select the required application from the "**App Type**" pull-down list. For simple 'single user' configurations all of the information required for the applications is auto-generated[69].

Since **'Server'** is required for everything else it is automatically defined first. **'RigControl'** and **'Cluster'** are the other most likely applications for simple (single machine) environments, plus **'Rotator'** if one is present.

After applications are defined they are by default always all started. To avoid this use **"Logger Only"** in the startup splash screen or, on a more permanent basis, clear the **"Enabled"** box for the application in its definition, (remembering that Server is required to be active if **any** other application will be active).

To store the configuration Click **"Save and Close"** (this will write the information to **'MinosConfig.ini'** and exit the window) or **"Save and Start Apps"** which will then require the **"Cancel"** button to exit the display.

By default **"Hide Server Apps"** is ticked for MqtServer to reduce the number of panels and icons displayed. Depending on configuration it may be advantageous to tick this for other applications.

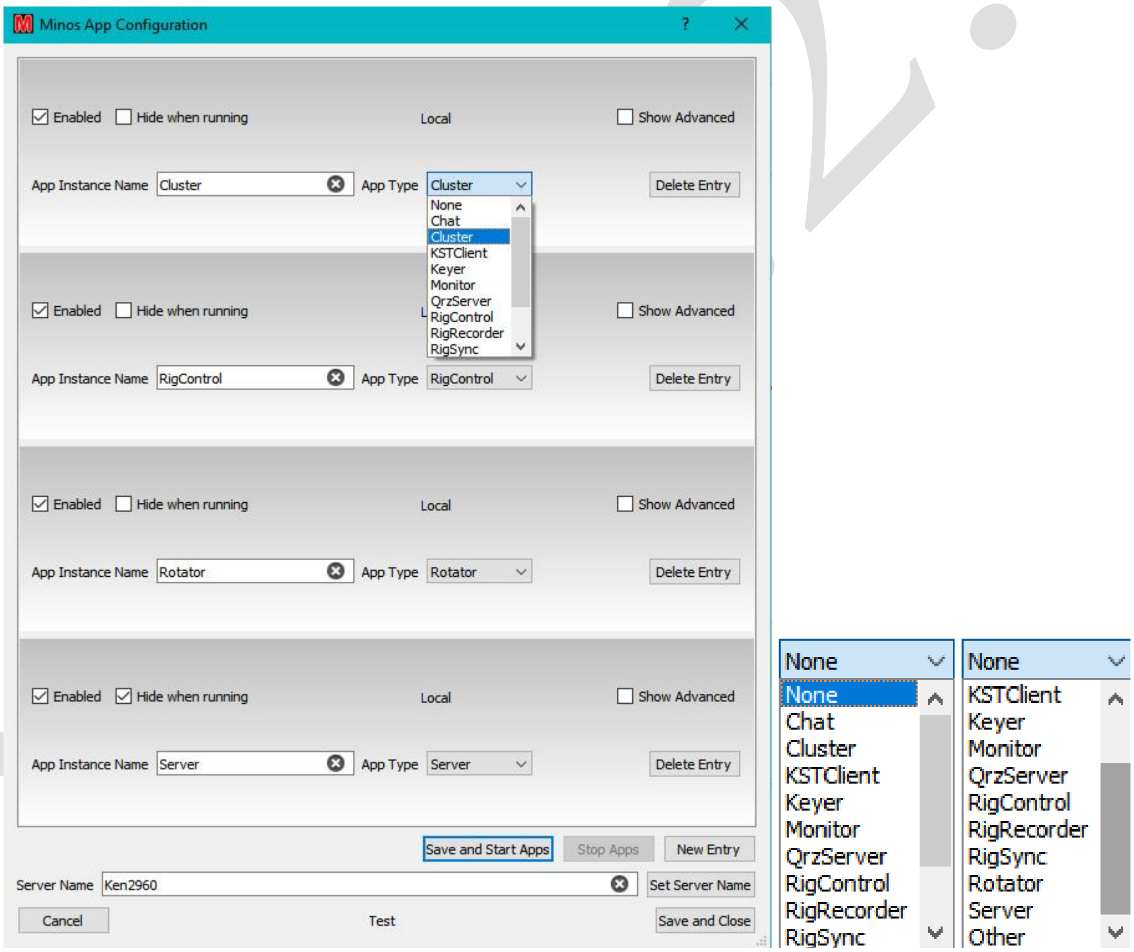


Figure 71 – Typical set of applications (and possible options)

[69] For details of more complicated configurations see section 4.3.6.1.

“Delete Entry” can be used to delete erroneous information, after which **“Save and Close”** stores the changes.

When Server first runs, it requires Firewall configuration as shown in Figure 72.

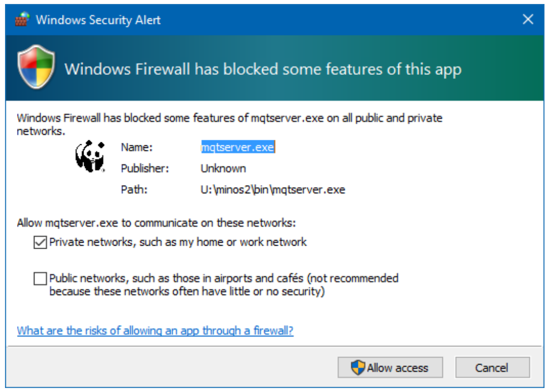


Figure 72 – Firewall message

If **“Hide when running”** has been unchecked for MqtServer then a window will appear that gives statistics on the active software components (Figure 73)[70]. Clicking **“Show”** will give a display of the server state in far more detail than any non-developer is likely to want!

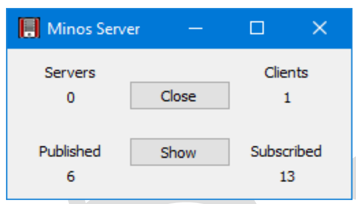


Figure 73 – Server statistics

When the Logger runs, a number of icons will also appear on the Taskbar for those applications that are not hidden. In Figure 74 the icons from left to right are:

- Main Logger
- Server (not normally displayed)
- Chat (not displayed when running as a Logger panel)
- KST Client
- Rig Control
- Rotator
- Cluster
- Monitor
- QRZ Server
- Rig Sync
- Rig Recorder

Figure 74 – Icons on Taskbar

[70] The ‘Servers’ count is more accurately ‘remote Servers’, i.e. excludes self.

The control applications defined in this manner are then linked to individual contests using the pull-down lists from the Logger or from the Contest Details panel[71] (see Figure 75), either when creating a contest or by selecting **"File/Contest Details".**

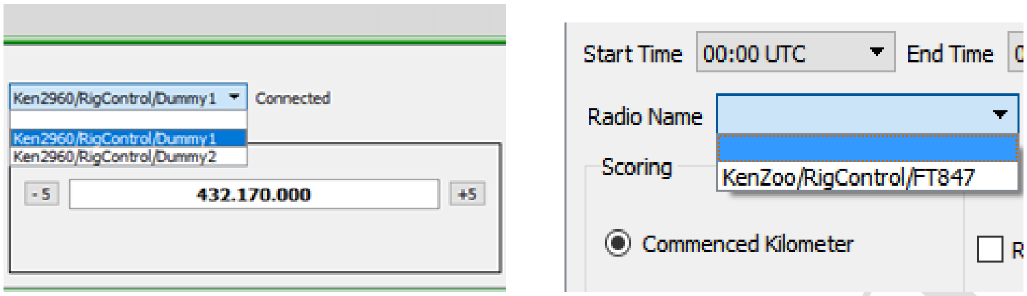


Figure 75 – Selecting a rig from Logger and Contest Details

4.2 Control Applications

These applications allow the control of rotator(s) and radio(s) using separate rotator and rig control applications communicating via (possibly USB connected) serial interfaces or, subject to finding someone who has one so it can be tested, full USB to their respective devices. The applications operate as servers for, and under the control of, the Logger.

While the programs can control a range of rotator controllers and radios using the Hamlib library, it is not possible to test all of these devices as they are not available to the developers. Some controllers may not support all the commands available or a particular controller's operation may not be fully understood.

A list of tested configurations for Rigs (and Rotators) is maintained at https://minos.groups.io/g/users/files
including information on the the IC-9700 which several people have had difficulty connecting.
This location also contains a help file to give guidance with system configuration problems.
Feedback is welcomed on any information or changes that may be required to support as yet untested rotator and radio models.

The control applications can be connected to one or more contest log pages, selection of the rig in use (or rotator, if you have more than one) for a particular contest being via the drop-down lists as described above.

More complex arrangements such as different rigs for different bands are covered in section 4.4.

4.2.1 RigControl

This application connects to multiple rigs[72] (from those supported by the Hamlib library, or via the OmniRig package) using CAT interfaces via serial ports (which may be simulated via a USB device), or via direct USB connection if your rig supports it. The program will also support network radios, for example FlexRadio 6xxx. When you select a radio that supports a network interface, a network address and port number must be entered.

It provides direct control of the selected rig and extends the functionality of the memory Auxiliary panel.

Before the Logger can connect to the RigControl application one or more rigs must be defined (so there is content in the pull-down list). Transverter configurations can also be defined (see section 4.2.1.3) to adjust the frequencies seen in Minos to reflect the actual transmit frequencies, rather than the driver frequency.

[71] Both examples are for rigs, the equivalent mechanism is used for rotators.
[72] More than any normal operator would want to use/can afford.

Selecting **“Configure/Setup Radios”** within RigControl invokes the **‘Radio Setup’** dialogue (Figure 76) that allows the radio(s) to be defined.

“Configure/Configure Rigctld” opens a File Explorer window to allow a different RigCtld to be selected (see section 4.2.3).

“Configure/Trace Data Comms” enables more detailed trace information for diagnostic purposes.

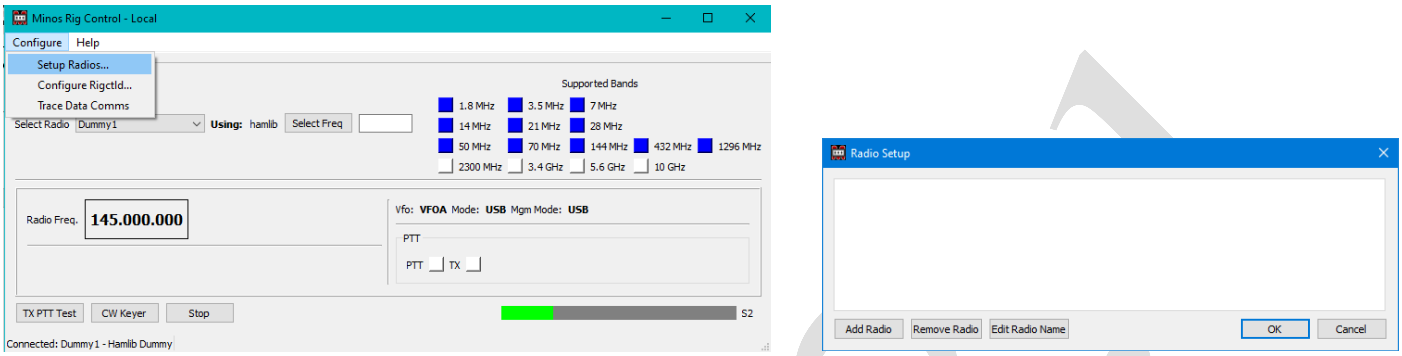


Figure 76 – Starting configuration of RigControl

Clicking **“Add Radio”** starts the definition of a rig (Figure 77). Note that Omnirig only appears in the pulldown list if it has been installed in the default location.

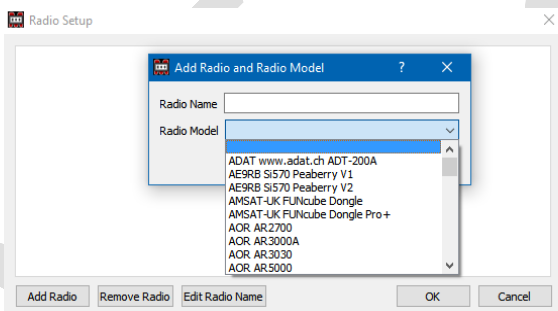


Figure 77 – Selecting the radio model

The **‘Radio Name’** field is used within a contest to select (or display) which rig is currently being controlled (e.g. in Figure 87). Once defined, it may be changed using **“Edit Radio Name”** or removed (provided it is not currently active within the Logger) using the **“Remove Radio”** button.

‘Radio Model’ is a pull-down list containing an extensive range of rigs supported by Hamlib. Note that one of these is **‘Hamlib Dummy’** which can be quite useful for testing/learning purposes. If OmniRig has been downloaded and installed (in **C:\Program Files (x86)\Afreet\OmniRig**) then OmniRig Rig 1 & 2 will also be included in the list. Using an OmniRig entry requires the OmniRig configuration to be done externally and the requisite **“Supported Native Bands on Radio”** to be enabled (see Figure 78). Be aware that OmniRig can sometimes take up to 60 seconds to start which will delay the frequency being displayed in Minos. (This happens to other programs that use OmniRig and is nothing to do with Minos.)

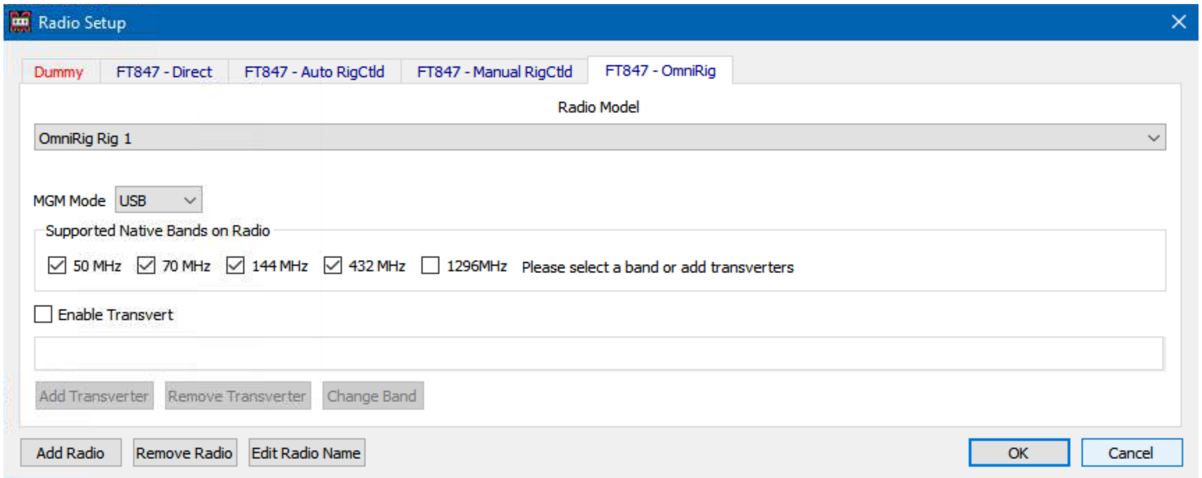


Figure 78 – OmniRig Configuration of supported bands

Once the name and model are defined clicking "**OK**" continues to the setup of the interface to the device (Figure 79)

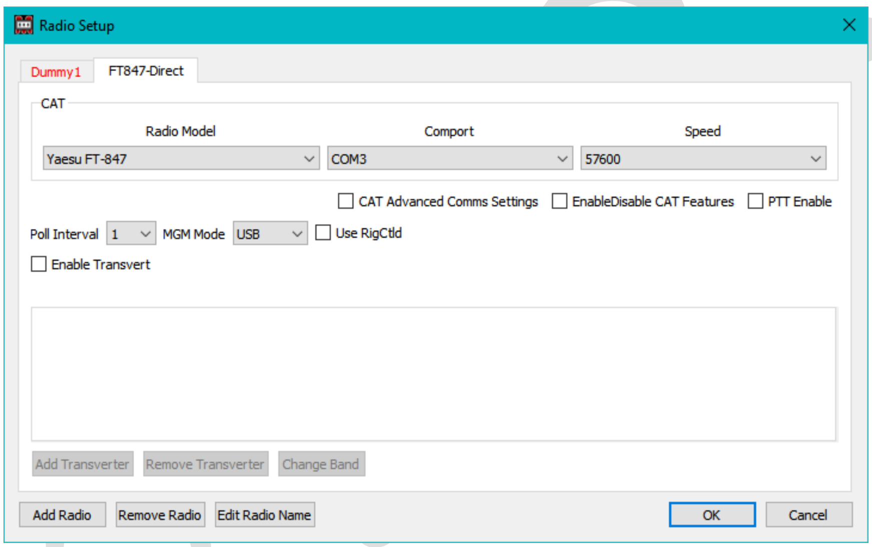


Figure 79 – Simple Configuration of a serial connected rig

Note that the **'Dummy'** tab is red, indicating that the 'rig' is in use by a contest and should not be changed. It may be possible, but the results are unpredictable.

If there is a need to change settings other than "**Comport"** and **"Speed"** then clicking the **"Advanced Comms Settings"** box (Figure 80) allows access to:

"**Data Bits, Stop Bits, Parity**" drop-down lists configure the serial communications[73] with the CAT interface of the radio selected from the "Radio Model" drop-down list.

"**Handshake**" Controls the flow control setting (e.g. XON/XOFF or CTS/RTS). Ensure this is correct for your interface (generally "None"). Some USB interfaces need CTS/RTS to be enabled to power the device.

[73] Tip: check the settings on anything else already being used, e.g. WSJT-X.

"Force: DTR" and **"Force: RTS"** allow the user to select **High** or **Low** from the drop down list to force the state of those lines on the COM port[74]. This is typically used to avoid problems with configuring Rigcontrol when enabling power to USB serial adaptors[75].

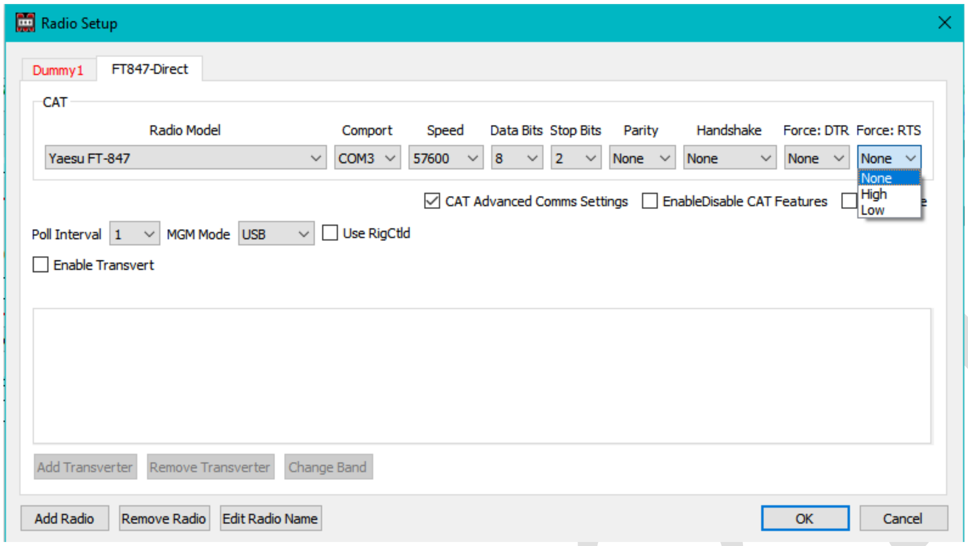


Figure 80 – Advanced comms settings

If mis-configurations have occurred error messages such as "**Please select a Com Port**" may appear.

Ticking **"EnableDisable CAT Features"** introduces a new group of features (Figure 81) that allow individual facilities to be enabled (or disabled).

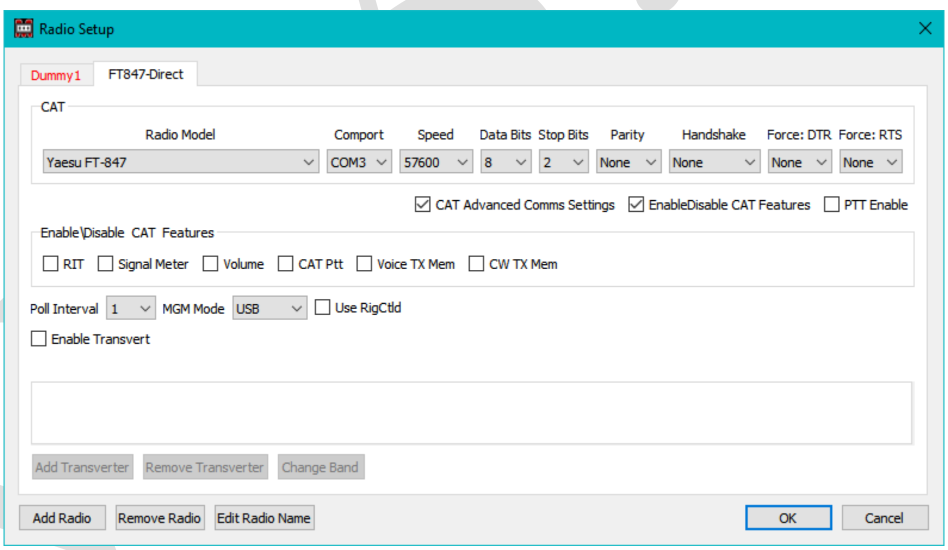


Figure 81 – Control of CAT Features

"PTT Enable" Enables PTT via CAT (for radios that support it), which can be used in the voice and cw Keyer message to determine end of message (see Section 4.3.3).

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[74] Some radios (e.g. Icom 7300 with non default configuration) may use these lines to control transmit switching.
[75] Using this removes the need to enble Hardware Flow control.

"**Poll Interval**" defines the interval in seconds at which RigControl interrogates the rig for information.

"**MGM Mode**" selects the rig mode to be used when MGM (data) communications are used.

"**Use RigCtld**" enables the device sharing mechanisms, see section 4.2.3.

"**Enable Transvert**" starts the transverter configuration dialogue (see section 4.2.1.3) and would be required if the rig is HF only.

If directly controlling an Icom rig then an additional "**CIV**" box will appear (Figure 82). Either leave the field blank to use the default CIV address of the radio, or enter the CIV number e.g. 0x66.

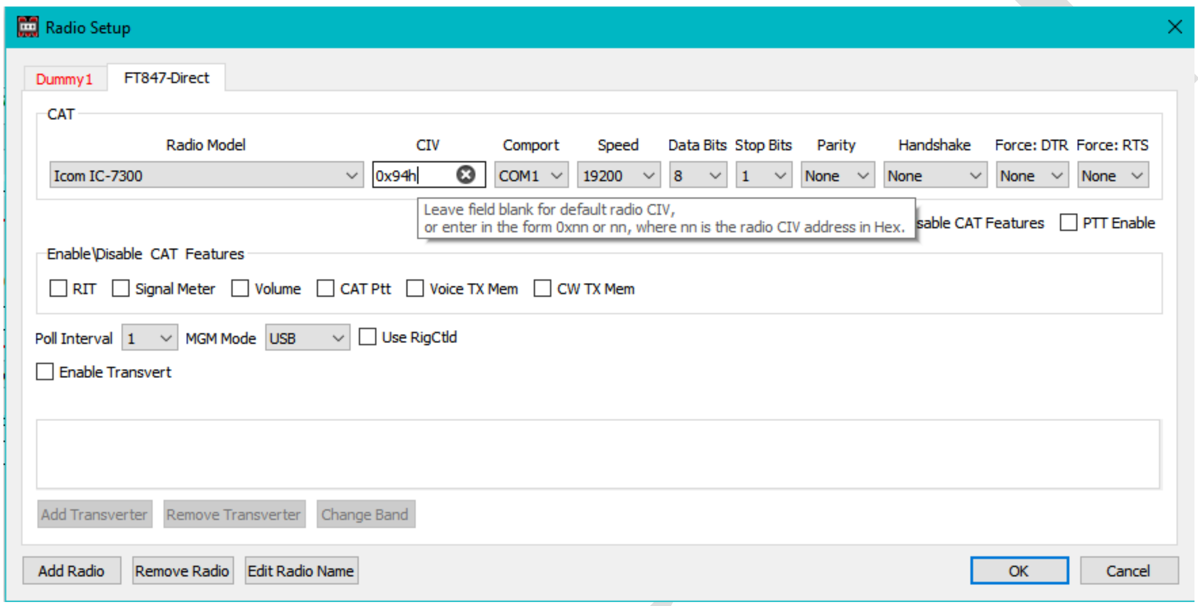

Figure 82 – Configuration of an ICOM rig

Note: With the IC7300 you need to ensure that **"CI-V Echo Back"** is **'ON'** in the settings[76].

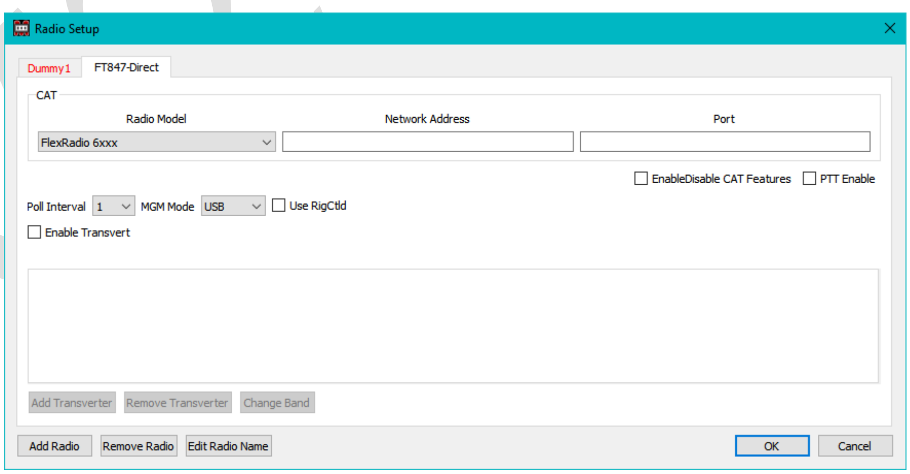

Figure 83 – Configuration of a network connected rig (FlexRadio)

[76] All Icom rigs need Transceive OFF and USB Port Link to remote.

A network connected radio (e.g. a FlexRadio) requires the target **“Network Address”** and **“Port”** to be entered (Figure 83).

If there are multiple rigs to be defined configure everything on one instance of RigControl and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

4.2.1.1 Log Radio Settings

The **‘Tools/Options’** panel **‘Log Radio Setting’** tab has three further sub-option tabs:

- In **‘Radio General Settings’** (Figure 84) there are three options[77] to control the usage of band presets and per contest frequencies/modes:

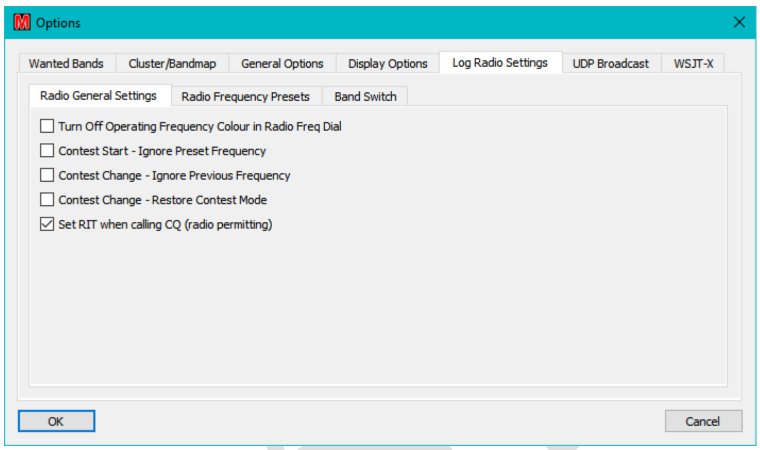

Figure 84 – Radio General Settings

- **“Turn Off Operating Frequency Colour in Radio Freq Dial”**: Disables the red display (within the RigControl panel) forxperimental Indication
- frequencies that are contrary to the Bandplan.

- **“Contest Start - Ignore Preset Freq”**: Inhibits Minos from switching the rig frequency on startup/contest open.

- **“Contest Change - Ignore Previous Freq”**: Inhibits frequency changes when switching between contests.

- **“Contest Change – Restore Contest Mode”**: If set, the mode is set to the contest default when switching to a contest. Otherwise the mode remains at what it was the last time the contest was selected.

- **“Set RIT when calling CQ (radio permitting)”**: If the connected radio supports RIT then it is enabled during CQ calls to allow fine tuning on incoming signals.

- The **‘Radio Frequency Presets’** tab (Figure 85) displays the table of ‘base frequencies’ with a tab for each mode of CW/Phone/MGM. These define the frequency the rig goes to when a contest is opened or when a “**Quick Sel**” band change is made from the Rigcontrol panel of the Logger (Figure 87). The default

[77] Note that there are some complex unresolved issues if the first two of these options are used.

frequencies are chosen to avoid such things as calling channels. This option allows you to change the start frequency to your favourite start frequency on a band[78]. Minos must be restarted for any changes to take effect.

The displayed bands are those selected in **'Wanted Bands'**

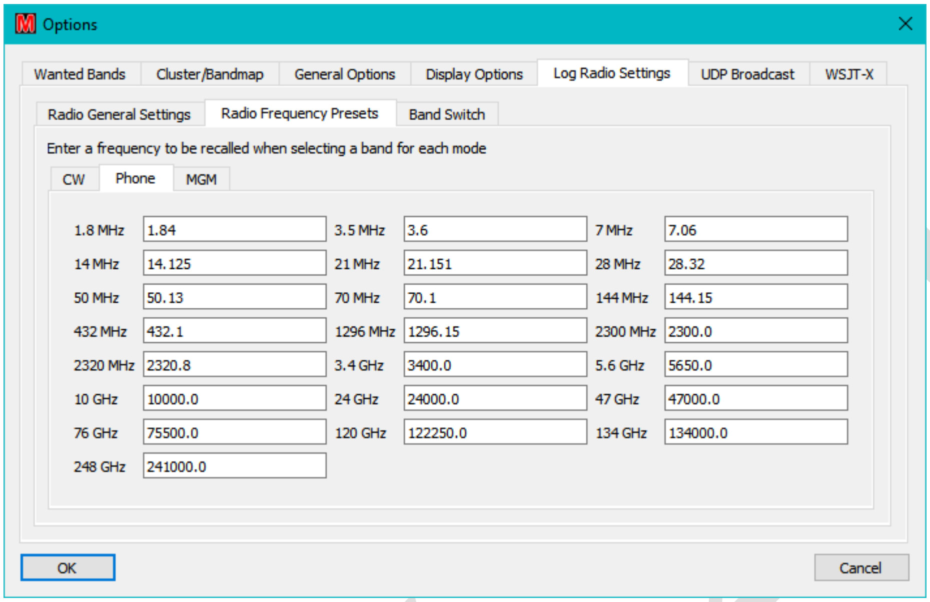

Figure 85 – Setup Band Frequency

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

- The **'Band Switch'** tab is relevant to Transverter Configuration and is explained in Section 4.2.1.3

4.2.1.2 Connecting to the Logger

Once the configuration has been done, the radio to be connected to the Logger can be selected either via the Rigcontrol panel pulldown, which shows which radio is in use at any particular time, or the Contest Details boxes (see Figure 75).

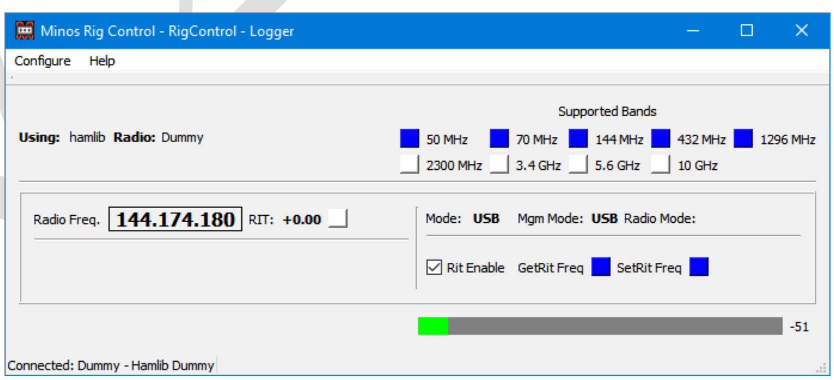

Figure 86 – RigControl operational

[78] It does not prevent you from specifying calling channels etc., so be careful.

A status line at the bottom left of the panel (Figure 86) shows the Connected/Disconnect status[79], and which radio has been selected, with what CAT configuration[80].

Note the **'Supported Bands'** boxes towards the top right. Blue indicates that a band is supported directly whereas yellow would indicate the use of a transverter.

At the bottom of the panel there is an **'S-meter'** bar, although in the example the indicated magnitude is lower than S1 and hence has a numeric value on the right.

Once configured to operate in conjunction with the Logger a contest may utilise the rig selected within the application, and the RigControl panel becomes visible in Logger (see Figure 87). As can be seen, because the application is connected to the Logger, it shows 'Logger' in the title, together with the name assigned to the instance of the app being used they are using (RigControl, by default).

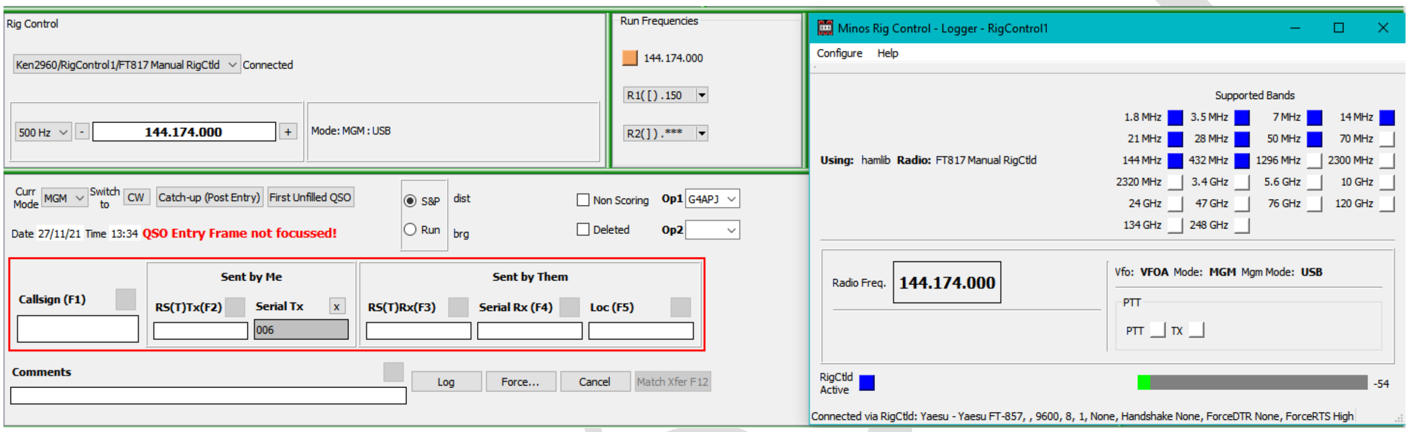

Figure 87 – RigControl connected to Logger

Within the RigControl panel (see Figure 90) on the top left is the name of the selected radio and its status which should normally be **'Connected'** for correct operation[81].

The current frequency[82] of the rig is displayed and may be changed by selecting characters inside the box and either typing in characters or using the mouse wheel to increase/decrease (band limits are enforced)[83]. If the mouse wheel is used then the changes are made immediately; if characters are typed then **'Return'** causes the rig to be updated.

The border will turn mauve to show it is in edit mode.

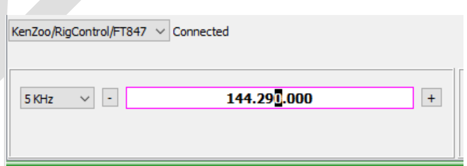

Figure 88 – Frequency change in progress

[79] Selecting a radio within the RigControl panel of the main Logger initiates the connection.
[80] The bulk of the Rigcontrol content displayed will only appear once a contest has connected to the rig (when 'Disconnected' changes to 'Connected'), subject to the CAT connection working.
[81] If the target of a remote connection is not active when log started then it loses the connection and the target is not in the list. It reconnects automatically when the target becomes available.
[82] The calculated frequency is displayed if a transverter is in operation.
[83] Minos doesn't control the VFO selection, it always sends the commands to the 'current-VFO'.

Ctrl-f selects the KHz digit, or use the mouse to select digit, as the tuning step. With the cursor in the "Freq" box you can now use the mouse wheel or the up/down arrow to tune. Left/Right arrow change the selected digit, hence the step.

Hit **'Enter'** to select the new frequency or escape to cancel. The frequency may be 'nudged' by the amount selected from the left hand drop-down box[84] using the +/- buttons to allow calling near the frequency of a station being received[85].

Figure 89 – Frequency out of contest band

If the frequency is 'nudged' out of the band (or perhaps the rig band is changed manually or via WSJT-X) then a warning appears as shown in Figure 89. Clicking **"Reset Frequency in Band"** resets the frequency to be in the current contest band, based on the selected contest tab that is open– using the band selected in **"HF Band Switching"** if appropriate.

If the selected radio supports RIT (through CAT) it can be enabled through the RigControl panel (see Figure 90) and it will then become visible in the RigControl panel within Logger (see Figure 16). Turning it on (Figure 87) then allows incremental tuning using the mouse wheel in the same manner as for the main frequency. A specific RIT frequency can also be entered, although that is unlikely to be very useful. There are a number of shortcut keystrokes available - listed in Figure 177.

Similarly, if the rig supplies volume control information this will be added to the display under the Band Select list/Mode display.
In multi-band contest with multiple tabs the Logger initially displays the current rig frequency on all tabs using a particular RigControl, switching to the frequencies from the **'Band Frequencies'** list as contest tabs are selected (or "**Band Select**" changed). The use of the "Band Frequencies" list can be disabled using **"Contest Start - Ignore Preset Freq"**.

During operation any changes to that frequency are only propagated to the active tab; those not in focus do not get updated.

When an inactive tab is selected and put into focus the frequency (and mode) used the last time the tab was active is restored to the rig (whatever band that happens to be, i.e. if the 2M tab has a 6M frequency from the startup then selecting the 2M tab will cause the rig to go to a 6M frequency). This may be disabled via the **"Contest Change - Ignore Previous Freq"** option, in which case the rig stays on the same frequency and mode when you change contests.

The current mode (USB/CW/FM/MGM) on the radio is displayed[86], and is changed as selected within the QSO data entry frame (since it may be changed in the Logger when RigControl is not in use).

The rig frequency is adjusted to compensate for changes between USB and CW.

[84] The sizes available for the 'nudge' are mode dependant.
[85] Useful to call a station near a frequency that is in use.
[86] At startup the radio is always switched to USB

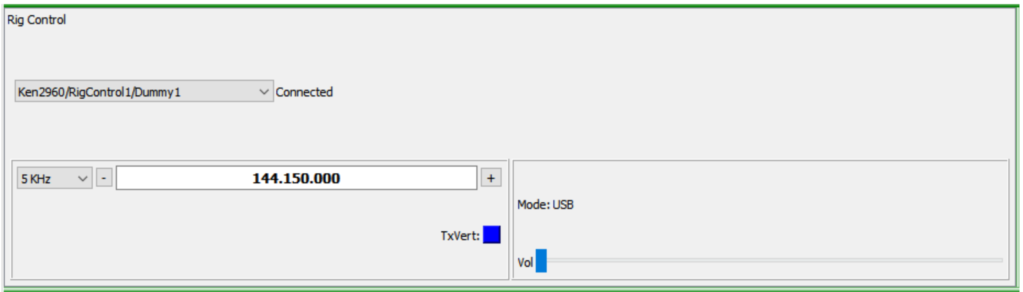

Figure 90 – RigControl panel within Logger

When RigControl is in operation the memory functionality described in section 3.2.3 is fully realised, as well as the R1/R2 button functionality described here. These buttons represent the 'run frequencies' (and mode) selected by the operator and are configured using the same R/W/E/C dialogues as the standard memories (see Figure 91). Additional functionality appears when Bandmap is active - see section 4.3.5.3.

Each button displays the frequency if set and "***" if nothing is stored. In Figure 91 (right) R1 is set to (144.)174[87] and R2 is empty. The square brackets indicate the CTRL shortcut symbols that can be used. Run memories are held within, and retained with, each log.

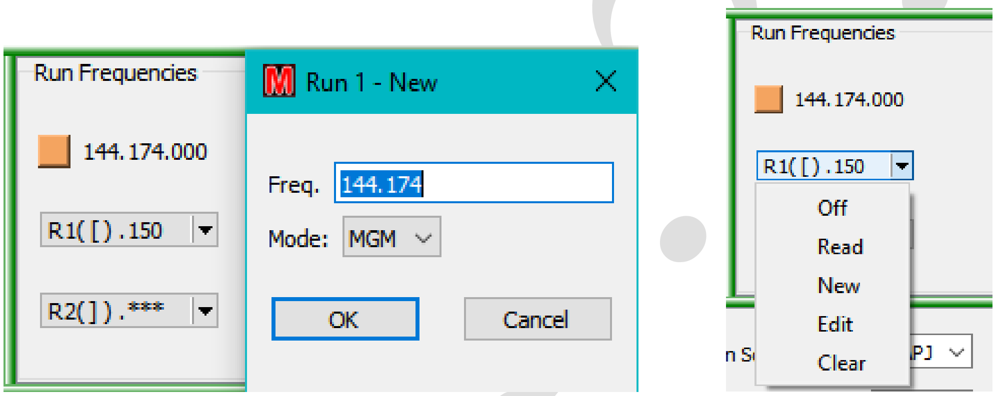

Figure 91 – Storing a run frequency

Once a run frequency has been selected from the R1/R2 Run Freqs buttons, the rig will be tuned to that frequency and mode and the selected button turns orange to indicated that it is enabled (see Figure 92 [1] where the cursor is hovering over the button). Pushing the button will then toggle run mode between on and off[88]. If the rig is tuned away from the run frequency (out of run tolerance[89]) the button will turn yellow (see Figure 92 [2]) and any details in the QSO entry panel disappears. **Such content may be recovered by pressing the <ESC> key**.

When you tune off the run frequency and the button is on (yellow) you will be in S&P mode[90]. Clicking R1 again returns the rig to the original run frequency and retains the S&P frequency marked 'Restore' (see Figure 92 [3]). Clicking the button again switches back to S&P mode on the stored frequency (see Figure 92 [4]).

[87] It is presumed to be **144**.230; if you have changed bands it will still be on the original band.
[88] If no run frequency is stored in the button then nothing happens.
[89] Fixed at +/- 300Hz.
[90] Logged contacts will be added to the BandMap when in S&P mode (see 4.3.5.1), but **not** in Run mode.

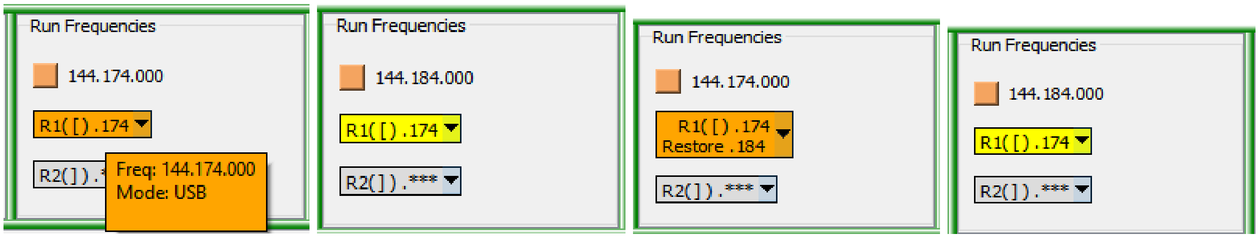

Figure 92 – Run mode in operation

If the button is on the run frequency (orange), with no Restore data held, clicking the button will turn run mode off. If the button is yellow (i.e. off the run frequency) the button menu **"Off"** function (Figure 91) must be used.

Under some circumstances (most likely comms failures) errors may be returned from the Hamlib routines controlling the rig (Figure 93).

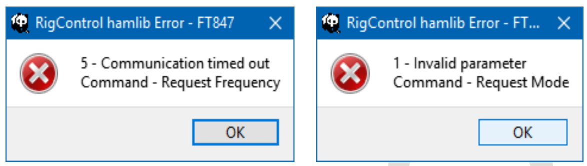

Figure 93 – Hamlib error messages

When **"OK"** is clicked the error is reported back to the Rigcontrol and Bandmap panels which then disconnect (Figure 94).

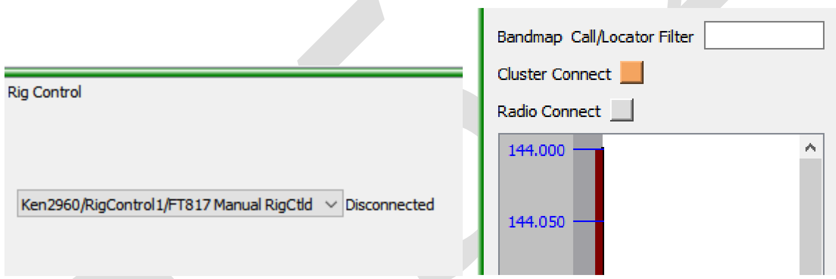

Figure 94 – Disconnect after Communication errors.

A **"Reconnect"** button appears of the relevant instance of RigControl (see Figure 95). Clicking this button causes an attempt re-establish communication and if successful the Rigcontrol and Bandmap panels will reconnect. If this does not happen, investigate the comms settings and cables.

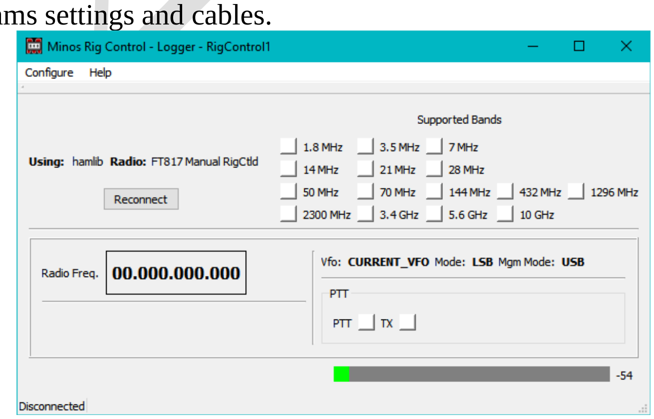

Figure 95 – Reconnect to rig

4.2.1.3 Transverter Configuration

If **“Enable Transvert”** is ticked the rig being defined/modified will be used with a transverter and the buttons are enabled to allow this definition (see Figure 96), including making the **“Enable Transvert Switch”** option visible.

“Add Transverter”, “Remove Transverter”, and **“Change Band”** are used to control the definition of possibly multiple transverters/bands.

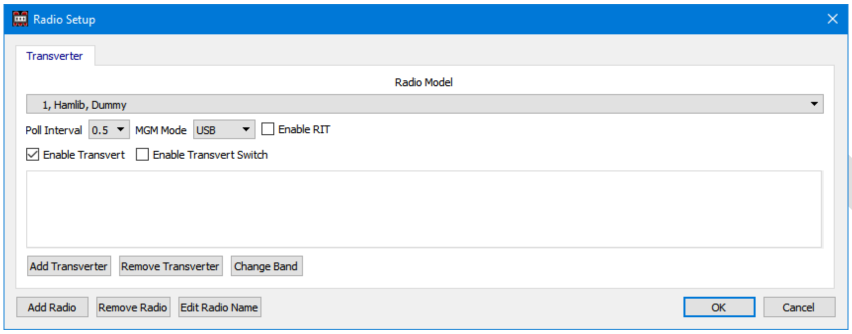

Figure 96 – Transverter configuration

When you add a transverter, a list of bands are displayed (Figure 97). The radio may support the band directly, but a transverter can still be selected for that band if for instance its RX performance is better than the radio.

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

If the radio supports a band for which a transverter is configured, the transverter frequency combination will be used in preference to the radio band. Configure a different radio definition (or don’t configure a transverter for that band) if this is not what is wanted.

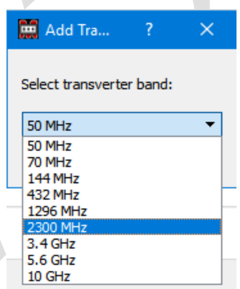

Figure 97 – Transverter band selection

Having created a transverter tab for one or more bands (Figure 98) the frequency definitions govern how it operates.

“Radio Freq” is the frequency the rig actually tunes to get the desired output

“Target Freq” is actual operating frequency

“Offset Freq” is the difference between them, which the transverter produces[91]

[91] Currently the focus must be moved from the entry boxes to get the calculation displayed.

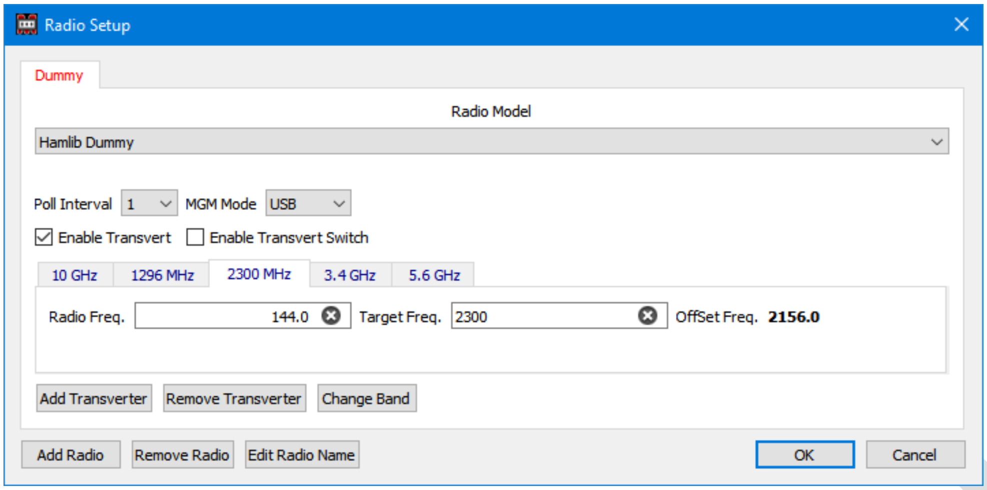

Figure 98 – Transverter definition

These frequencies are then reflected in the Rigcontrol display and passed to the rig. So, with the above definitions for 2.3 GHz if 2,300.1 MHz is entered in the Rigcontrol panel of the Logger the Rigcontrol display will appear as in Figure 99 and a frequency of 144.100 will be sent to the rig.

If you have a K3, and/or are having problems with 2300MHz, then talk to G4RQI who has extensive experience in configuring VHF/UHF transverters to handle this situation. The K3 is set to output the final frequency on its display and CAT. Transverters for each band are then added in Minos, but the radio and target boxes are blank (0.0 in them).

This could be useful for other rigs set to display and output the transverter frequency, or for a radio where the Hamlib profile doesn't support the band. (There is rumoured to be an Icom radio that had 1296, but the profile didn't include that band).

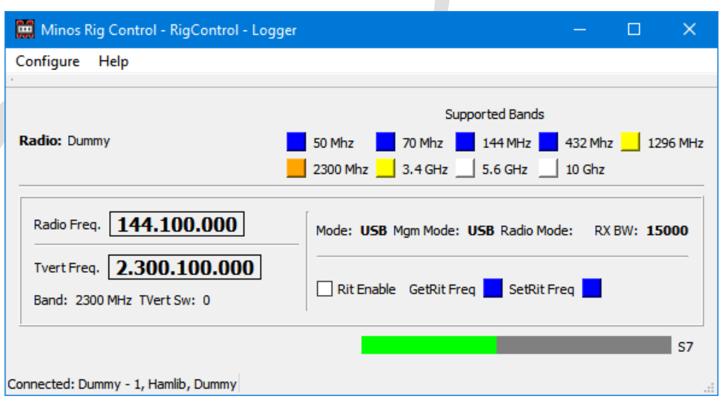

Figure 99 – Transverter in Rigcontrol display

The **"TxVert"** indicator within the RigControl frame of the Logger (Figure 100) shows blue when the rig is directly accessing the band, and orange when a transverter is in use (a band such as 70MHz might possibly use either).

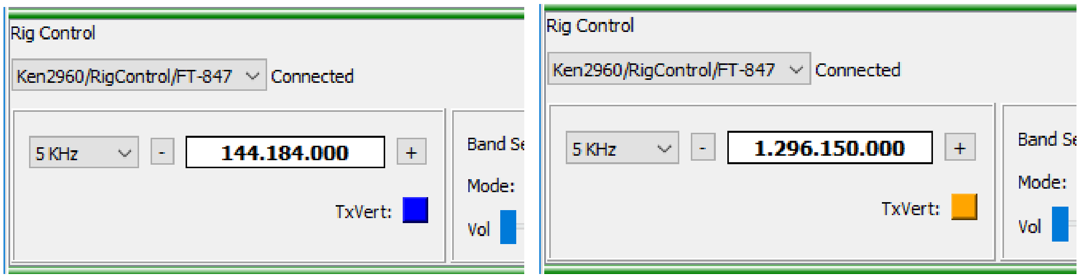

Figure 100 – Logger RigControl transverter indicator

There is a facility to drive external switching of the tranverter either via an external com port or via a network message to an external control system[92]. Within the **"Options"** panel (invoked from the **"Tools"** pulldown list) select the **"Band Switch**" tab within **"Log Radio Setings"** (see Figure 101).

Ticking the **"Enable Band Switch"** box enables this facility for all defined transverter bands.

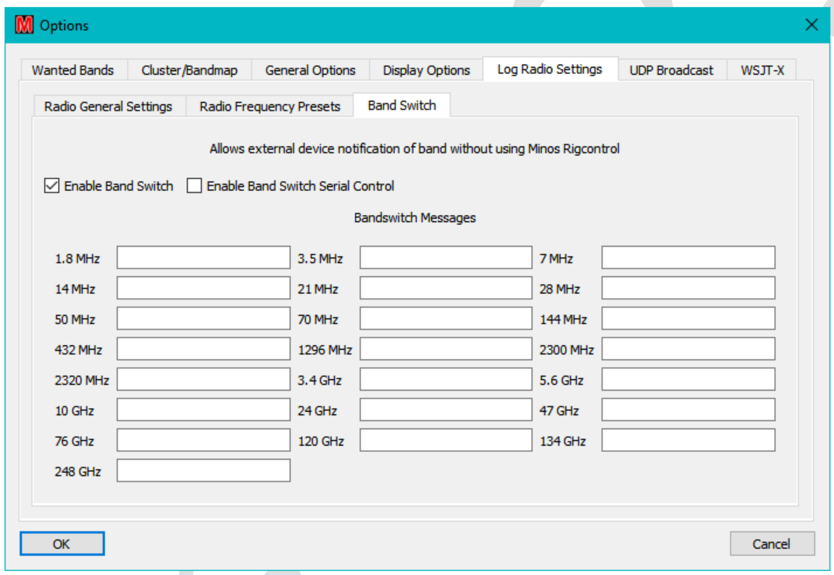

Figure 101 – Bandswitch Messages

To use an external com port, enable **"Enable Band Switch Serial Control"** which then displays the pull down list to select the com port to be used (Figure 102). The message transmission will be via the selected com port. The 'standard' default set up is assumed (9600baud, 8bit, 1 stop bit, no parity, no flow control. It does not require RigControl to be active.

The intention is that the message will be read by an Arduino or a Pi based 'switch box' (see Appendix section 7.5) .

[92] Still under development, but see section 7.5.

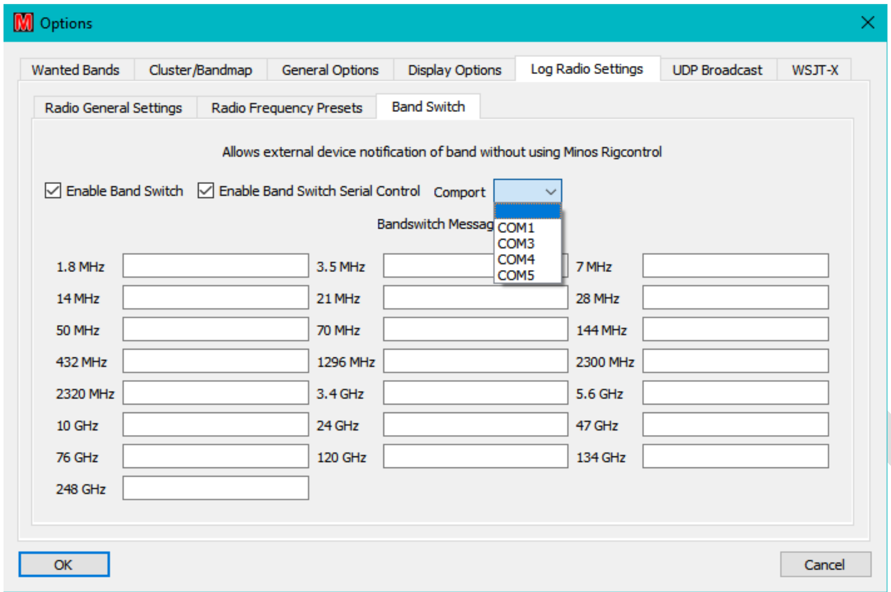

Figure 102 – Band Switch Serial Control

Regardless of the path used, the message to be sent is entered in the box for the relevant band. It is free format (numbers or letters, which might typically be just a simple number), but control chars should be avoided - things like \r to send control chars are not parsed. The defined text is 'wrapped' in a ":" as start and "LF" to terminate, so if "1" were entered in 1.8MHZ the string ":1\n" would be sent when triggered by a change of contest tab.

4.2.2 Rotator

The application has many similarities to RigControl. It can connect to multiple rotators[93] (from those supported by the Hamlib library) via serial ports (which may be simulated via a USB device), or via direct USB connection if your rotator controller supports it.

PstRotator can also be used (see Figure 105 [4]), with any of its supported rotators. Users should ensure they have the latest versions or at least 15.94 of AZ/EL version and 13.78 of the AZ only version. If the PST software is installed into a non-default location then use the **"Configure/PST Rotator Config"** option to define that location else the **"Add Antenna"** dialogue will not show PSTrotator. There have been one or two 'funnies' with the interconnection to PSTrotator, possibly caused by not having a licence for the software.

The Rotator application is intended to be used under control of the Logger.

Once the application has been connected to a particular contest (see above) the rotator(s) in use need to be configured so that the selected controller operates under control of the Logger. Figure 103 shows the startup with Rotator defined to a contest but with no controllers yet defined.

Selecting **"Configure/Setup Antennas"** allows the rotator(s) to be defined.

Figure 105 shows an example configuration of a rotator.

[93] Once again, more than any normal operator would want to use/can afford.

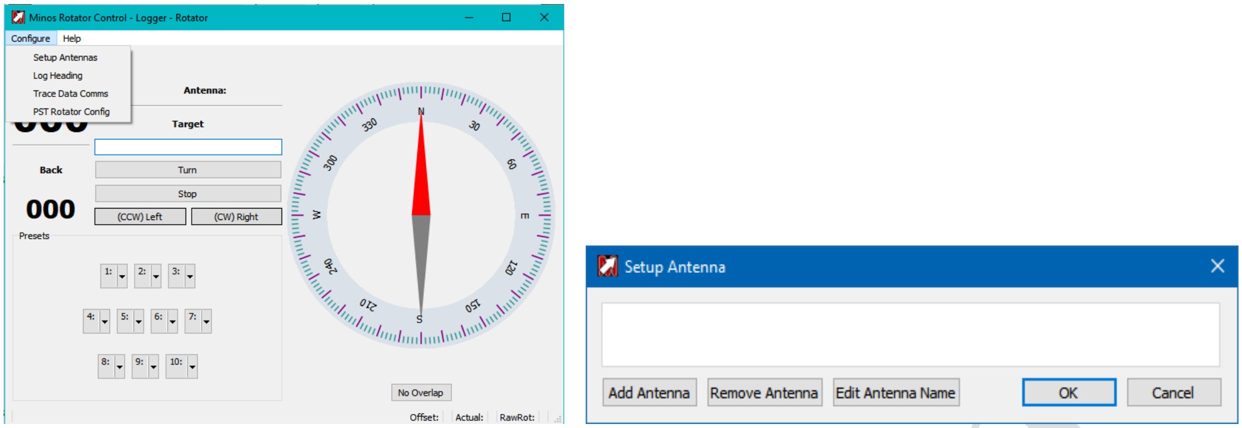

Figure 103 – Starting configuration of Rotator

Clicking **"Add Antenna"** starts the definition of a rotator (Figure 104). Remember, PSTrotator will only appear in the pulldown list if the software is installed in the default location or its actual location has been configured.

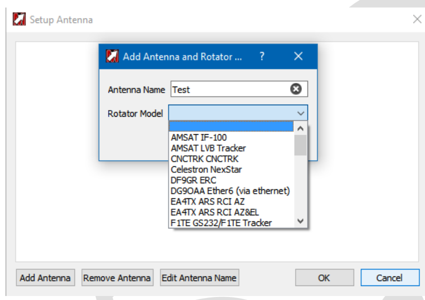

Figure 104 – Selecting the rotator model

The **"Antenna Name"** field is used to select (or display) which rotator is currently being controlled (e.g. Figure 111). Once defined, it may be changed using **"Edit Antenna Name"** or removed (provided it is not currently active within the Logger) using the **"Remove Antenna"** button.

Depending on which rotator is selected different options will appear (Figure 105).

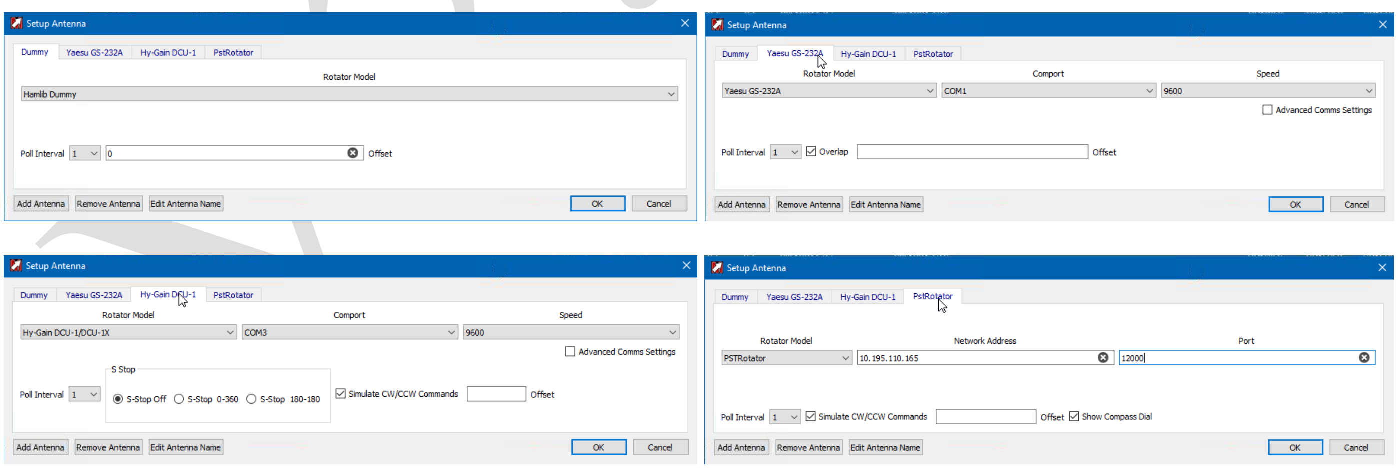

Figure 105 – Defining various rotators

If there are multiple rotators to be defined configure everything on one instance of Rotator and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

The **'Antenna Name'** is the method by which the particular rotator is selected from the pull-down list.

"Rotator Model" is a pull-down list containing an extensive range of rotator models supported by Hamlib. Note that one of these is **'Hamlib Dummy'** which can be quite useful for testing/learning purposes.

If there is a need to change settings other than "**Comport"** and **"Speed"** then clicking the **"Advanced Comms Settings"** box (Figure 105 [2] & [3]Figure 80) allows access to:

"**Data Bits, Stop Bits, Parity**" drop-down lists configure the serial communications with the rotator controller selected from the "Rotator Model" drop-down list.

"**Handshake**" Controls the flow control setting (e.g. XON/XOFF of CTS/RTS). Ensure this is correct for your interface (generally 'None').

"**Poll Interval**" controls the interval between polls of the rotator; the default of '1' should be correct for most installations.

"**Overlap**" will only appear, and will be ticked by default, if the Rotator being controlled supports overlap capabilities. The capability for overlap and the amount of overlap allowed is derived from information in the Hamlib library.

Some rotator models do not support overlap and the rotator module in Hamlib may not support overlap either.

The box should only be unchecked[94] to turn off overlap when using an overlap-capable protocol with a rotator that does not support overlap. The rotator will then only turn between 0 - 360 degrees. If Overlap is unchecked then the Overlap box will not be displayed on the Rotator Display.
Beta testers are required to check out this functionality.

The **"Offset"** to support an antenna mounted offset from the rotator bearing a value between +90/-90 may be entered in the **"Offset"** box[95].

Rotator bearing is always shown in Actual on status line. With the offset (from Rotator definition) the pointer indicates the 'real' beam heading, not the rotator current setting.

With some rotators such as the Hy-Gain in Figure 105 (or if **'Overlap'** is unchecked on Yaesu rotators[96]) a number of settings appear to allow a rotator with a north-stop to be mounted with the mechanical stop to the south while the sensor is sending 0 – 360, instead of -180 to +180:

"**S-Stop Off**" the rotator will act as normal

"**S-Stop 0-360**" Minos inverts the reading of 0-360 that the controller sends

"**S-Stop 180-180**" is used for rotators such as some old Stolle models, that are mounted with a south stop, but use a compass sensor that sends 180 - 180[97].

---

[94] If Overlap is unchecked and an antenna is sitting in an overlap region, you may not be able to rotate the antenna. The antenna will need to be manually moved < 360 or > 0
[95] There is also an antenna offset field in contest details to allow for any skew between antennas in the same stack, or inaccuracy in the direction setting of the antenna.
[96] The Yaesu protocol is probably the most commonly emulated.
[97] Actually the stop is set to 181 to 179.

"Simulate CW/CCW Commands" is for rotators that don't support turn-right/left commands in their protocol. When the turn right/left buttons are pressed, a 'rotate to end stop' bearing command is used. If that causes a problem unchecking the tick box will cause the turn right/left buttons to be removed.

If using PstRotator the **"Network Address"** and **"Port"** information (see Figure 105 [4]) must be configured into PstRotator Settings (**"Communication/UDP Control Port"** and **"Setup/UDP Control"** must be enabled. Also in **"UDP Control Port"** the box **"Automatically positions reporting when rotator is moving"** [sic] should be cleared.). The **"Network Address"** can either be the local loopback address (127.0.0.1) or the actual address of the local machine, **not the PstRotator default of .255.** Access to a remote machine is not supported.

If the **"Show Compass Dial"** box (enabled by default) is disabled the rotary dial is removed from the Rotator application display. Perhaps better is to mark Rotator as **"Hidden"** within the Application definition once configuration is done.

"Configure/Log Heading" allows a location and filename to specified for logging bearings when "Logging Enabled" is ticked (see Figure 106). Bearings will continue to be appended to the specified file[98] every time Minos Rotator is started until **"Logging Enabled"** is cleared. The **"Bearing Difference"** is the incremental change required before a new bearing is logged.

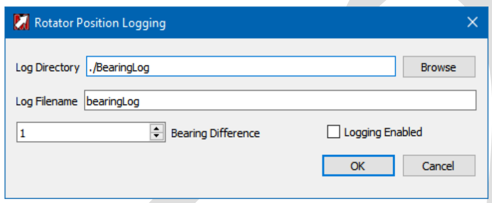

Figure 106 – Configuring Rotator Position Logging

```
Mon Oct 30 11:33:57 2017 Bearing is 19 Degrees
Mon Oct 30 11:33:57 2017 Bearing is 22 Degrees
Mon Oct 30 11:33:58 2017 Bearing is 25 Degrees
Mon Oct 30 11:33:59 2017 Bearing is 26 Degrees
```

Figure 107 – Example Bearing Log

"Configure/Trace Data Comms" enables the tracing of communications between Minos and the Radio[99]. It is recommended that this is only turned on for diagnostic purposes. If checked the state is saved and it will be enabled when RigControl is restarted, so remember to turn it off when done. See section 5.3 for mote information on trace logs.

Once the configuration has been done, using **"Antenna Select"** to specify the rotator in use connects the application to the Logger. The rotator in use can be changed by selecting a different entry from the pull down list; the top left of the Rotator panel in the Logger will indicate which is being used (see Figure 111). The current status of the rotator (e.g. **'Turning CW'**) is also shown adjacent the connection information.

[98] If you want a new file, either specify a different name (saved through restart) or move/rename the existing file.
[99] Standard trace messages are now recorded at all times.

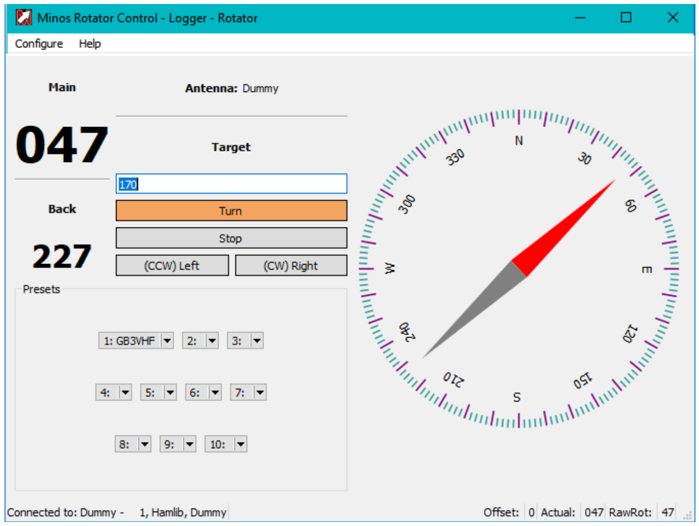

Figure 108 – Rotator control operational

A status line at the bottom left of the panel shows the Connected/Disconnect status, and which rotator has been selected.

"Configure/Edit Presets" allows the definition of the ten memories visible in the bottom left of the Rotator panel (see Figure 108). Each consists of a name (typically a callsign) and the relevant target bearing. These memories are effective for **all** rotators on the host machine, and are retained across restarts[100].

The presets are replicated in the Rotator Control panel of any Logger contest accessing the rotator[101].

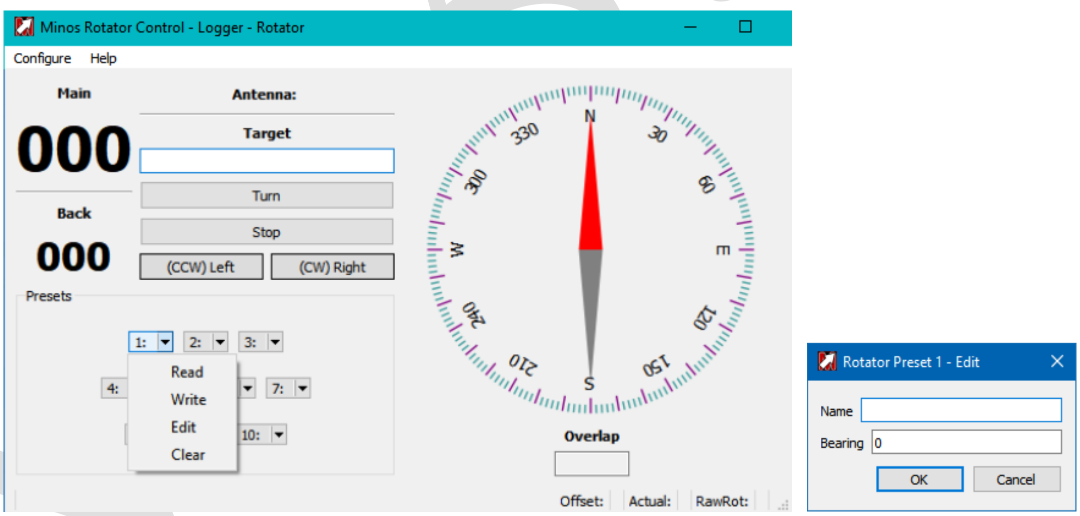

Figure 109 – Configuring Rotator control pre-sets

[100] Entries may be removed by going into "Configure/Edit Presets" and deleting the individual content.
[101] On any machine accessing the host in complex configurations.

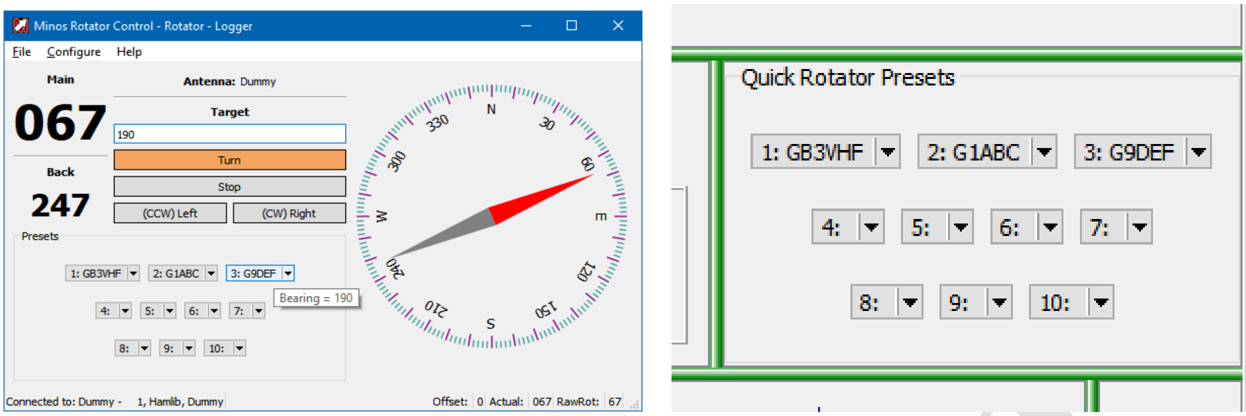

Figure 110 – Configuring pre-sets

Clicking one of these entries initiates rotations to the requisite bearing.

Once configured to operate in conjunction with the Logger a contest may utilise the rotator selected within the application, and the Rotator panel becomes visible in Logger (see Figure 111). As can be seen, because the application is connected to the Logger, it shows 'Logger' in the title, together with the name assigned to the device currently selected to be used.

With Rotator in operation there are enhancements to the memory functionality described in section 3.2.3. The memory content is extended to include bearing information. When a memory is read the bearing is inserted into the Target field for the rotator and rotation can be initiated by **"Turn"/'Return'/ 'Ctrl-T'**.

Extra data is also written to the log for reference purposes: **'Rot heading'** gets filled in with the current bearing[102].

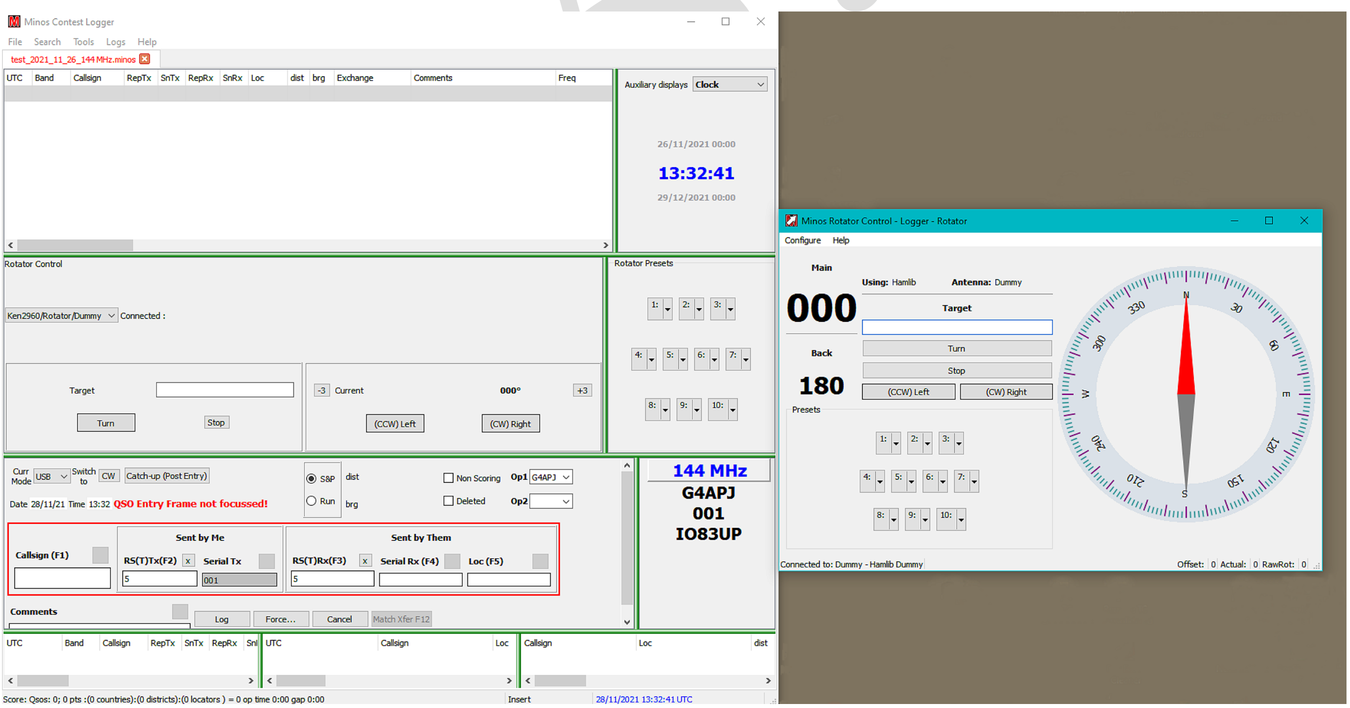

Figure 111 – Rotator connected to Logger

[102] If Rigcontrol is operational and connected to a rig then the rig name and current frequency get filled in as well.

The antenna can be controlled from the Rotator application or from the buttons in the Logger. If actions are initiated from the Rotator application the panel in Logger shows what is happening.

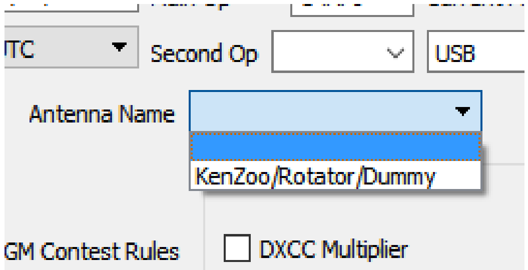

Figure 112 – Selecting the Rotator Application Instance

Within the Rotator panel in Logger (see Figure 113) the text at the top left shows the rotator in use ('Dummy2' in this example) and the status (e.g. **'Turning to bearing'**).

The left hand box shows the target bearing ('170') and the **"Turn"** button changes colour to show that an operation is in progress (it clears when complete). The rotation to the requested bearing will occur via the shortest valid movement using overlap.

The right hand box shows the **'Current Heading'**[103] for the rotator in use as well as the control buttons **"(CCW) Left"** and **"(CW) Right"** [104] and the 'Nudge buttons' **("-3"/"+3")** which nudge the rotator 3 degrees every time the button is pressed[105] (in the direction of the adjacent box, i.e. + is CW).

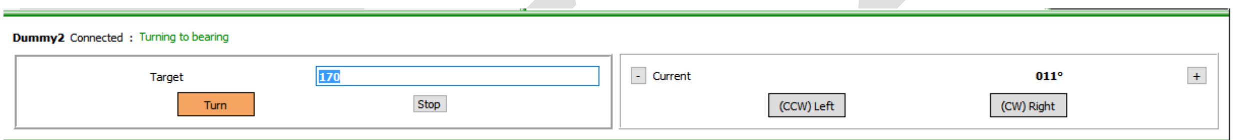

Figure 113 – Rotator panel within Logger

The **"Target"** (heading) may be entered manually, or is calculated from any QRA locator entered in the data entry panel in the Logger. This locator itself may be derived from an entered callsign found in the archive and transferred using F12 (or equivalent).

If a target bearing has been manually entered then rotation may be started by hitting **'RETURN'**, else click the **"Turn"** button or Ctrl-t to start rotation[106]. Rotation may be stopped by clicking the **"Stop"** button on the left or by using the shortcut (Ctrl-s). If the rotator backend does not support stop, the **"Stop"** buttons are hidden in Rotator and the Rotator Control panel in Logger. Since there is no way to determine if a rotator under control of PSTRotator supports a stop Button, the **"Stop"** button will always appear for PSTRotator.

There are a number of shortcut keystrokes available - listed in Figure 176.

While rotating the bearing is checked for end stop, if reached a stop will be automatically issued.

---

[103] Red indicates overlap > 360 degrees and blue for overlap < 0 degrees.
[104] Left and Right are from the perspective of physical buttons on the rotator.
[105] If a button is selected when rotator is moving, a stop command is issued before the selected command is issued.
[106] A bearing may also be selected, and rotation initiated, by clicking on the rotator ring within the Rotator Control panel.

Logging a contact while the rotator is turning logs the ‘incomplete’ bearing (i.e. the bearing at the moment of logging).

The Rotator application (see Figure 108) displays the bearings (Main and Back), together with a ‘compass bearing’ schematic of the current bearing heading (actual, calculated using any offset defined). A status line at the bottom left of the panel shows the Connected/Disconnect status and the name assigned to the rotator that has been selected.

There are control buttons to **“Turn”, “Stop”**, **“(CCW) Left”** and **“(CW) Right”**.

Some rotator controllers do not support a Rotate Left (CW - ClockWise) or Rotate Right (CCW - Counter ClockWise) Command. To allow the CW and CCW buttons to function, the command rotate to minAzimuth (typically 0 degrees) is sent for CCW command and the command rotate to maxAzimuth (typically 360 degrees) is sent for CW. The CW and CCW buttons are latching, selecting the buttons again will stop rotation. Rotation will also stop when the rotator reaches the endstop if a stop command is not sent first.

If a control button is selected while the rotator is moving, a stop command is issued before the selected command is issued.

If you change any value in the Current Selected antenna, you will be prompted to reload the values. You can choose “No” to skip and the values will be loaded when the program is restarted.

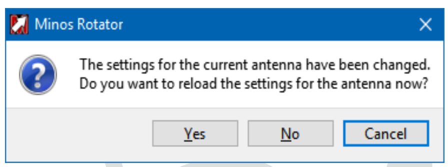

Figure 114 – Rotator settings changed

4.2.3 Device Sharing

A common requirement is to be able to share control of the rig(s), and possibly rotator(s), that Minos is using with other programs such as WSJT-X.

Hamlib provides a mechanism for doing this; it is called rigctld (or rotctld for rotators[107]). These network servers or ‘daemons’ control the actual radio and use a TCP/IP based call/response protocol to handle requests from multiple client programs. OmniRig is also available, although that supports a maximum of two rigs.

In principle Minos could treat a rig as local but control it using remote access to rigctld on the target server, but this approach has been rejected. Despite the fact that rigctld and its companion utility programme rigctl implement no security measures it can be a bit tortuous to successfully configure such remote access in a Windows environment[108].

It is much simpler to handle the rigctld access locally and ‘advertise’ the result(s) through the Minos server communications.

To enable this operation, click the **“Use RigCtld”** box in the radio definition (Figure 115, left). When the rig is selected from a contest the rigctld daemon will be started with the requisite parameters, and connection established through it to control the rig. Once the connection is operational the **‘RigCtld Active’** indicator turns orange in the RigControl Window (Figure 115, right).

[107] Currently not, maybe never, supported in Minos.
[108] This is still being investigated.

“RigControl/Configure/Configure RigCtld” (see Figure 76) opens a File Explorer window to select the daemon to use.

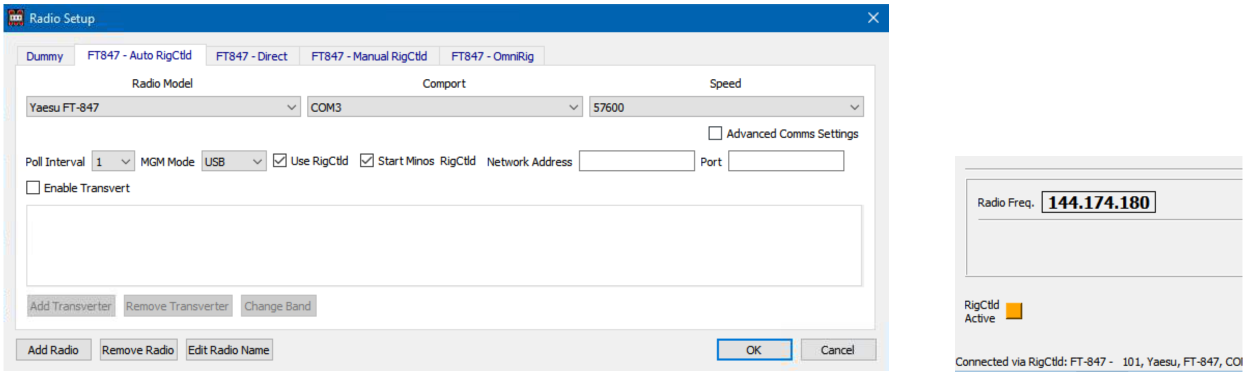

Figure 115 – Defining rigctld connection to Rigcontrol

The rigctld connection is established on the local loopback address (127.0.0.1) using port 4532 by default[109] (if **“Port”** is left blank in the radio setup). Other applications accessing the rig through rigctld must use these parameters (see WSJT-X example below).

Any such applications must be started **after** Minos has create the rigctld daemon, and terminated before Minos[110] is disconnected from the rig (hence terminating rigctld). This restriction can be avoided by starting rigctld externally (typical from a BAT file, section 4.2.3.1) prior to starting Minos[111], which also allows the user to select the location and specific version of rigctld that will be used.

If startup is enabled, RigControl by default initiates the daemon using **rigctld.exe** from the **/Bin** folder. The Minos installation process places a copy of the Hamlib version of this programme in the folder.

For WSJT-X to share a rig the same information is supplied via the **“File/Settings”** dialogue, although the layout is somewhat different (Figure 116); other applications will have their own methods of defining the parameters.

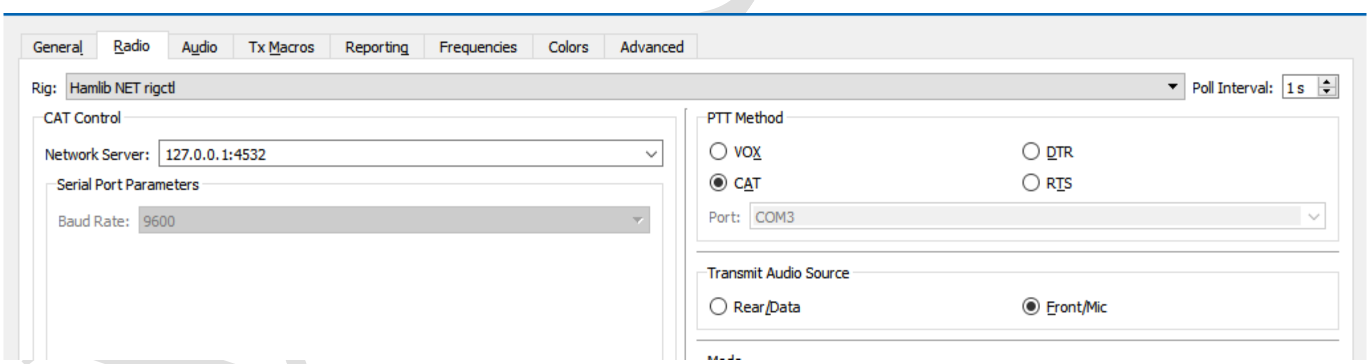

Figure 116 – Defining the rigctld connection to WSJT-X

If Minos has started Rigctld and then terminates, Rigctld will also terminate and WSJT-X will produce an error.

[109] If multiple devices are in use it is recommended that even port numbers be used in sequence from this default (one instance of rigctld, and hence port, is required per concurrent connection to a rig), but it doesn’t really matter as long as there are no clashes (and the right ones are connected).
[110] This is also the preferred sequence to synchronise the various views of what frequency the rig should be using!
[111] The **“Use RigCtld”** box must still be ticked within the rig definition. Note that the **‘RigCtld Active’** indicator will not light under these circumstances.

4.2.3.1 Rigctld commands

There are several different versions of the software. Minos is built on the premise of using Hamlib V4.5 (not earlier versions), however WSJT-X includes a special, 'heavily modified' version of rigctld thought to have been created because Hamlib versions prior to 3.3 were in some way deemed unsatisfactory.

No problems have been identified with using either version with either programme. The only identified difference is that WSJT-X maintains the CAT connection all the time whereas the Hamlib 4.0 and later version drops the CAT connection once all client connections have terminated (thus allowing a direct connection from another programme, but raising the risk of contention).

If you have WSJT-X on your system it is probably better to use that version (and hence avoid installing Hamlib).

In you wish use the Hamlib version then to install it get the installation programme from https://sourceforge.net/projects/hamlib/files/latest/download or search around https://sourceforge.net/projects/hamlib/files/hamlib/4.5/ for 32 and 64 bit versions (zip and exe).

Run hamlib-w64-4.4.exe (or hamlib-w32-4.4.exe) to install into <target folder>; only the DLL and program options are required to use the software although the documentation *may* be of interest to some users, particularly if multiple rigs are to be shared.

Before the daemon can be used it must be started. This can either be done from Minos as described above, or from an external .BAT file, perhaps at system startup, containing a single command line of the form:

<WSJTX location>\bin\rigctld-**wsjtx** -m 1001 -r com3 -s 9600 -T 127.0.0.1 -t 4532

This example connects to the FT-847[112] using COM3 at 9600 baud and makes the service available on the loopback address, port 4532, which is the easiest way to use it with Minos.

The equivalent command for the Hamlib version merely changes the folder from which the program is invoked:

<target folder>\hamlib-w64-4.4\bin\rigctld -m 1001 -r com3 -s 9600 -T 127.0.0.1 -t 4532

For an ICom rig the CIV number (typically 66 hex) has to be expressed in decimal, e.g.:

< folder location>\rigctld -m3046 -rCOM6 -s19200 **--set-conf=civaddr=102** --set-conf=data_bits=8 --set-conf=stop_bits=1 --set-conf=serial_parity=None --set-conf=rts_state=ON --set-conf=dtr_state=ON

In the above example (for an ICom 746PRO) some handshake lines are set to enable the USB interface with power.

Rotator sharing is only done via manually configuring rotctld (Figure 117) and has not been tested. Volunteers wanted!

[112] The numbers for the device type (-m) are the Hamlib values which are different in V4.0+ from what they were in earlier versions.

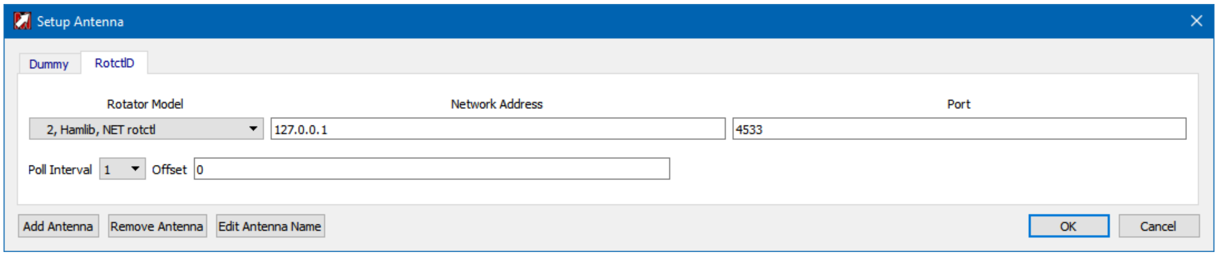

Figure 117 – Defining rotctlD connection to Rotator

The equivalent command to start the rotator daemon[113] for the above Antenna definition to use would be:

<target folder>\ hamlib-w64-4.4\bin**rot**ctld.exe -m 1 -T 127.0.0.1 -t 4533

Note the different port number! If multiple devices are in use it is recommended that even port numbers be used for rigs (one instance of rigctld is required per physical connection to a rig) and odd numbers for rotators, but it doesn't really matter as long as there are no clashes (and the right ones are connected).

[113] WSJT-X does NOT include a version of rotctld.

4.3 Information Applications

4.3.1 Chat

Chat is similar to an IRC application allowing text messages between stations. It can now be run in either (or both) of two configurations: as a separate window or, with suitable Screen Layout definition, as a panel in the Logger.

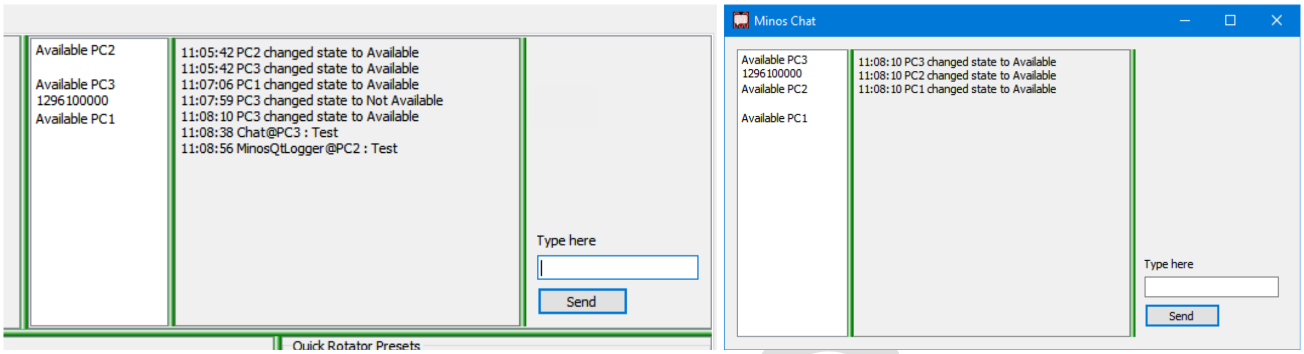

Figure 118 – Examples of Chat usage (Logger Panel and Standalone app)

The application is self configuring as machines come Available (chat application is visible) or go NotAvailable (chat is not visible, machine may or may not be active). To send a message (to all available stations) enter it in **"Type here"** and click **"Send"** (or hit **'Return'**).

4.3.2 Monitor

Monitor allows display of the logs on both remote machines and the local machine. The application is self configuring as Loggers come online/disappear and contests are opened/closed. The contests on a particular machine may be expanded/hidden using the chevrons on the left.

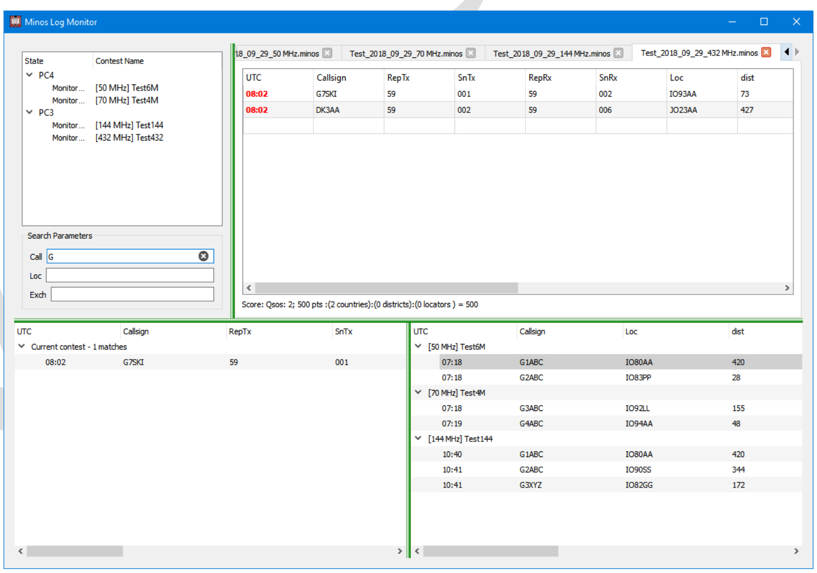

Figure 119 – Example of Monitor usage

To start monitoring a contest double click the contest entry in the top left hand panel. A new tab will be open, displaying data from the monitored log. Click **'X'** on the tab to terminate monitoring of that contest log.

There is a search facility beneath the log selection panel, the results of which are displayed into the lower panels (current contest to the left and other contests to the right). The search, and hence its results, can be cleared by deleting the search content or by pressing **'ESC'**.

4.3.3 Keyer

The Keyer facility is still experimental and this section is preliminary. Keyer provides both voice and CW capabilities, through a variety of methods, but the Rigcontrol interfaces are only supported on recent Icom and Yaesu radios. Indicators in the panel will show if the radio supports the feature. Functionality and operation will vary according to rig capabilities and the method used.

For testing and documentation an IC-7300 simulation is being used. Anyone with suitable real equipment who wants to volunteer to help would be welcomed.

Having added the Keyer panel to the Logger screen layout (see section 3.5), the pulldown list at top left allows selection of the method to be used (Figure 120):

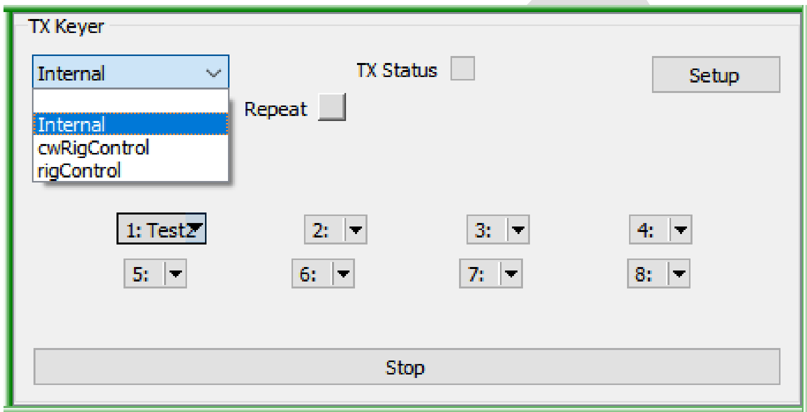

Figure 120 – TX Keyer panel in Logger

The function of the **"Setup"** button depends on the interface method selected, although **"Number of Buttons"** and **"Use CAT PTT for EOM"** are common to all three; the former determines how many message buttons are displayed, the latter enables the use the 'CAT Tx On' message to discover when the message has ended instead of using a timer.

Regardless of the keyer method used, the "**Message Buttons**" may be defined using the pulldown options on each button:

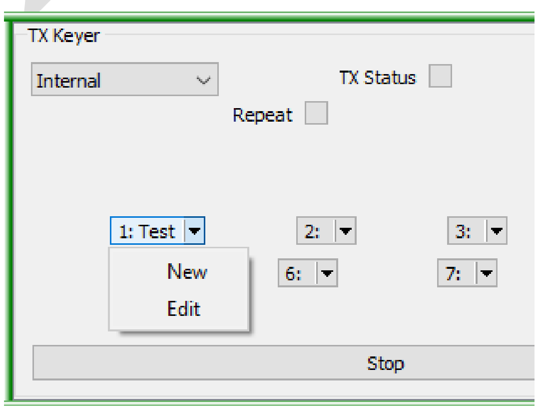

Figure 121 – Definition of button content

- **New:** Create a new user command (see Figure 127).
- **Edit:** Allows changes to be made to a command (if no command exists then no action)

Once the content is defined, clicking the button sends the message to the rig. Each button can be given an individual name and set to repeat, after a defined pause interval if required.

4.3.3.1 Internal

"Internal" uses the computer's sound card and thus is usable with any rig.

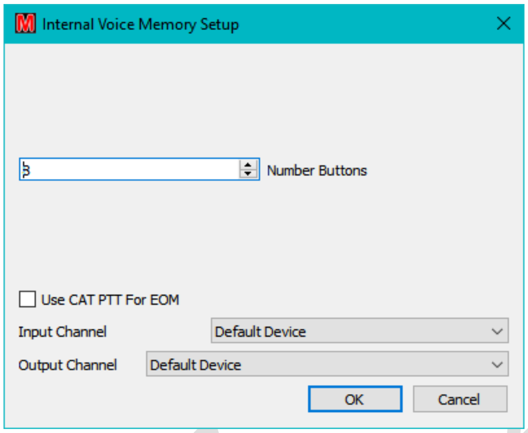

Figure 122 – Setup for Internal Voice Memory

"Input Channel" and **"Output Channel"** determine (via a pulldown list) which audio device is used for each function. Input is used when recording a message via the individual Keyer button's **"New"** or **"Edit"** function.

The "**Message Buttons**" may be defined using the pulldown options on each button in similar manner to Figure 121. **"New"** and **"Edit"** have very similar layout (Figure 123) with a number of fields:

- **"Name"** is what is displayed on the button (Figure 120).
- **"Record"** starts the recording process, coloured indicators will appear in the black box (Figure 124) according to the level set by the blue slider or numerical box.
- **"Stop"** terminates the recording.
- **"Test Replay"** plays back the recorded clip. It may be necessary to change the output device to be able to listen to it.
- **"Show"** provides a spectrum display of the recorded message(Figure 125)
- **"Message Duration"** gives an indication of the message length.
- **"Repeat Message"** flags that the message should continuously repeated.
- **"Repeat Pause Duration"** inserts a delay of the specified number of seconds between repeats of the message.

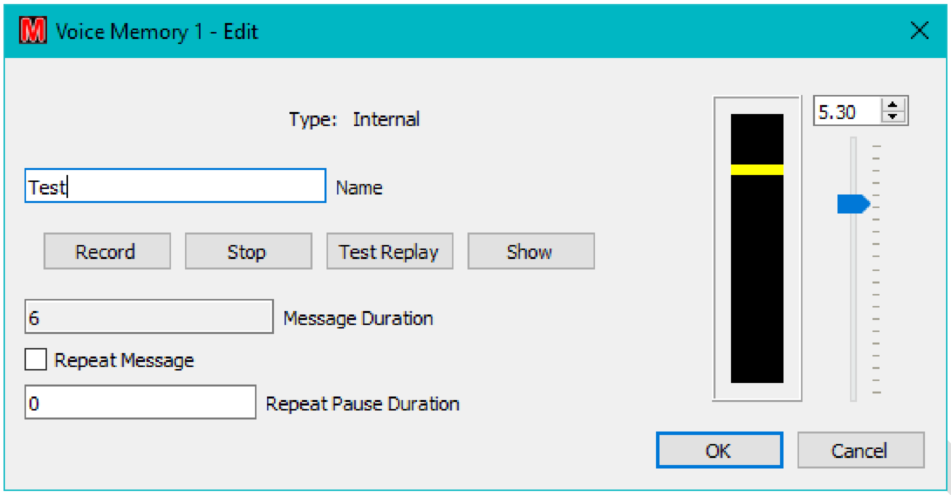

Figure 123 – Defining button content for Internal Keyer

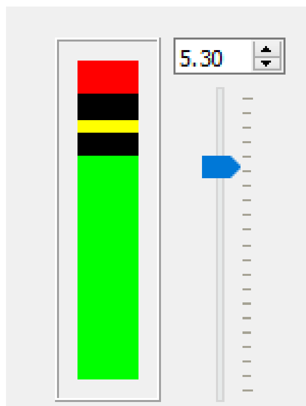

Figure 124 – Recording overloading input

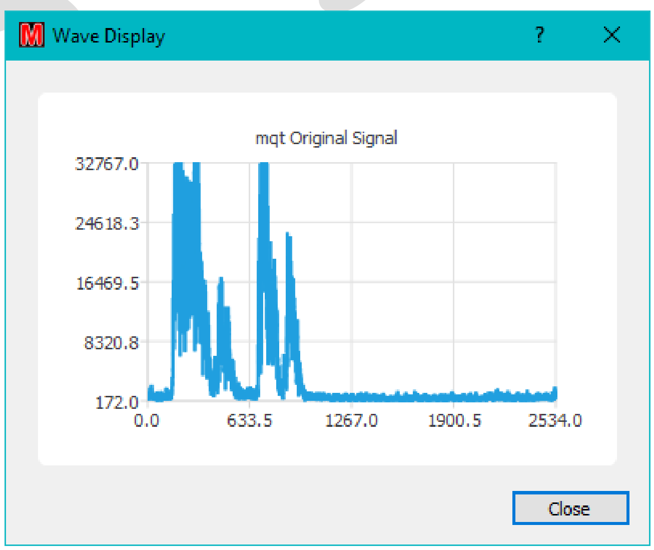

Figure 125 –Output displayed by the SHOW button

4.3.3.2 cwRigControl

“cwRigControl” provides the same functionality as RigControl below, but for CW messages. Icom and **Yaesu** work differently. On Icom the CW message is stored in the button as text which is sent to the rig. Yaesu pre-store the CW message in the radio, it should not be possible to enter the text in the button dialog.

There is no facility to stop a stored CW message on Yaesu radios.

There is a method of storing messages via CAT on the Yaesu, but the Hamlib API does not (at present) support it.

4.3.3.3 rigControl

“rigControl” recalls pre-stored messages from the voice memories of Icom and Yaesu radios . The voice key should work with the Icom radio, that has been tested for real. It is the CW and Yaesu that is not well tested.

<<Use CAT PTT For EOM>>

The **“Avail”** button will show orange if the radio supports the facility.
“TX Status” shows orange when the radio is in TX.
“Repeat” is orange if the message used is repeating.

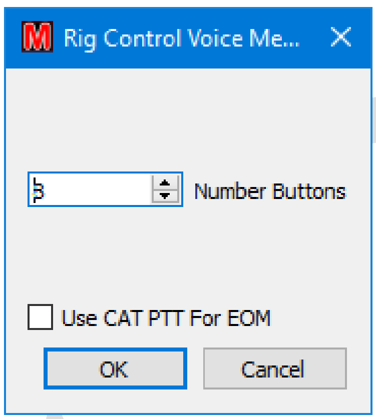

Figure 126 – Setup for RigControl connections

Shift-F1 to Shift-F8 may be used as short cut keys to start the relevant message, and Shift-F10 will stop the message.

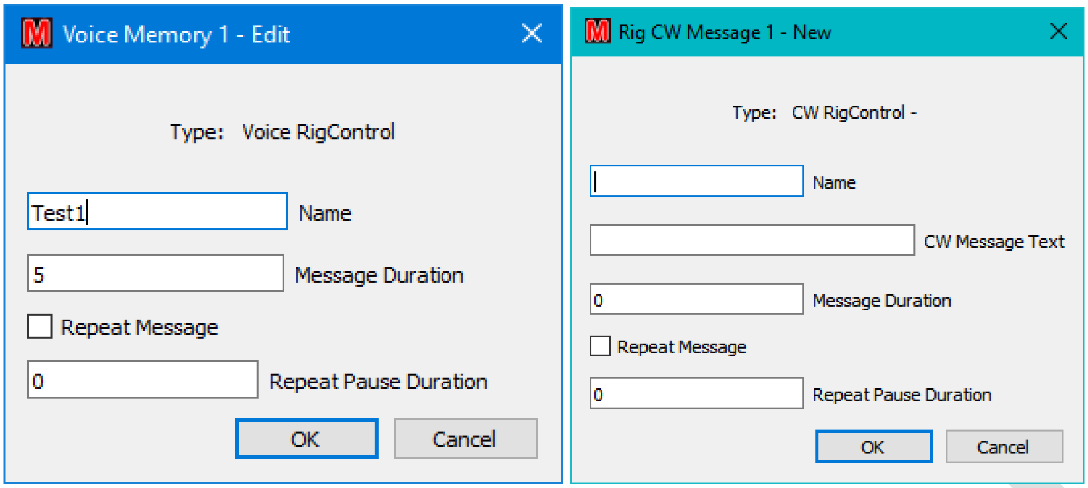

Figure 127 – Setup for RigControl connections

Once everything has been configured the Keyer may be initiated by clicking on the requisite button which will turn orange to indicate that it is in use. The Repeat indicator will light if appropriate (Figure 128), in which case transmission will continue until the **"Stop"** bar is clicked.

<<TX Status>>

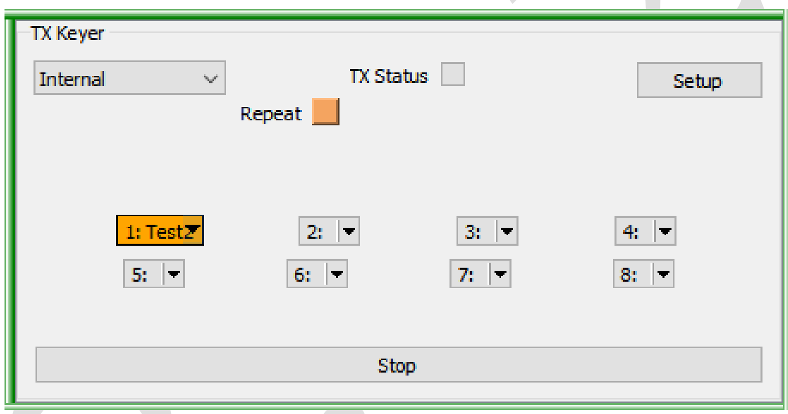

Figure 128 – Keyer activated

4.3.4 Cluster

This facility monitors a selected DX cluster node[114] and passes spots into any open contest tabs that have a Cluster Client frame configured. The spots displayed in each contest can be controlled by various filter mechanisms on a per-frame (contest) basis.

Before Cluster data can be displayed the Cluster Server must be operational.

4.3.4.1 Cluster Server

The Cluster Server Application is initiated in the same manner as any other Application, see section 4.1. It can be defined as operating remotely so that one or more Cluster Servers can provide data to multiple client machines, each with multiple contests.

[114] Using any filters defined in '**…\Configuration\Cluster\cluster_start.txt**' if it is used during Minos startup.

Once the Cluster Server is running, invoke the Setup screen from the Configure pulldown list (see Figure 129). This allows the cluster server to be configured to get the spots that are then filtered/displayed in the Cluster Window(s).

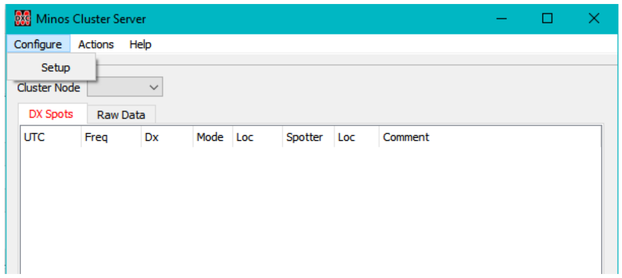

Figure 129 – Initial Cluster Server panel display and Setup

The General tab (Figure 130) allows adjustment of the **"Time to Live"** for spots[115].

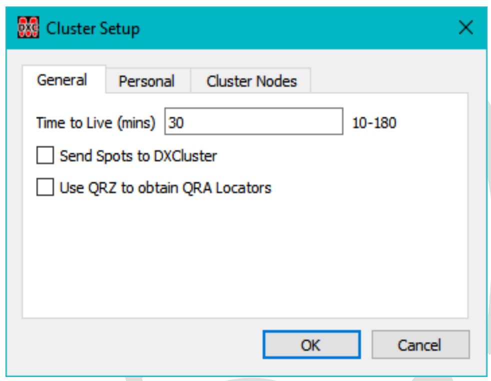

Figure 130 – Cluster Setup general parameters

The time to live must be between 10 and 180 minutes; the default is 30 minutes. Spots will be removed from the Cluster Client panel(s), including filtered call/locator entries, after the period specified.

Spots are not saved in the clients or server. It may be useful to have a predefined button (show/dx) to recall the recent spots if a restart of the program has occurred[116].

If **"Send Spots to DXCluster"** is ticked it enables spot sending as described in Section 4.3.5.4.

"Use QRZ to obtain QRA Locators" enables the logger QRZ button functionality (see section 4.3.7).

Information on the **'Personal'** data[117] tab (Figure 131, left) must be supplied before a successful connection can be made to a DXcluster server. The DXcluster server to be used is selected from the **"Cluster Node"** pulldown list in the main screen. The contents of this list is derived from the "**Cluster Nodes**" tab (Figure 131, right) which contains a suitable selection of nodes, but allows specific nodes to be added or deleted; password information can also be supplied if required.

Any such changes to the node list will be lost if the software is re-installed!!

---

[115] Also controls time to live for Bandmap entries (see Section 4.3.5).
[116] This can also be done through the DXcluster startup file; at least one DXcluster node (GB7RAU) is configured to do this automatically.
[117] The 'Locator' is not derived from the contest details and must be changed for different locations.

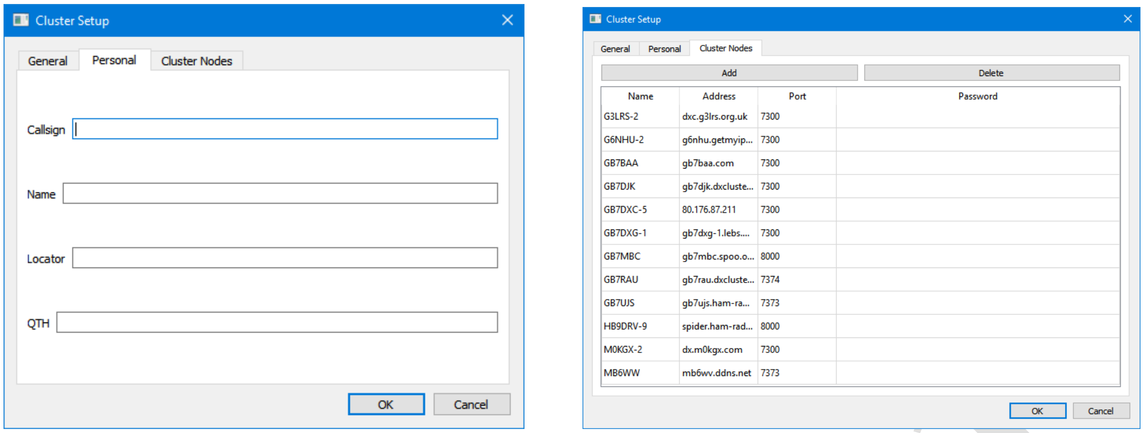

Figure 131 – Cluster Server configuration

Once the **'Personal'** data is fully defined, return to the main Cluster Server screen and select which of the defined DX clusters will be used as the source of data[118].

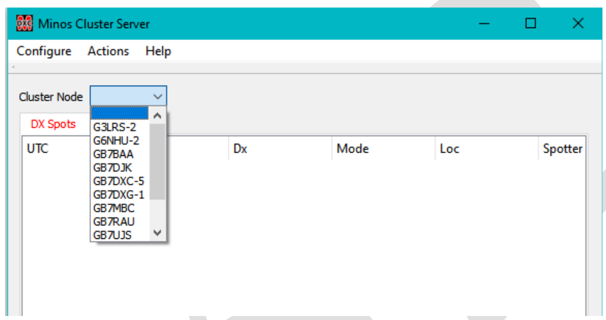

Figure 132 – Selecting the source DX cluster

Attempting to connect to a DX cluster before configuring **all** personal data will result in an error message:

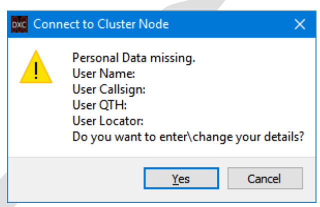

Figure 133 – Personal data missing for connect to DXcluster

Clicking **"Yes"** initiates the **"Cluster Setup/Personal"** screen to input the information. Once done, deselect then reselect the DX cluster node to retry the connect. Check the **'Raw Data'** tab to see that the connect has been successful.

Once the server has successfully connected to the DXcluster spots should start appearing in the server display for all (monitored) bands from 50MHz upwards (see Figure 134). The tab selected in the lower half of the panel is the one that was selected when the Cluster Server last ran.

[118] Note that at least some DXclusters only allow a single login under a given callsign so a different callsign may have to be used if multiple servers are invoked.

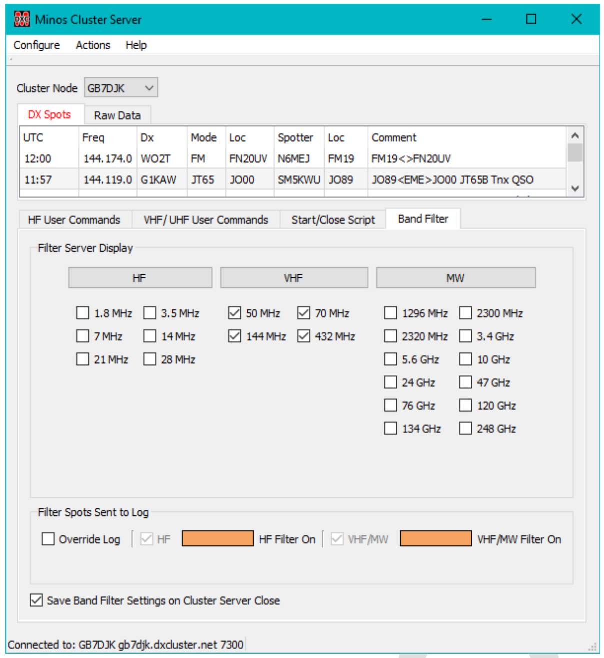

Figure 134 – Initial Cluster Server panel display

The **"Band Filter"** tab has two groups of controls:

> **"Filter Server Display"** (top) controls the display (immediately above) on the server. By default only the VHF/uWave bands are selected. Clicking the buttons above the bands, toggles the group either all on or all off.
>
> **"Filter Spots Sent to Log"** (bottom) filters the spots set to the Logger according to the contest(s) active. A VHF/uWave contest will set the VHF/uWave filter, an HF contest will set the HF filter. Thus if both types of contest are open, all spots will be sent to the Logger. The long buttons turn orange to indicate the active filter(s).
>
> **"Override Log"** allows a manual override to determine which filters are set. There is a warning if both are deselected.
>
> When **"Save Band Filter Settings on Cluster Server Close"** is ticked the current settings are preserved for Server restart. The last saved settings are always loaded.

The **'Start/Close Script'** tab controls the invocation of scripts when connecting to, or signing off from, the Cluster.

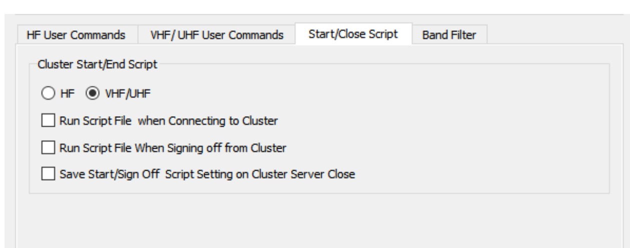

Figure 135 - Start/Close scripts

The **"HF"** and **"VHF/UHF"** buttons control which scripts are used, not the bands being used, so two different scripts can be run in each case.

If the **“Run Script File when Connecting to Cluster”** button is ticked then commands contained in **‘…\Configuration\Cluster\cluster_start.txt’** and/or **‘cluster_start_hf.txt’** are sent to the DXcluster when the connection is made[119].

Ticking **“Run Script File when Signing off from Cluster”** causes **‘cluster_end.txt’** and/or **‘cluster_end_hf.txt’** to be run before the connection is terminated, thus enabling filters (or other information) to be reset to their default values (although there is no display of any resulting data).

To preserve setings for restarts, tick **“Save Start/Sign Off Script Setting on Cluster Server Close”.**

4.3.4.2 DXspider Commands

Selecting the **‘Raw Data’** tab allows the operator to view the dialogue with the DX cluster. This is most likely to be of use when initiating **“User Commands”** (defined in the **‘User Commands’** tabs of the Cluster Server panel – see Figure 136Figure 134), including for diagnosis if no spots occur (there may not be any of course!)[120].

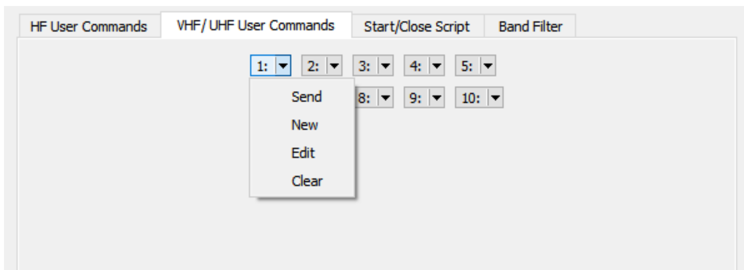

Figure 136 – User Commands

Up to ten **‘User Commands’** may be defined using the pulldown options on each button (Figure 136)[121]:

- **Send:** Send the command (if one is defined) to the DXcluster.
- **New:** Create a new user command.
- **Edit:** Allows changes to be made to a command (if no command exists then no action)
- **Clear:** Clears the content after a positive response to the confirmation message.

Figure 137 – Defining a User Command

Once defined, these commands are stored in **‘…\Configuration\Cluster\ClusterCommands.ini’** which provides another mechanism for defining/editing their content, and to ensure that they appear each time the Cluster Server is started.

[119] This is in addition to any DXcluster defined startup file, that may be there from years ago. Use **show/startup** to check.

[120] **Beware:** command syntax *may* vary by DXcluster; DXspider based nodes are assumed.

[121] A useful document for the adventurous can be found at http://www.dxcluster.org/main/filtering_en.html

The **“Actions/Clear All Spots”** facility deletes all spots in the server application; it does not affect the client displays.

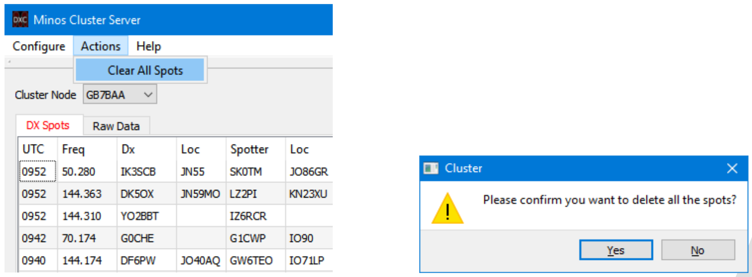

Figure 138 – Clearing spots.

Once the Cluster Server has been configured and is fully operational it **may** be advantageous to use Application Setup to change it to hidden, although the “DX Spots” tab does provide a useful summary across all bands if there is space on the screen.

4.3.4.3 Cluster Display

Once the Cluster Server has been configured (section 4.3.4.1) the Cluster Client (display) frame is added[122] using the **“Configure Screen Layouts”** capability (see section 3.5)[123].

One cluster frame is allowed in each open contest tab.

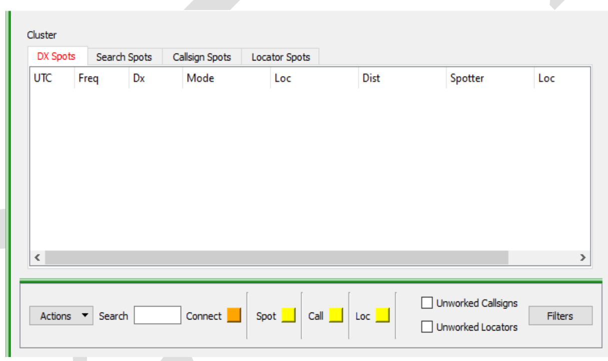

Figure 139 – Initial Cluster Client panel display

The **‘Connect’** indicator turns orange when communication has been established with a Cluster Server (local or remote) that has established communication with its designated DXcluster.

Ticking the **“Unworked Callsigns”** and/or **“Unworked Locators”** boxes will filter out from the display and entries that match logged contacts for Callsign or Locator respectively.

‘Search Spots’ tab (Figure 140) allows the Dx and Loc fields within the panel content to be searched for strings of interest. Enter the required data into the “Search” field and hit return. The display will switch to the **‘Search Spots’** tab to display any results found[124].

---

[122] Cluster Client forces a minimum height to the row (so don't put it with RigControl if space is limited).
[123] The layout including “**Cluster”** should be made the default if the facility is used with most contests.
[124] Currently only partial callsigns and four character Locators are processed.

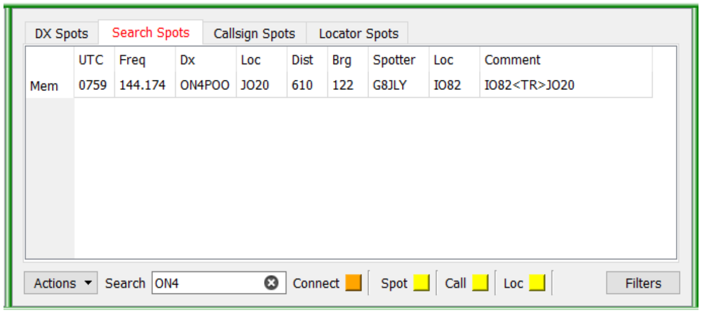

Figure 140 – Search facility

The content of the display can be controlled by clicking the **"Filters"** button and adjusting the Band/Mode of interest or the callsigns/locators to be watched and the distance constraints (see Figure 141).

Beware: if you don't see anything, check your filters.

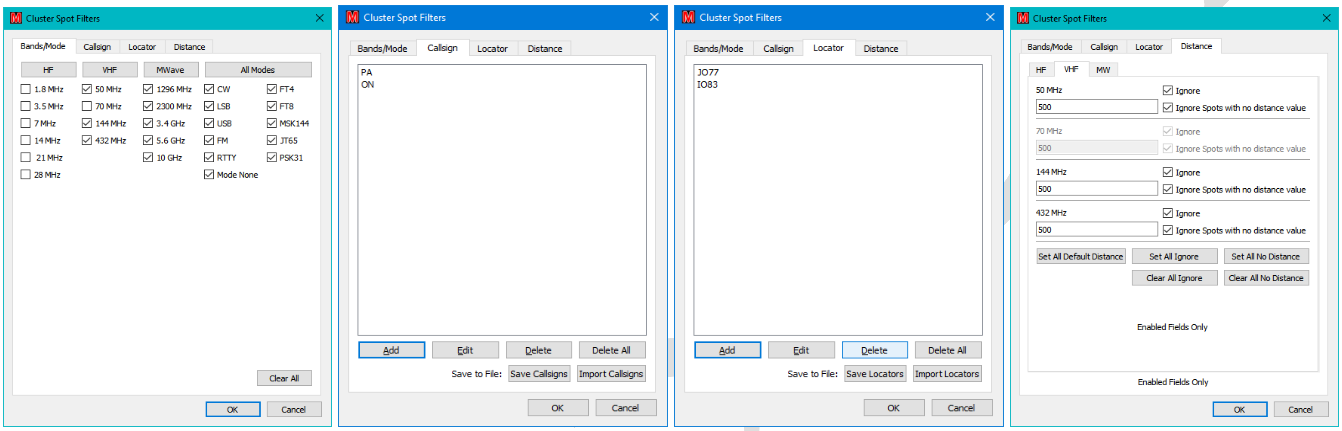

Figure 141 – Setting up filters

By default each frame is set to receive spots just for the band and current contest mode configured for the contest within which it is defined .

The Callsign and Locator tabs allow predefined callsigns and (four character) locators to be watched[125]. These Callsign and Locator filters only apply to spots that satisfy the Band (/Mode) filters.

The Save/Import Callsigns/Locators buttons can be used to export/load favourite callsigns or locators via file(s) (in a default location of **'…\Configuration\Cluster\callsignFilterLists'**). The lists are maintained within each contest context so this facility would typically only be used when a contest is created.

The Distance filters further restrict the spots displayed on the basis of the specified maximum ranges[126]. Note in Figure 141 (4) that the 70MHz box is disabled because 70MHz is not selected in in Figure 141 (1).

Once content is displayed in the Cluster frame a single click may be used to transfer information to other panels. Clicking on different parts of the entry have different effects:

---

[125] DX locators are a bit hit and miss, it depends upon the spotter sending them in the comments.

[126] Can be disabled by ticking the "Ignore" box for a given band.

- **Mem:** Sends the spot to memory panel and ‘tags’ the spot in blue. A spot may only be sent to the memory panel once, unless the Logger is restarted.
- **Callsign:** Load the logger QSO panel with the spot details. Note that in many cases this may only be a 4 character Locator and hence will be flagged as invalid in non MGM contests.
- **Freq:** If Rigcontrol is connected to a radio then it is set to the frequency of the selected spot.
- **Loc:** If Rotator is connected then rotation will be initiated to the bearing of the selected spot (provided it is not blank).

The **“Actions”** pulldown list also provides these functions, as well as allowing individual or all spots to be cleared (after a confirmation message). To use this feature it is probably best to click on the time to avoid activating the ‘field functions’ listed above.

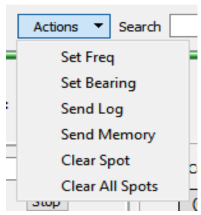

Figure 142 – Actions menu on spots

The callsign turns red if the station has been worked, the locator will turn red if the square has been worked on the current band. Callsign and locator matching is disabled for monitored bands other than the current contest band.

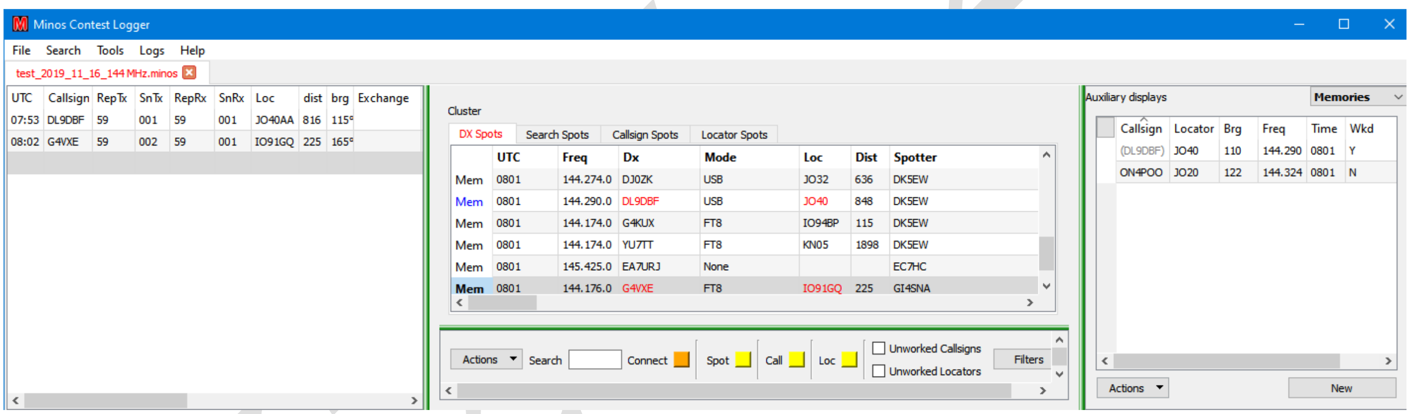

Figure 143 – Colour coding in spots panel

The indicators **‘Spot’, ‘Call’**, and **‘Loc’** at the bottom of Figure 144 are normally yellow, but will turn orange if a new entry appears in one of those views when that tab is not selected[127]. Selecting the tab will clear the indicator.

[127] Figure 139 shows that a match has occurred on a callsign filter.

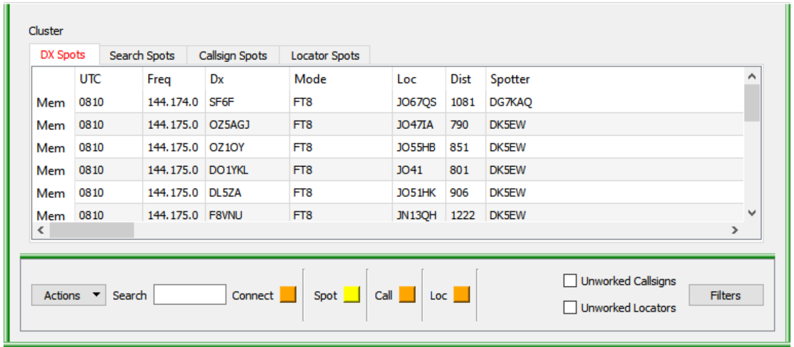

Figure 144 – New matching spots available

If a new spot arrives whilst the Callsign or Locator Spots tab is selected the Spot indicator will turn orange (Figure 145):

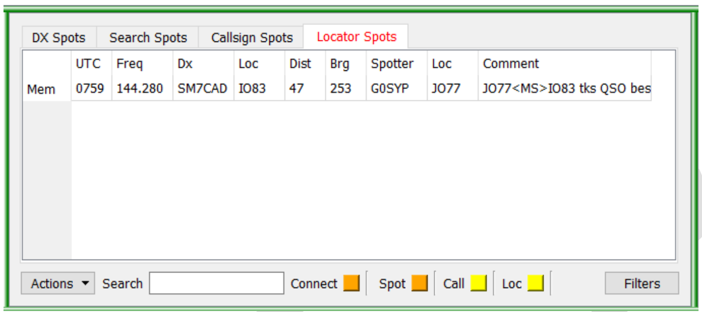

Figure 145 – Indication of new spot arrived

The Cluster Display will show 'Mouse within frame! Updates paused for 5 secs.' if appropriate. This is to warn that updates are inhibited to prevent problems chasing entries while trying to select one. If the mouse remains in the frame for an extended period of time, updates will occur every 5 seconds.

4.3.4.4 Options for Cluster

Within the Options panel (see Section 1.1.2) on the **'Cluster/Bandmap'** tab, the **'Cluster/Bandmap Distance'** sub-tab (Figure 146) is used to set the ***defaults*** for the distance filters. The distance must be in the range 100 to 50000 (!) otherwise an error message is produced.

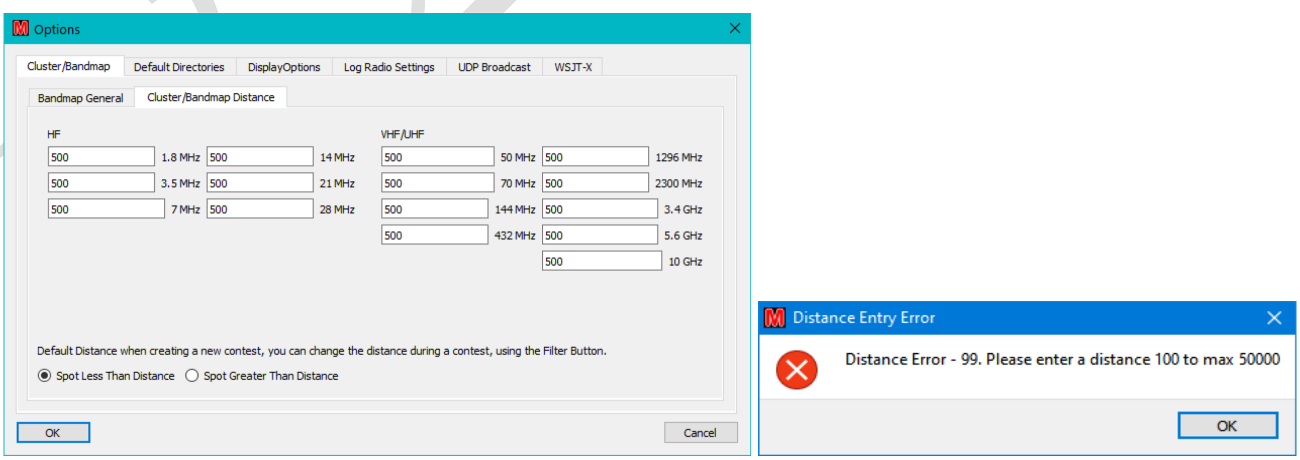

Figure 146 – Default distances for filters

The two buttons **"Spot Less Than Distance"** and **"Spot Greater Than Distance"** determine how the values are interpreted.

4.3.5 Bandmap

Bandmap provides an alternate, graphical, display of information from the DXcluster. It also allows information to be transferred from spots to the QSO Log panel, RigControl, and Rotator control, as well as logged contacts to be sent to the Bandmap display (and DXCluster, see Section 4.3.5.4).

Once the Cluster Server has been configured[128] (see Section 4.3.4.1) and Rigcontrol is operational (see Section 4.2.1) the Bandmap frame is added on a per-contest basis using the **'Configure Screen Layouts'** capability (see section 3.5)[129].

4.3.5.1 Display format

An example display is shown in Figure 147. **Beware**: if you don't see anything, check your filters

The frequency scale on the left of the panel will not contain frequency markers until the rig is connected to the contest. Once the rig has been connected and the frequency scale has been set up it will be retained even if the rig is disconnected or the band changed.

The red markings on the frequency scale, adjacent to the white space, indicate where the current mode is in contravention of the bandplan. No attempt is made to force compliance. The visibility of these markers is controlled by the parameters in **"Tools/Cluster/Bandmap Configure"** (see section 4.3.5.5/Figure 155). Currently only CW and USB modes are implemented, and are subject to checking.

The Display works only on the band of the open contest, although content is retained when switching between contests. The Bandmap data does not persist from session to session, so after a restart it will lose the data up to that point. <<This *may* be rectified in a future release>>.

[128] To receive the DXcluster spots. Cluster Client is not required, indeed its presence *might* prove distracting.
[129] The layout including "**Bandmap"** should be made the default if the facility is used with most contests.

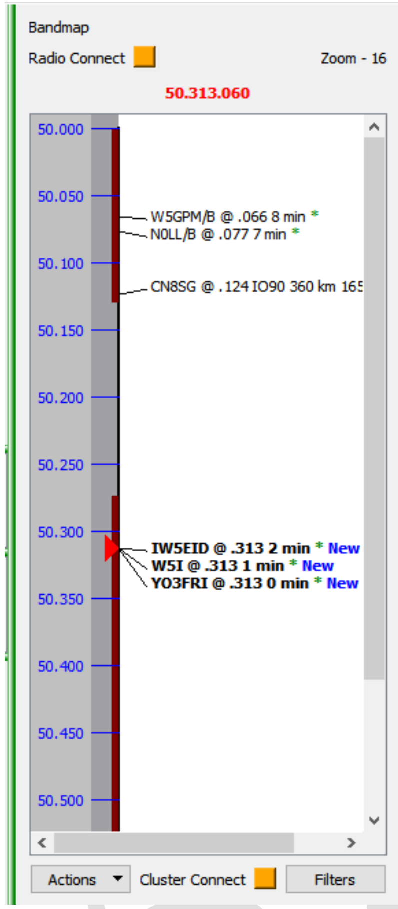

Figure 147 – Bandmap display

The default range of the display defaults to the size of the band, but can be controlled using the **'bandmap_limits.ini'** file in the configuration directory.

To vary the zoom in order to focus on areas of interest use the scroll wheel[130] while hovering over the grey frequency bar. The current zoom setting (- 16, the least zoom, in Figure 147) is dislayed in the top right of the panel. The scroll bar on the right can be used to move within the extended display thus created, or the scroll wheel while hovering over the white portion of the panel[131].

The **'Connect'** indicator turns orange when communication has been established with a Cluster Server (local or remote) that has established communication with its designated DXcluster.

Cluster spots[132] will mark the freq of the station, show callsign, locator, distance and bearing if the spot includes the distant station locator (see example in Figure 148).

By hovering the mouse over spot a tool tip is displayed showing additional information where available (the full frequency, the spotter's callsign and locator, computed mode of spot[133], and comments in the spot (see Figure 148).

---

[130] It is also possible to use the keyboard – see 'Keyboard options' below.
[131] If the dial cursor goes out of view while zooming or scrolling, clicking the frequency display at the top will bring it back into view.

[132] If no spots are appearing, check the filters.
[133] The mode is deduced from the comments if they exist.

Max of three spots shown on the same frequency; the most recent spots are retained.

Cluster spots will be purged using the same time to live duration as set in the cluster server. Logger spots are not removed.

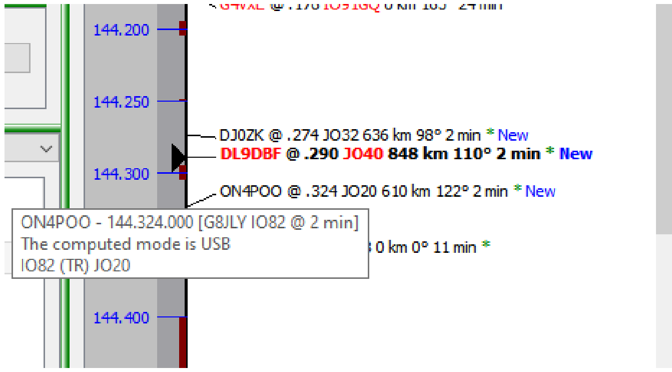

Figure 148 – Marker info

A variety of markers and symbols are used within the Bandmap display:

- When a new spot is added the suffix New is appended. After three minutes this suffix is removed.
- If the spot is from the DXcluster it will be marked with green asterix '*'.
- If either the Callsign or Locator have been worked in the current contest they will be marked in **red**.
- Any spots of particular interest can be 'marked'[134] (see 4.3.5.3) which is shown with a '**#**'. The symbol is removed if the entry is then unmarked[135]. A marked spot will not be removed after timeout.
- If a spot is in bold it shows that the rig is currently set to the frequency of that spot.
- A spot that has been selected by clicking on it is marked grey (it may also be bold).
- If a station displayed on the Bandmap is worked then it is removed and a new entry is created[136] containing the information from the Logger (which may differ from the original spot).
- The current rig frequency is indicated by the black 'play' symbol ('dial cursor' on 144.290 in Figure 147). If this symbol turns red it indicates that it is a frequency/mode pairing that does not match the bandplan.
- If operating in Run mode, the current Run frequency is marked on the display[137] as 'CQ Frequency', with the fractional frequency appended (see Figure 149); the marker changes colour if the rig moves off the run frequency.

If the Bandmap needs to be updated because a worked station has changed frequency, tune to the new frequency then enter their callsign in logger. It will show in red (already worked) in the logger contact box. Now click the **"Save"** button, or tune off frequency and the marker (spot) will be moved to the new frequency.

[134] Only cluster spots can be marked.
[135] Manually initiated markers from the Mark/Save Freq buttons in the QSO logger cannot be unmarked, they must be deleted (Cleared).
[136] The new entry becomes selected.
[137] It is removed when run mode is off or the rig is tuned off frequency.

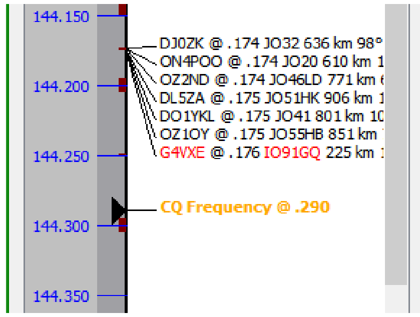

Figure 149 – Run frequency marker

4.3.5.2 Available actions

Various actions are available from the display, either by the Actions/Filters buttons at the bottom, via clicking, or from the keyboard.

The **"Filters"** button (see Figure 151 right) allows mode or distance-based configuration of the spots that will be displayed. The mode is computed from the frequency reported, but is overridden by information noted in the comment. The mode-filter settings are initially set to the contest mode by default[138]; any changes are saved with the contest.

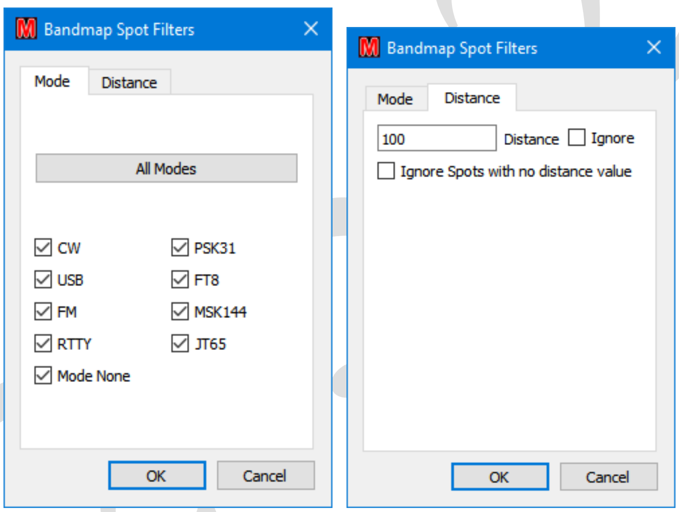

Figure 150 – Bandmap Filter options

If an entry is selected, the **"Actions"** button (Figure 151 left) provides a context menu of items:

- **"Mark Spot"** – converts cluster spot (**only** cluster spots, no error message is given) to a saved spot which will not be purged.
- **"Umark Spot"** – removes the marking ('#') and makes the spot eligible for purging at the end of the time to live. Since marking/unmarking does not affect the time to live it could be deleted immediately.
- **"Set Freq"** – tunes radio to the spot's frequency.
- **"Set Bearing"** – turns rotator to spot bearing.
- **"Send Log"** – sends marked spot details to log.
- **"Send Memory"** – sends marked spot to memory
- **"Save Zoom Level"** – saves the current zoom level on the frequency bar to the current contest. This will become the default zoom level when the contest is (re-)opened.

138 MGM will enable all the MGM modes.

- **"Restore Saved Zoom Level"** – restore the zoom level to what it was when the contest was opened or the last **"Save Zoom Level**" was done.
- **"Clear Spot"** – deletes the selected spot (after confirmation).
- **"Clear All Spots"** – deletes all the spots (after confirmation).

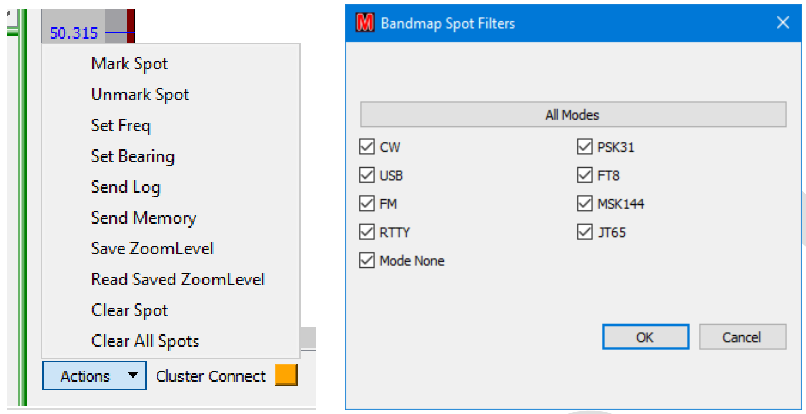

Figure 151 – Actions and Filters buttons

The clicking options are:

- Single click a marker to select it and to tune to its frequency[139].
- Click on white space to remove selection.
- Double click loads marker details to QSO log panel, but does **NOT** replace report and serialRx.
- The rig can be set to a particular frequency by clicking on the (dark) grey bar.
- Click a frequency label (blue chars) to move the rig to that frequency. Clicking between blue chars does not do anything.
- If the dial cursor is not viewable, click the dial frequency at the top of the panel (144.290 at top of Figure 147) to bring that frequency back into the viewport.
- Right clicking any marker (unselected or selected) allows the same options as the **"Actions"** button (apart from 'clear all spots).

There are a number of shortcut keystrokes available (listed in Figure 178).

As with the Cluster Display, Bandmap will show 'Mouse within frame! Updates paused for 5 secs.' if appropriate[140].

4.3.5.3 Related QSO log panel functions

When the Bandmap panel is operational extra functions become available in the QSO logger panel (see Figure 152).

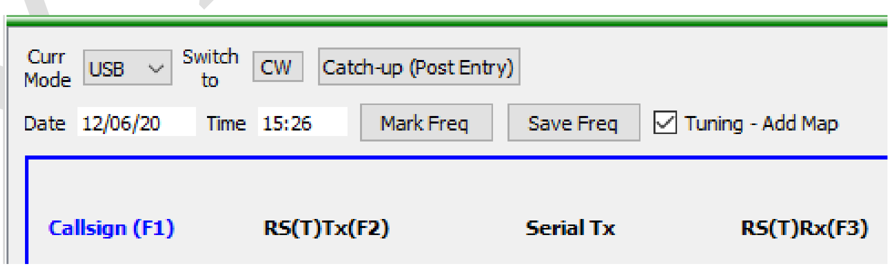

Figure 152 – Bandmap functions in QSO log panel

[139] If the frequency display turns red the current rig mode is wrong for that frequency.
[140] Similarly, updates will occur every 5 seconds if the mouse remains there.

There are two buttons associated with Bandmap in the QSO Log Panel:

- **"Mark Freq"** – Marks the freq with "???" to show the freq was occupied, but the callsign was not identified, even if present in the QSO panel. If rotator is connected, then the heading will be marked with the spot.

  Logged contacts will be added to the map when in S&P mode.

- **"Save Freq"** – **Provided there is a valid Callsign** in the QSO panel a marker is written to the Bandmap containing the Callsign, and any Locator information present. Distance and bearing are calculated from that Locator information and added to the marker. If no Locator information is present, but there is a connected rotator then its current heading is used and the distance information is suppressed.

- If another "**Save"** is issued for the same callsign on a different frequency then the Bandmap marker is moved to the new frequency. If a cluster spot arrives with the same Callsign as a marked spot, but with a new frequency, the marker will be moved to the new frequency.

  The mode of new markers is calculated on the basis of frequency, and filtered on the basis of any active mode filters.

  The marked entry will have the symbol # at the end to show it is from the Logger. Any markers thus tagged will not be purged.

These two buttons are disabled in run mode.

There is also a tickbox:

- **"Tuning – Add Map"** – If set, and a Callsign is present in the QSO panel, then a Bandmap marker will be automatically generated if the rig is tuned off frequency. Locator, distance, and bearing information will be added in the same manner as the "**Save"** button. The QSO panel will be cleared, but its contents can be restored by pressing the ESC key.

4.3.5.4 Sending Cluster Spots

Providing the server app has been configured to send spots (see Section 4.3.4.1) it is now possible to manually send cluster spots from the QSO log panel. If you are in Run mode and on a run freq, the Spot button will be disabled.

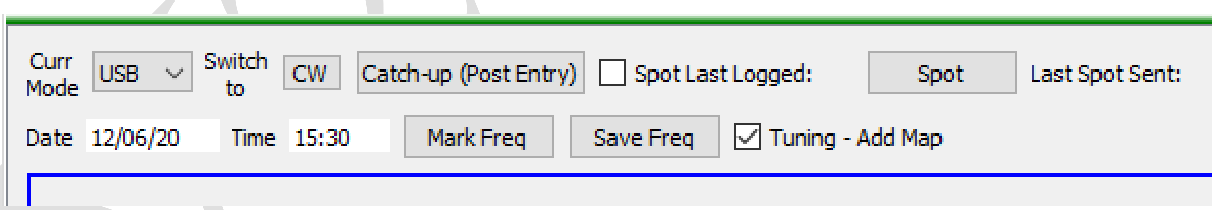

Figure 153 – Spot controls

The **"Spot"** button initiates the sending of the spot. This can be from either of two sources, controlled by the setting of a tickbox **"Spot Last Logged"**:

- **"Clear":** the spot data is sent from the QSO panel. The Callsign, Frequency, and Locator (if entered) will be sent.

- **“Set”**: the spot data is sent from the last logged QSO (which is indicated by the **“Spot Last Logged:”** field) which will normally contain all of Callsign, Frequency and Locator[141].

In either case, the spot comment with contain a comment of the form IO91SN< >IO92JL where the first locator is the spotters locator and the second is the DX locator (see Figure 154).

The callsign of the last logged contact is displayed between the **“Spot Last Logged”** tickbox and the **“Spot”** button.

Once the spot has been sent the Callsign and Frequency will be displayed to the right of the **“Last Spot Sent”:** label and an entry will be added to the **‘Sent Spots’** tab of the Cluster Server (see Figure 154).

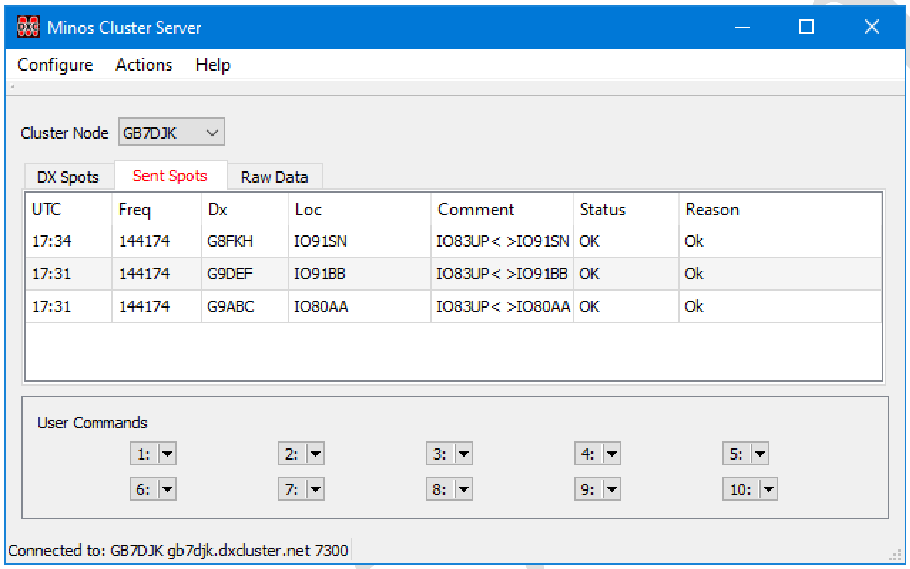

Figure 154 – Spots that have been sent

The table of Sent Spots contains details of the sent spot and their status. OK if spot was sent ok, or Failed if the spot failed to be sent.

Sent spots will not be forwarded to the Cluster Client or Bandmap.

4.3.5.5 Options for Bandmap

Within the Options panel (see Section 1.1.2) on the **‘Cluster/Bandmap’** tab, the **‘Bandmap General’** sub-tab (Figure 155) is used to define general parameters for Bandmap.

[141] It is possible to spot a logged (forced) entry which has no Locator information.

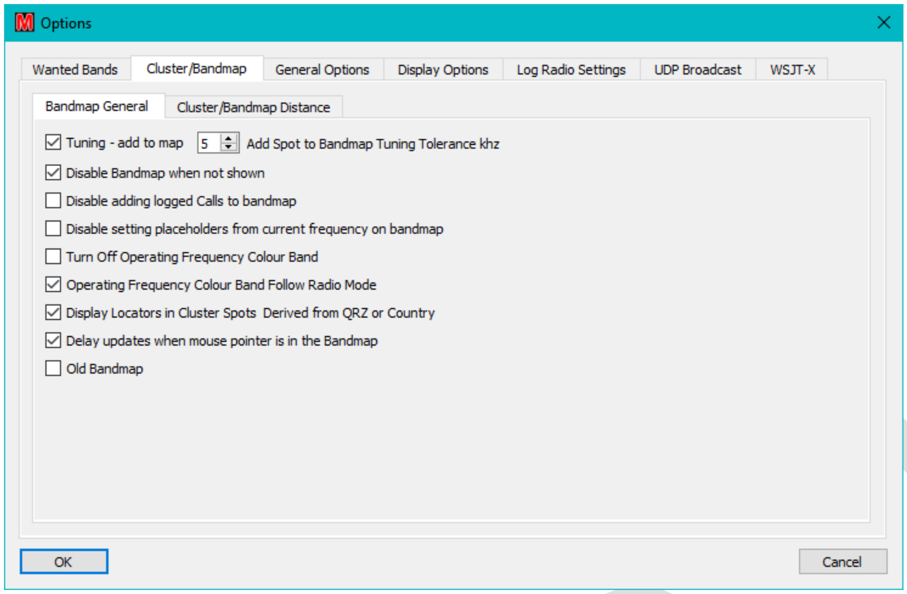

Figure 155 – Bandmap general options

which consists of a number of options:

- **Tuning – add to map** –If enabled (default off), allows Tuning Tolerance additions to take effect.
- **Add Spot to Bandmap Tuning Tolerance khz** – (in the range 1-10) Controls the allowable frequency variation before the Callsign (and Locator) information in the QSO frame is added to the Bandmap when operating in Run Mode – see Page 69. The requested value is only changed on Logger startup or Contest open.
- **Disable Bandmap when not shown** – When on (default), stops any bandmap interactons unless there is an active bandmap panel
- **Disable adding logged calls to bandmap** – If enabled (default off) calls are not added to the Bandmap when being logged.
- **Disable setting placeholders from current frequency on bandmap** – (Default off) If set, tuning across a spot on the bandmap doesn't set placeholder text for it.
- **Turn Off Operating Freq Colour Band** – If set, disables the colour band on the right hand side of the frequency bar.
- **Operating Freq Colour** Band **Follow Radio Mode** – If set, causes the content of the colour band to change dependent on the current mode of the contest.
- **Display Locators Derived from Country** – If no locator is found in the spot, and none can be found via QRZ Server (if active and enabled in Cluster setup), then the prefix extracted from the callsign and used to calculate a locator. Locators that are not from the spot are displayed in italics to signify they should be used with caution.
- **Display Updates when mouse pointer is in the Bandmap** – Don't "pause" updates when the mouse is in frame.
- **Old Bandmap** – Uses the "old style" display which is not centred on current frequency **.**

4.3.6 KST Client

KST client provides an interface to the ON4KST amateur radio chat facility. The interface is similar to, but not the same as, KST2Me. Multiple instances may be run provided they have different names in the Minos Application Definition. Each will have its own parameters (Login details).

The application may also be run standalone, separate from Minos, but once again separate configuration information must be provided which will then be used for all the standalone instances.

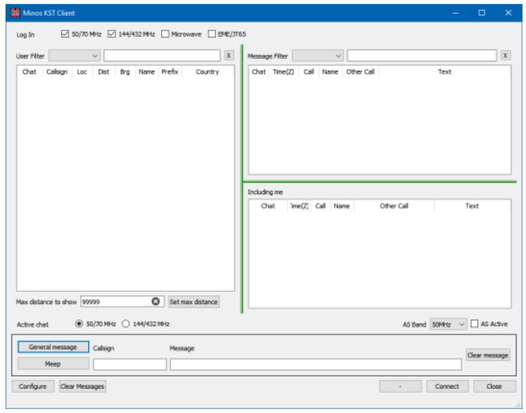

Figure 156 – KST Startup

“Configure” – invokes the Configuration panel (Figure 157) which then requires:

- **Callsign, Locator,** and **Password** for access to KST to be entered. The Locator can subsequently be changed if required, which will be acknowledged in the ‘Including’ panel (bottom right of the KST screen).
- **First Name** allows the name to be changed from that specified in the account (and could be used to indicate /P for example). **Note that if changed it is user’s responsibility to reset it!**
- **Automatically connect on load** – Allows the client to log in to the server without having to click **“Connect”**
- **“Ok”** – Can only be clicked if the **Callsign, Locator,** and **Password** fields have been filled.
- **“Cancel”** – Allows changes to be discarded.

The AirScout configuration information in the bottom half of the panel is described in Section 4.3.6.1.

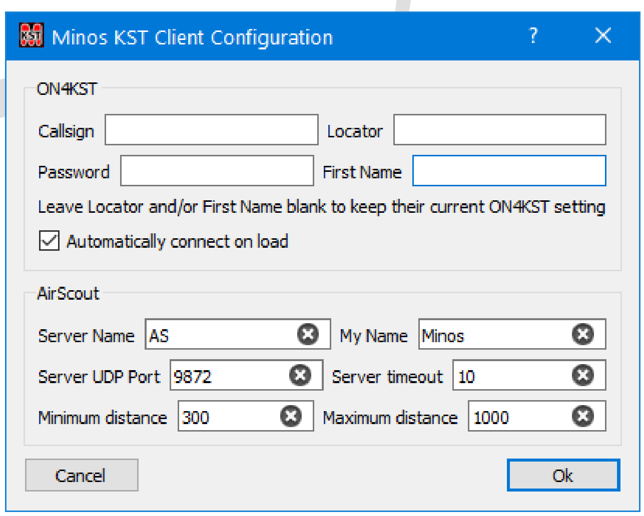

Figure 157 – KST Client configuration

Once the configuration information has been specified the client can connect and generates a display of the form shown in Figure 158.

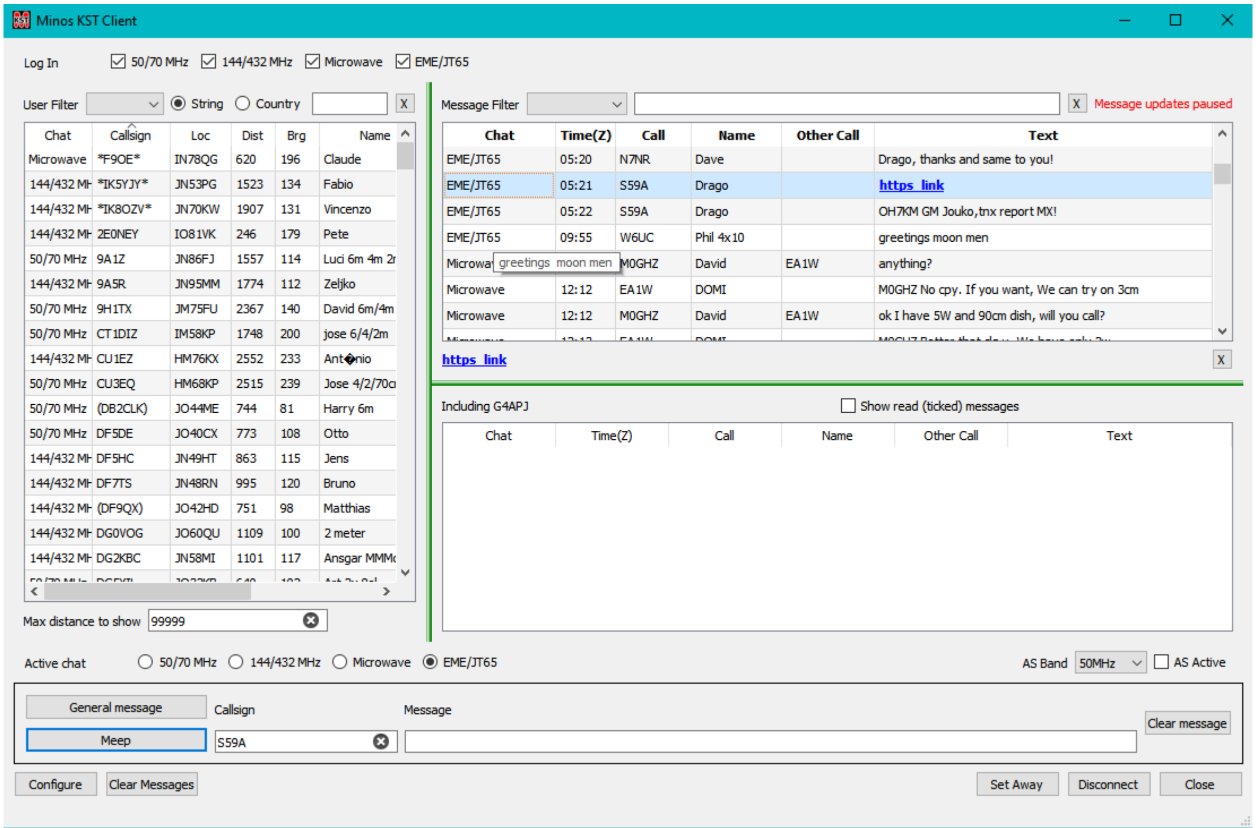

Figure 158 – KST Client in operation

Clicking the column heads allows the display order to be changed. (Figure 158 shows the information sorted by Callsign).

If there are a lot of messages scrolling past the display can be temporarily paused by hovering over it. While the display is paused, a message is displayed (top right of Figure 158).

Hovering over a message displays the text of the message (useful if panel size is minimised) and clicking on a message transfers the content to the space below the panel, enabling enclosed http links to be followed (**https_link** in the middle of Figure 158).

The **left frame** displays the list of users who are logged on, or a sub-set thereof on the basis of the filter(s) just above the display panel:

- **Log In** – The tick boxes select which chats the user is logged in to. Unless filters are used this also determines which bands will be used to provide content to the display panels.
- **User Filter** – The pull-down list selects which chat the user list should be selected from (blank means all). Any string entered into the text box is used as a filter to select and display records containing that string in any of the Callsign/Locator/Name/Prefix fields. Multiple strings may be used, separated by a space.

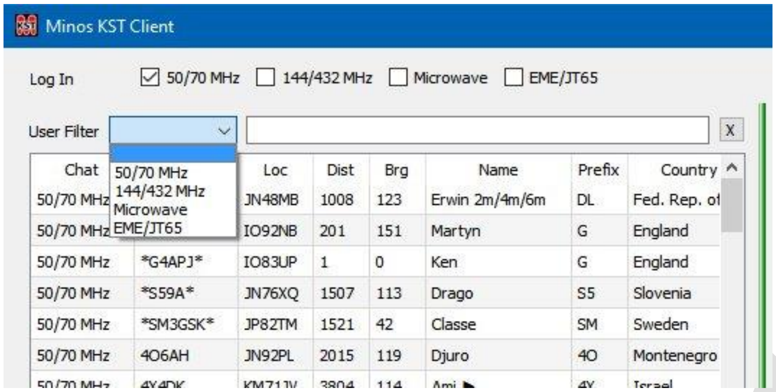

Figure 159 – KST Block Filters

A time ordered list, or sub-set, of user messages (from the selected conversations) is displayed in the **top right frame**, under control of:

- **Message Filter** – A pull-down list that selects the chat of interest in the same manner as for the user list.
- **Text Box** – Any string entered is used as a filter to select and display records containing that string in any of the Call/Name/Other Call/Text fields. Multiple strings may be used, separated by a space.

Note that for performance reasons messages may be truncated with no scroll bar appearing. The only option at present is to make the window wider or hover the cursor over the message of interest to see the full text. It is hoped a solution can be found eventually.

The **bottom right frame** displays any messages containing the callsign used to log on to ON4KST.

Below these three frames there are further controls:

- **Active chat** – Selects which chat a message is sent to or a 'Set Away' command is applied to.
- **AS Band and AS Active** – See section 4.3.6.1
- **General Message** – Send the message text without any callsign specific identifier.
- **Meep** – Sends the message text to the specified callsign. <CR> can also be used.
- **Callsign** – The destination Callsign for a targeted message. It can be set by clicking on an entry in the Users or Messages display.
- **Message** – Type in the text to be sent.
- **Clear message** – Clears the **Callsign** and **Message** boxes.
- **Configure** – invokes the Configuration panel (Figure 157) as described above.
- **Clear All** – Clears the content of all three display panels. This content can be (more or less) recovered by restarting.

- **Set Away/ Back** – Marks the user as away or back on the chat (and chats the button text to the opposite option).
- **Connect/Disconnect** – Switches the state of the connection to ON4KST **for all selected chats**.
- **Close** – If running under control of Minos this restart the client. If running standalone the KSTclient exits completely.

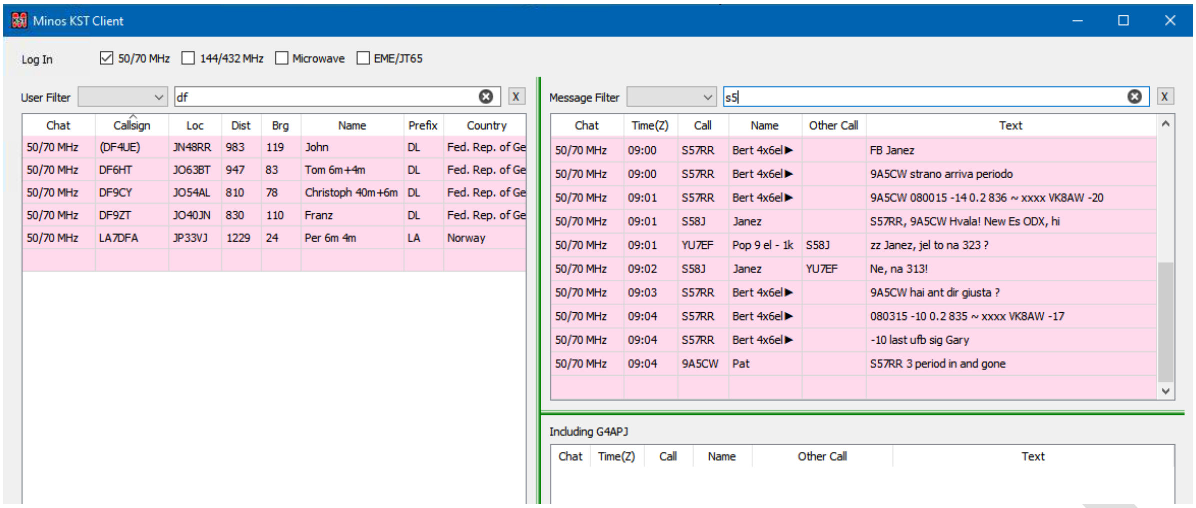

Figure 160 – KST Filters in operation

4.3.6.1 AirScout Integration

Airscout needs to be installed and active for this functionality to work. If it isn't then no error message is displayed, but nothing of interest happens.

If the **"AS Active"** box is ticked (just above the 'Clear Message' button on the bottom right of Figure 156) then AirScout integration is started, subject to the requisite configuration information having been supplied (see Figure 157):

- **'Server Name'** and '**Server UDP Port'** must match those defined in the **'Air Scout Options, Network tab, UDP Server Settings'**. AirScout is assumed to be running on its default **'UDP Server Port'** (9872) on the same machine.
- **'My Name'** = Can be any arbitrary name, the default is "Minos".
- **'Server Timeout'** The number of seconds to wait for a response.
- **'Minimum distance'** and '**Maximum distance'** define the range of entries to be considered.

The **'AS Band'** pulldown selects the band to be used for airscout calculations; it is held separately (and with no linkage) to the current KST chat.

Once the connection is established[142] additional information (AS) is displayed in the User panel (on the left), and a new panel appears bottom left (Figure 161).

[142] There may be Firewall re-configuration as per Figure 72

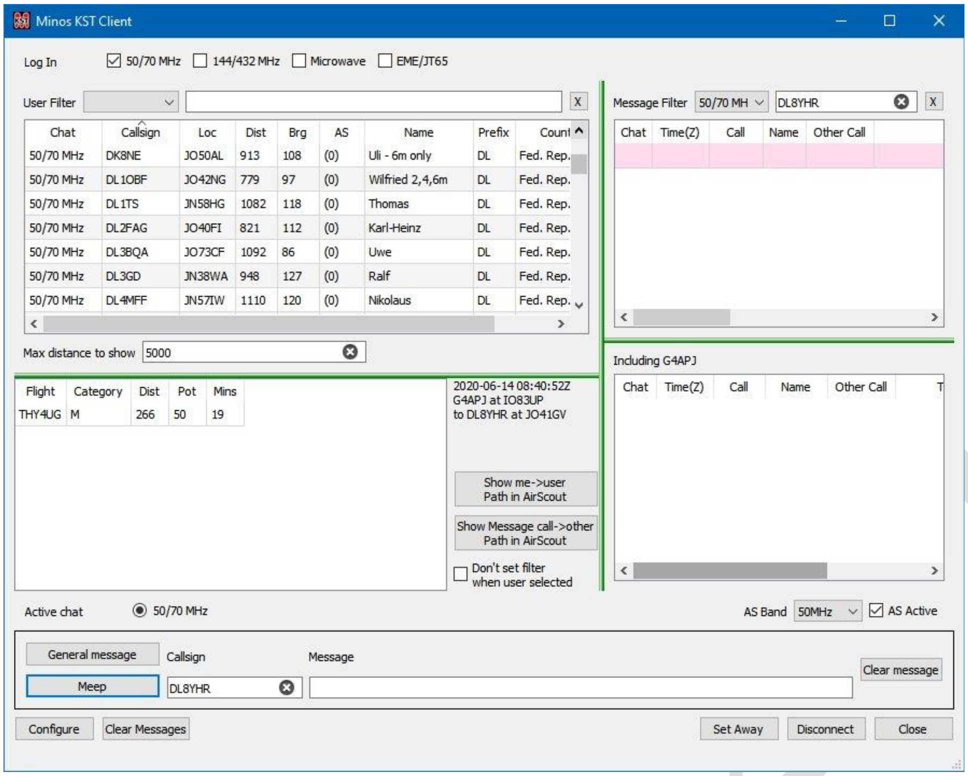

Figure 161 – Airscout operational in KST Client

The **'AS'** column in the User panel contains information of the form n(m). The first number (in magenta) is a count of the planes currently scattering; the second number (in brackets) is a count of all the planes at any time.

All users on KST between the minimum and maximum distance specified, and poll airscout for their nearest planes. Various markings are used: "<" (too close), ">" (too far), "n(m)" (e.g. 1(15) for n planes actually in place, m total (so m - n still getting closer).

Clicking on a specific entry in the User Filter activates the AS display for that user. Within the new AS panel the display has several columns:

- Flight: The Flight Designator (colloquially known as the Flight Number).
- Category: Category of aircraft according to AirScout's category code (L/M/H/S)
- Dist: The calculated distance of the flight from the ideal reflection point
- Pot: An estimate of the usability of the flight in 25% increments.
- Mins: Time in minutes until likely to become effective.

There are three options down the right hand side of the AS panel:

- **'Show me->user Path in AirScout'**: Brings up the AS map (including planes) of the path from the home locator to the selected user.
- **'Show Message call ->other Path in AirScout'**: When clicked on a user in the Message Filter display, the AS map between that user and the station that logged the spot[143].

---

143 The idea is that when someone says to another "plane in 5 mins" or some such it is possible to see what they are talking about.

- **‘Don’t set filter when user selected’**: Under normal circumstances, clicking on a user filters the messages to just those to from that user. Selecting this option ensures that no filter is applied and all messages are displayed.

A Watchlist is sent to AirScout in order to keep in sync with KST

4.3.7 QrzServer

QRZ server is used by Cluster server to lookup the QRA of the spotted station when none can be determined from the spot comments. If the lookup fails then Cluster server uses the callsign prefix to determine an approximate locator. Any such locator will appear in italics to indicate that it may not be accurate.

When first invoked, the server will request authentication information to access QRZ.COM and confirm the login result in its main window.

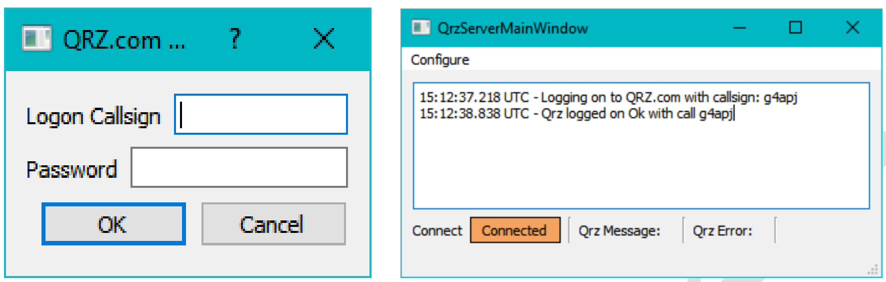

Figure 162 – QrzServer configuration and login

The login information can be modified later via **“Configure/Set up QRZ”** from the main QRZ window.

To enable look up of cluster spots, the **“Use QRZ to obtain QRZ Locators”** option must be ticked within Cluster Setup (see Figure 130).

The QRZ display panel should be added to the Logger (via screen layout). When this QRZ ‘client’ is showing, a **“QRZ”** button appears above the F1 Callsign box in the QSO panel (Figure 163).

With a callsign in the F1 field, clicking on this QRZ button causes a lookup and display (if available) of the station details, with the lookup being logged in the QRZserver main window. Suffixes such as /P should be removed during processing.

Note: In order to perform lookups of QRA Locators an XML Callbook Service subscription is required at QRZ.COM

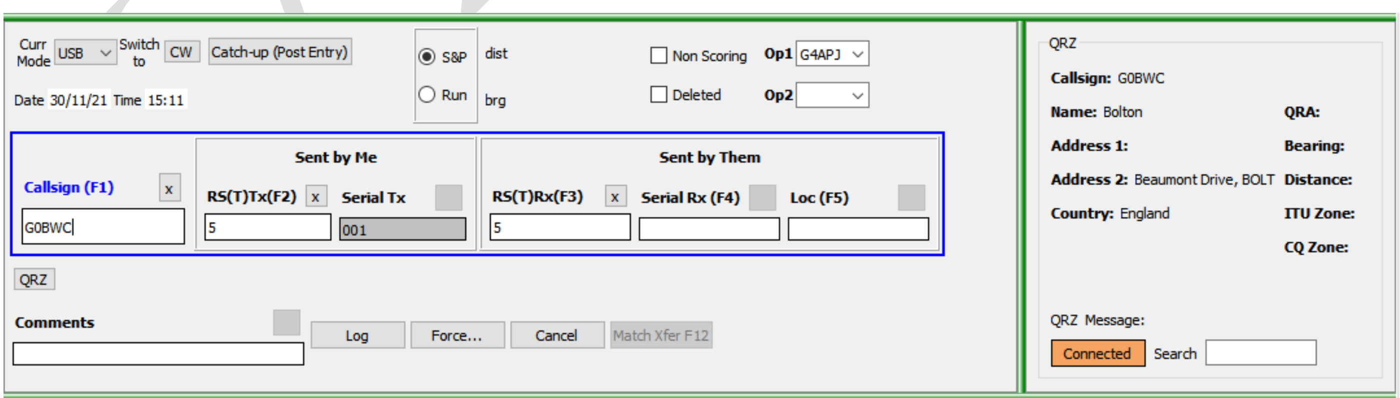

Figure 163 – QRZ lookup in the QSO panel

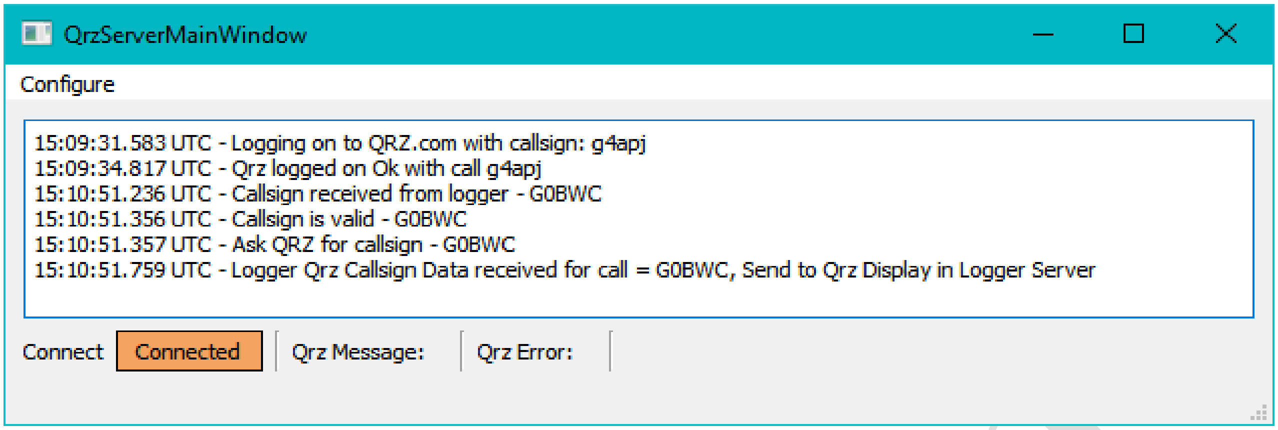

Figure 164 – QRZ button lookup logged

4.3.8 RigRecorder

The audio input from the rig can be recorded using this application.

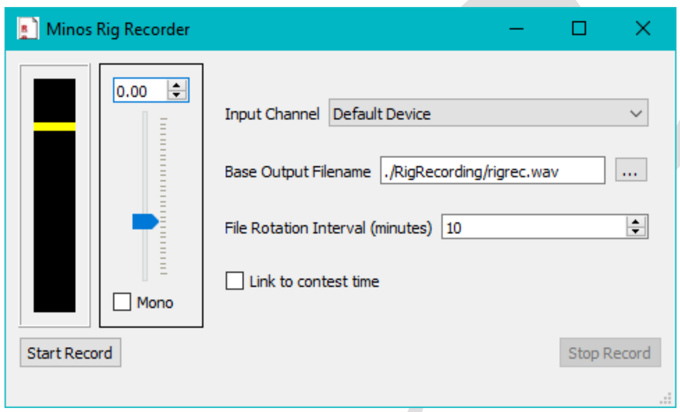

Figure 165 – RigRecorder screen

The box at top left gives an indication of level of audio input. The yellow bar is really only a reference point on an otherwise rather bland display

There are a number of options that can be defined:

- The blue **Slider** is used to adjust the level of audio input to the recorder.
- **"Mono"** takes a single channel.
- **"Input Channel"** allows the audio source device to be selected.
- **"Base Output Filename"** is parsed to provide the root directory (which is created if necessary) into which daily sub-folders of the form 2021-09-22 are created. The base filename then has the date and time appended to create WAV files of the form rigrec_2021-10-15 12-12-28.wav in the relevant sub-folder.
- **"File Rotation Interval"** defines the number of minutes for each (completed) recording (default 10).
- **"Link to contest time"** automates the start/stop of recordings. If the current time is within the time limits of ANY open contest then recording will be enabled and stopped once the (last) end time is reached. An additional 60 secs padding is added at each end.
- **"Start Record"** manually initiates recording (if idle).
- **"Stop Record"** manually terminates recording (if active).

4.3.9 RigSync

RigSync is intended to provide synchronisation between the main rig and an SDR, although it can be used to synchronise two ‘ordinary’ rigs should that be required.

A TS-2000 emulation is used as the interface to both SDR#[144] and SDR Console[145]. Because this is a second ‘rig’ that will be active at the same time as the main rig a second instance of RigControl is required.

The RigControl instances to be used are selected via the Configuration panel (Figure 166) invoked from **“Configure”** at the top of the main screen (Figure 167) and the pulldown lists within the main screen. The instances can be on separate machines[146].

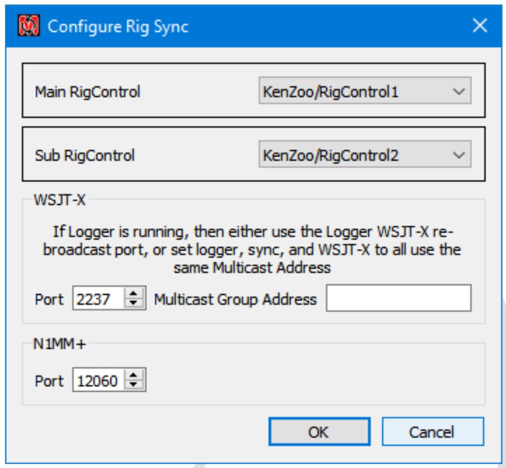

Figure 166 – RigSync configuration of rigs

Usually the SDR is set to track the main rig so that its display is centred on where the main rig is tuned. After a click on an SDR trace the frequency can be transferred back to the main rig.

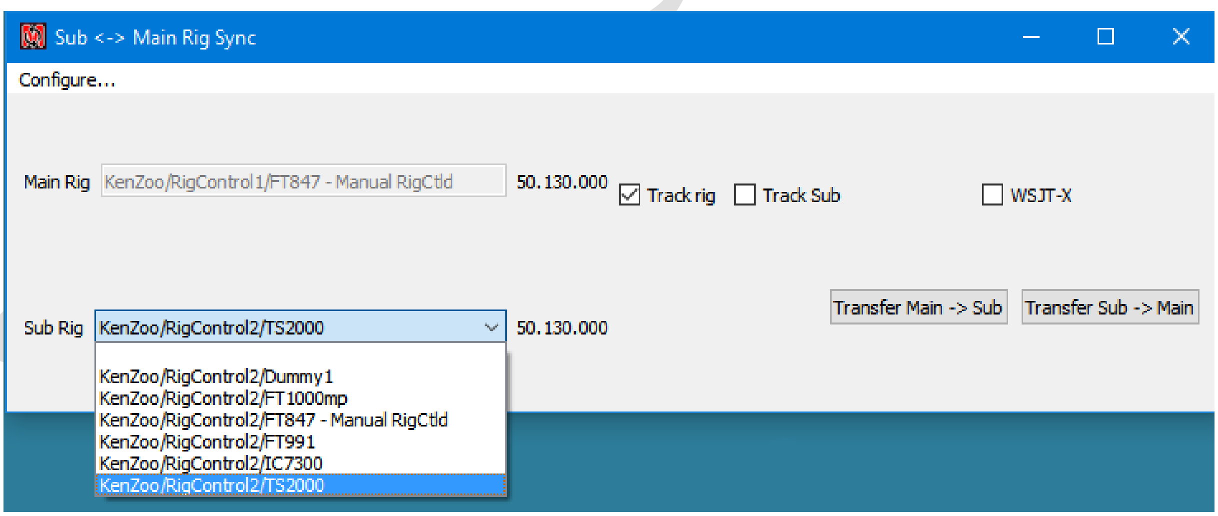

Figure 167 – RigSync main screen

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144 Via the Calico plug in from https://www.rtl-sdr.com/calicocat-new-serial-cat-control-plugin-for-sdr/
145 Configured via Tools/Options within SDR Console.
146 On first invocation RigSync may require Firewall access to be granted.

There are a number of displays and options on the main RigSync screen:

“Main Rig” displays the RigControl instance and rig for the selected principal rig. The current frequency of that rig is displayed to the right of the box.

“Sub Rig” is the RigControl instance and rig for the slave rig, typically the SDR. The current frequency is displayed to the right of the box.

“Track rig” (the default) causes the slave rig to follow the band/frequency of the principal rig.

“Track Sub” reverses the direction of tracking so the principal rig follows the sub-rig.

“WSJT-X” Cause both the main and sub (if active) rigs to follow the WSJT-X frequency, so Minos doesn't need to connect to the main rig. WSJT-X will also track frequency changes initiated from the main rig.
There is also N1MM that allows the frequency broadcast from N1MM.

4.4 Advanced Configuration

Advanced configuration is required for several possible activities:

- Development and testing
- Multi-PC environments necessitated by limitations on space, ports, or PC performance
- Multi-operators

Advanced configuration is invoked by clicking "**Show Advanced**" on one or more applications within App Configuration (see Figure 168)[147].

Complex/multi-machine environments involve other components and careful planning is required, particularly of the names used.

The defaults are quite sufficient for single machine operation with multiple Rigs or Rotators attached, with manual selection of the one being used at any particular time.

4.4.1 Development/debug

For development/debugging it is a simple matter to change the "**Program**" invoked for a particular function. Use the selection button on the right to browse to the development/test version of the application; the "**Parameters**" passed to it and the "**Home Directory**" used may also be modified.

It may be necessary to change the **"App Instance Name**"[148] (**"Rotator-debug"** in Figure 168) in the **"Apps Configuration**" in order to link the contest being used for the testing with a particular (debug/development) application.

Some test scenarios may require the use of a remote machine, and thus may be considered in the same manner as complex/multi-machine environments

[147] Any applications configured for remote access will always have "**Show Advanced**" ticked.
[148] The App Instance Name must be unique on any one machine.

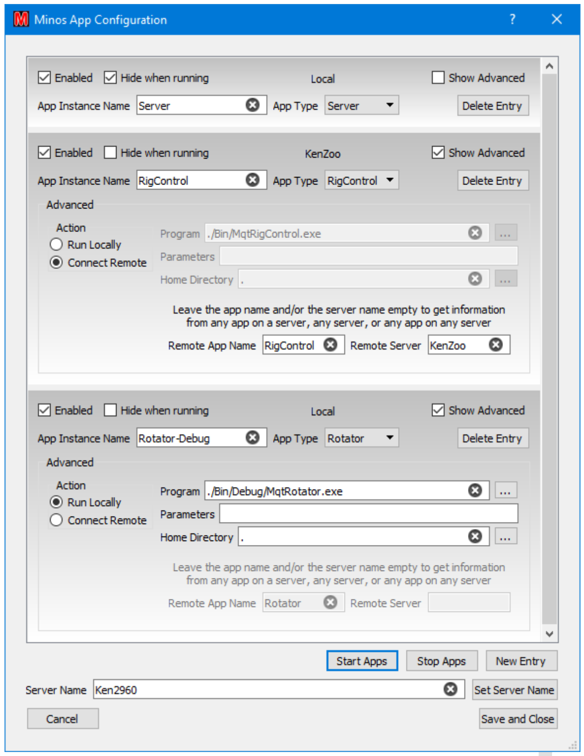

Figure 168 – Advanced configuration

4.4.2 Multiple rigs

The rigs defined through **"Configure/Setup Radios"** are common to all instances of RigControl on any given PC[149]. This means that a given rig can be accessed via multiple routes such as **'PC/RigControl1/Rig1'** and **'PC/RigControl2/Rig1'**.

A number of factors may force the use of a multi-computer environment:

- Number of available ports on any one PC
- A single PC is not powerful enough to handle everything
- The number of concurrent contests, hence:
  - Operators
  - Available monitors
  - Contest logs open
  - Active instances of Minos (only one per machine)
- Physical separation

For information on configuring multiple machines see section 4.4.3

Within each contest that uses a specific rig, change the Contest Details **"Radio Name"** definition (or use the pulldown list in the Logger) to select the requisite rig; the result is stored within the contest, but can be changed at any time.

Configuration information is shared between all instances of RigControl on a single machine. If the wrong entry from the **"Select Radio"** pull-down list was inadvertently selected it would cause problems that might be difficult to diagnose (especially under contest pressures) [150].

[149] The same would be true for Rotator if so configured.
[150] There is a degree of protection due to contention for the comms port(s).

The same technique would apply if multiple rotators were in use (but changing the Rotator App names obviously), and of course both techniques can be used together.

4.4.3 Multi-machine environments

In multi-machine environments all the machines have to be in the same LAN broadcast domain; no intervening routers.

The server name for a given PC defaults to the machine name, but can be changed by setting the new name in the "Server Name" box at the bottom of the App Configuration panel and clicking **"Set Server Name"**.

The default instance of RigControl controls the locally connected rigs, and acts as the target for remote access to the local systems.

In order to have different rigs on different machines, multiple instances of RigControl must be defined; to avoid confusion these instances should have meaningful names. Probably the easiest is to append the server name[151] to which the instance connects, e.g. Rotator-PC2, or the functional area, e.g. Rigcontrol-VHF. Similarly, the target band could be appended, e.g. RigControl-2M and RigControl-70cm as shown in Figure 169, although this implies that each target PC only handles one band (possibly through multiple machines).

The choice will depend on the requirements/topology of the target environment, but make sure the planning is done before the contest, the result is documented and tested, and the final list is adhered to[152]!

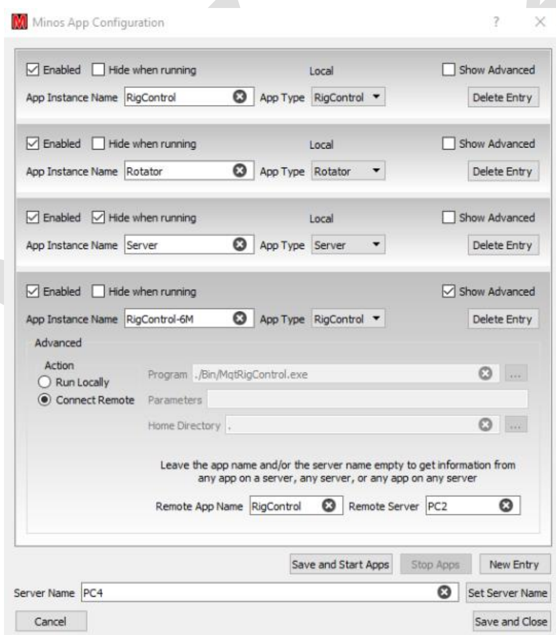

Figure 169 – Configuring two rigs, using remote RigControl

It might be useful to stagger the configuration process; get one working before doing the next.

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[151] Which need not be the same as the PC name, if changed through the Apps Config panel.

[152] You can, if you want, leave the **"Remote App Name"** and/or **"Remote Server"** fields blank within a 'remote' Rigcontrol definition. This will result in **all** instances becoming visible and possibly resulting in confusion, so once again; plan it!

"App Instance Name" is the name by which the connection is known on the local machine. **"Remote App Name"** is the name on the target machine, typically the default. Consistent naming helps – especially in more complex configurations (see section 4.4.4).

If the app is set to **'Connect Remote'** then **'Remote App Name'** and **'Remote Server'** must match the definitions on the remote machine. It is probably desirable to change the names on both machines to indicate that the app is remote/being used by a remote machine.

Obviously the name of the server on which the instance runs must match the name defined in that machine's server definition.

Beware: there is nothing to stop multiple contests sharing a rig or rotator; any such access must be coordinated!

4.4.3.1 AppStarter

If running a machine with no Logger (e.g. a Pi, acting just as a server device) then the Application Starter is used *instead* of the Logger to control the applications.

If the server machine is a Windows PC then the Application Starter can be started using the command[153]

>start /d C:\Minos2 C:\Minos2\Bin\MqtAppStarter.exe

(within the system startup file) to ensure the correct default directory is used. AppStarter itself does not display much information (Figure 170) but the configuration dialogue is the same as the Logger EXCEPT that appstarter has an "autostart" checkbox.

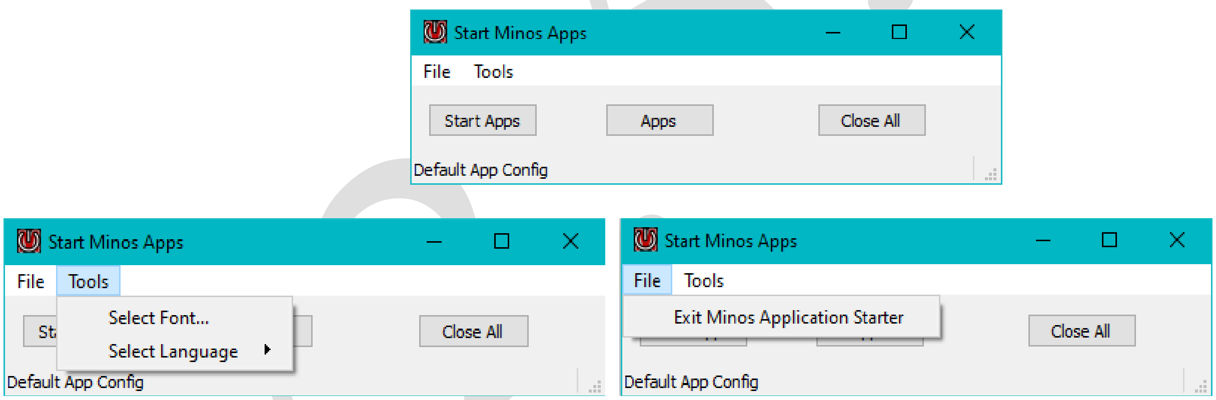

Figure 170 – AppStarter

If the server machine is a Raspberry Pi then Aplication Starter is run from the login script, invoked by having the machine auto-login. It then runs headless, with any access through SSH.

4.4.4 Example complex configuration

This configuration is intended to show the methods and capabilities rather than reality; it is overly complex!

Analyse requirements for a particular situation (e.g. cable lengths?)

[153] Assuming Minos has been installed into **C:\Minos2**

Figure 171 shows an example is a sort of 'super contest group' combining VHF NFD with the 432MHz-248GHz contest, and a possible way such a hypothetical situation might be configured:

- PC1 Runs the Logger for 6M and 4M with a rig for each band, both connected to and controlled by PC2. It has a single rotator with two Antennas, one for each band, serving the two contests.
- PC2 Runs the Logger for 2M, which can use either of two rigs - one of which is controlled by PC4. It has two rotators, one of which is controlled by PC4. It also runs the 6M & 4M rigs for PC1.
- PC3 Runs the Logger for 70cm 'and up' (23, 12, 9, 6, 3), with a separate rig for 70cm (controlled by PC4) and locally controlled rig/transverters for the other bands. It has two rotators; the one for 70cm is controlled by PC4, the locally controlled rotator handles all the antennas for the GHz bands.
- PC4 runs the AppStarter to control the 2M rig/rotator from PC2 and the 70cm rig/rotator from PC3.

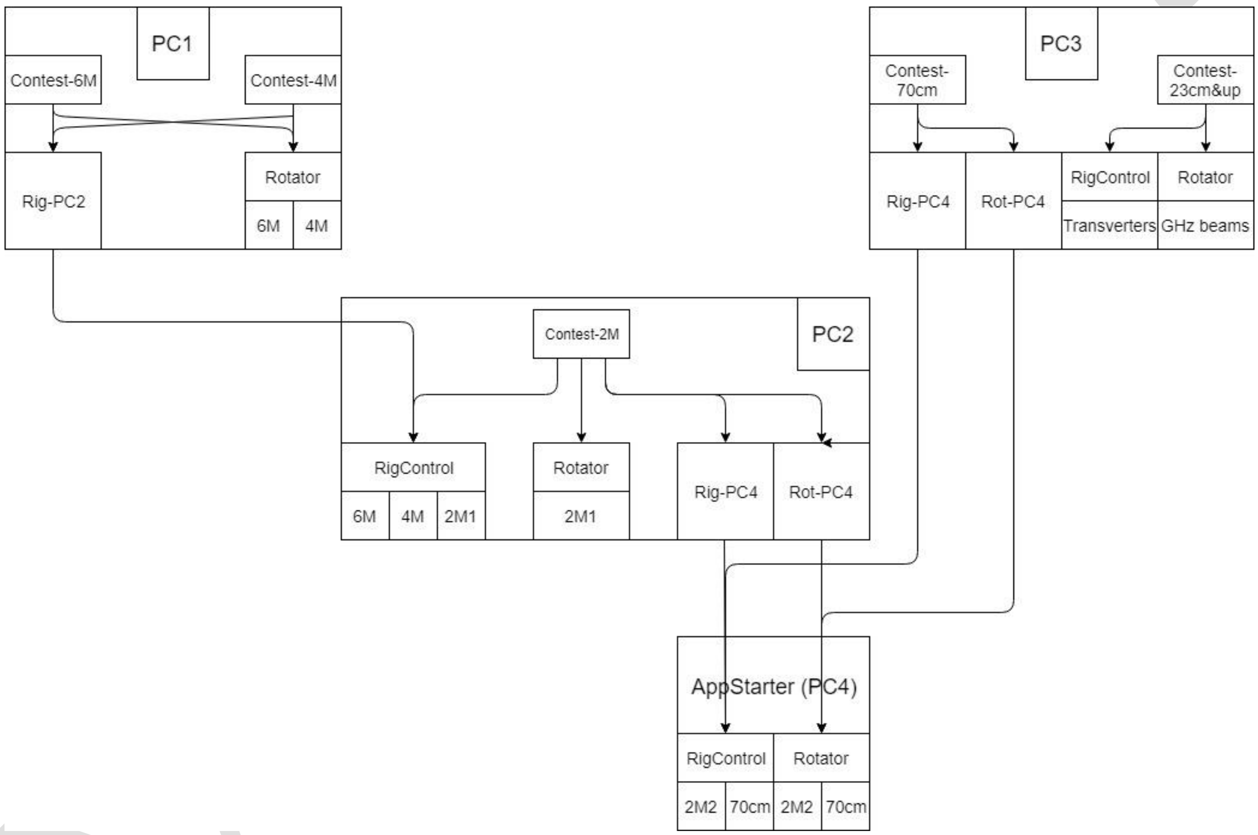

Figure 171 – Example complex configuration.

An example naming plan is shown in Figure 172

	Log	Appname	Connects to
PC1	6M 4M	RigControl-PC2	PC2/RigControl/6M PC2/RigControl/4M
		Rotator	6M & 4M aerials
PC2	2M	RigControl	6M, 4M, 2M1 rigs
		RigControl-PC4	PC4/RigControl/2M2

		Rotator	2M1 aerial
		Rotator-PC4	PC4/Rotator/2M2
PC3	70cm	RigControl-PC4	PC4/RigControl/70cm
	23cm-3cm	RigControl	23cm-3cm rigs (transverters)
	70cm	Rotator-PC4	PC4/RigControl/70cm
	23cm-3cm	Rotator	23cm-3cm aerials
PC4	(AppStarter)	RigControl	2M2 & 70cm rigs
		Rotator	2M2 & 70cm aerials

Figure 172 – Naming plan.

5 Internals

This section is still very much work in progress, and likely always will be. In fact it may even be removed, or at least substantially reduced since significant changes have taken place and almost all of the parameters are now set via the Options panels.

Minos is written using the open source Qt framework. If you want to get involved, then ask on the GJVList reflector. Details of how to build the software are documented in the main source repository.

When you do anything that makes a log change Minos appends a record to the file with the changes; one of the fields is a QSO sort key so that records can be correlated. The exact details are not documented here; anyone who really wants to 'go inside' it can look at the code (or ask G0GJV).

There should be no requirement to edit the file; indeed, it is highly recommended that you do **not** do so, in case of corrupting it.

5.1 Folders and files

The overall folder structure is:

- **Bin:** programs and DLLs (MqtLogger.exe is the main Logger programme)
- **Configuration:** configuration and INI files
- **Docs:** Documentation
- **Help:** <<Placeholder>> for help file(s)
- **Lists:** the default location for CSL file(s)
- **Logs:** the default location for storage of the .minos log files. They are not particularly interesting reading in raw format; certainly ***don't even consider*** editing them!
- **Tracelog:** diagnostic traces are written to this folder

5.1.1 Configuration files

- **Bandlist.xml:** Describes the amateur bands and the various formats possible. This file should not be edited!
- **B2Mults.xml:**
- **Cty.dat:** The DXCC countries and prefixes list. Documentation of the CTY.DAT format for CT9 can be found on the K1EA website (http://www.k1ea.com/). CTY.DAT should be used **unchanged**; any modifications required should be made to CTY.SYN. This file is updated at the same time as downloading the calendar[154].
- **Cty.syn:** As countries keep adding to their valid prefixes this file may need to be modified to keep up to date. Each line starts with a primary DXCC prefix, followed by synonym prefixes, separated by commas. You can use # on the start of a line to signify comments.
- **District.ctl:** Postcode information.
- **DISTRICT.SYN:** Synonyms for single character postcodes.
- **keyerConfig.xml:**
- **MinosConfig.json:**
- **MinosKeyer.json:**
- **portConfig.xml:**
- **Prefix.syn:** Allows postcodes to be checked for correct country and also duplicate checking if a station changes country in the middle of a contest.
- **ScreenConfigs.json:** Screen layout information
- ***contests.*:** Contest calendar(s) downloaded from the RSGB web page.

[154] The URL is http://www.country-files.com/ .

5.1.2 INI files

The content is fairly self-explanatory, but if you want detailed information then look at the Open Source code. Some of the INI files are only created when the program is first run or as various applications are configured; others (not documented here, see main text) are created by the user for special purposes. **In general there should be no need to touch any of these files directly;** use the provided interface to make changes!

The configuration options have now moved from **"File/Options"** and **"Tools/Advanced Options"** into a command panel accessed from **"Tools/Options"**. These options are described in the main text. The **"General Options"** tab is used to set the Log and List directories' default locations.

\Configuration

- **AppConfig.ini:** the template for application definition.
- **Bandmap_limits.ini:** determines the initial extent of the Bandmap display for given band(s); the default is the whole band.
- **BandSwitchData.ini:**
- **clusterBandmapFilter.ini:**
- **Display.ini:** holds the selected statistics options and screen layout information.
- **Entry.ini, QTH.ini, Station.ini, Contest.ini**: contain the various bundles of information used to define a contest entry (as described in section 1.2).
- **ListPreload.ini**: contains the list of CSL files to be read on startup.
- **LocSquares.ini:** maps locator squares to countries to help validation**.**
- **LogsPreload.ini**: details the Contests and Contest Sets to be loaded on startup.
- **MinosConfig.ini**: defines the configuration of the applications (see Section 4).
- **MinosLogger.ini**: holds a pointer to the logs location and the autofill setting.
- **MinosRigControlConfig.ini:** tracelog and MGM mode control.
- **MinosRigSyncConfig.ini:**
- **MinosRotatorConfig.ini:** the memory pre-sets from Rotator.
- **RigRecorder.ini :**
- **UKACBonus.ini:** definition of the bonuses to be used for various squares within UKAC.

\Configuration\Antenna

- **AvailAntenna.ini:** which antennas are available, and the settings to access the rotators.
- **RotatorCurrentAntenna.ini:** the currently selected rotator

\Configuration\Radio

- **AvailRadio.ini:** which rigs are available, and the settings to access them.
- **FreqPresets.ini:** the list of initial frequencies for each band
- **RigControlCurrentRadio.ini**

\Configuration\Radio\Transverter

- **<RadioName>TransVertRadio.ini :** the definition of the bands and settings for the transverter(s)

\Configuration\Cluster

- **ClusterSettings.ini :** Personal name etc from Cluster Server configuration
- **ClusterSites.ini:** A list of possible DXcluster sites (this is where password information should be provided if it is required.
- **ClusterCommands.ini:** Contains the user defined commands that have been defined.
- **cluster_start.txt, cluster_end.txt:** The DXspider commands sent at logon and logoff if those options are enabled.

5.1.3 CSL format

Archive lists (.CSL) are now proper CSV files, and the fields may be quoted, plus there is now a possible extra field in the list for “number of contacts” with the call/locator/exchange combination.

```
# Any line with a hash as the first character
# is considered a comment and is ignored
# Format is
# <Callsign>,<Locator>,<Exchange>,<Count>
#
2E0BMO,IO83PO,WN,
2E0CNJ,IO83TO,,
2E0DOD,IO83XI,,
2E0DOD/M,IO83XI,,
2E0DOD/P,IO83XG,,
2E0LAE/P,IO93BJ,SD,
2E0NEY,IO81VK,,
2E0RFX,IO83WO,OL,
2E0UAC,IO92FJ,,
2E0UOG,IO83PN,WN,
2E0XLG/P,IO83WV,,
2E0XLG/P,IO84WC,BD
…
```

The ‘count’ field is the ‘number of contacts’ which is used in various ways by the archive list maintenance tools.

5.2 Matching Rules

Matches are evaluated with an embedded space character representing any character (single character wild card). Matching is soft - if no exact matches are found, the program will progressively soften the match criteria - first, remove any /x (suffix, e.g. /P) from the callsign, then ignore the locator and QTH fields, and finally ignore the callsign prefix and number (e.g. G0GJV/P will be looked for as GJV). At each stage, if any matches are found the softening stops. The match display varies - for the current contest it is the same as the main display; for other contests it loses irrelevant information - date/time, reports and serial numbers, giving more room for more important information. As the match process progresses the header of the match list will indicate progress. Typing any character in these fields will kill the current match process and restart it.

In addition to the above match process the callsign is subject to a rigorous duplicate check. A duplicate (if present) will always appear first in the match list.

5.3 Trace Logs

Trace logs can be very useful for diagnosing problems. In the event of finding a problem please try to supply a detailed list of steps leading up to the fault, and the appropriate trace log.

Rigcontrol and Rotator log trace messages all the time apart from those passing over the comms link to the radio(s) or rotator(s)[155].

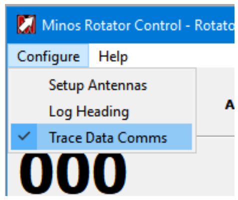

Figure 173 – Enabling extended trace log for Rotator

Tracelogs can be generated for diagnostic purposes. This can be enabled from the Configure menu where the TraceLog line can be enabled or disabled by checking the menu item. It is suggested that this should be disabled in normal operation.

Example tracelog with current antenna settings[156].

09:28:07.731 *** Antenna Updated ***
09:28:07.731 Rotator Name = Test Antenna
09:28:07.731 Rotator Model = 603 Yaesu,GS-232B
09:28:07.731 Rotator Number = 603
09:28:07.731 Rotator Comport = COM8
09:28:07.731 Baudrate = 19200
09:28:07.731 Databits = 8
09:28:07.731 Stop bits = 1
09:28:07.731 Handshake = 0
09:28:07.731 Antenna Offset = 0
09:28:07.731 Current Max Azimuth = 450
09:28:07.731 Current Min Azimuth = 0
09:28:07.731 South Stop Flag = 0
09:28:07.731 Overrun flag = 1
09:28:07.731 Rotator Max Baudrate = 9600
09:28:07.731 Rotator Min Baud rate = 150

5.4 Registry entries

All registry content is collected under [HKEY_CURRENT_USER\Software\Minos2Qt]

Don't even consider editing it, even if you think you understand it. It can be helpful to reset defaults when re-installing the programme for test purposes. This may be done by using the option on the file pull-down list.

[155] The logs are retained for 10 days and then automatically deleted.
[156] These settings can also be obtained via "Help/About Rotator Config" in the Rotator application. RigControl has the same capability.

6 CSL Management Utilities

6.1 Archive Utilities (G4AUC)

These four utilities can be used to create and populate archives (CSL files), check for possible errors in archives and logs, and merge archives. They keep track of the number of times (and the dates when[157]) a station has been worked. All use standard browse mechanisms to locate (or create) the files being processed.

Download **'setup_ArchiveUtilities_2.4.0.6.exe'** from https://sourceforge.net/projects/minos/files/ to a convenient location and then run it.

It uses the standard install dialogue to install the utilities to a default location of **'C:/Program Files (x86)/Archive Utilities 2.4.0.6'**

The .NET Framework 4 (full version) is required, but should already be installed on Windows 10.

After successful installation, the following facilities will be available (via desktop icons if you have chosen to create them):

- **Archive Maker**
  Optionally creates, and then populates, a .CSL file with the content of a contest entry (.EDI) file.

    - **"Create A New Archive (.CSL file) and add .EDI file(s) to it"** allows you to select a location for the Archive file (typically in the /Lists folder) and enter a suitable name for the file before saving (creating) it. A blank file is created; if this is all you want to do then click on **"Cancel"** instead of selecting any .EDI files.

    - **"Add .EDI file (or files) to an Existing Archive"** Open lets you select an existing Archive to which the .EDI file(s) will be added.

  In both cases, a new window opens to select the **'.EDI'** files. Browse to, and select, the required file(s) then click **"Open"**. The main programme window then shows all the contacts that are added to the file.

- **Archive Checker**
  The selected archive is scanned and entries logged to the main programme window. Possible matches (where the Callsigns appear to be very similar, or Locators the same[158]) are displayed.

  Ticking the **"Show Similar Locators"** box will additionally display contacts where the Locator is similar rather than exact match.

- **Contest Reporter**
  This program is intended to suggest where there may be typographic errors in your **'.EDI'** file (such as typing G9BAC instead of G9ABC). Remember that there may already be erroneous entries in the archive from earlier mis-typed QSOs.

  The **'.EDI'** file for checking is selected, and then the .CSL file which will be used as a reference. The selected log is scanned and entries logged to the main programme window.

[157] There must be some limit to the number of dates retained; it is not known at present what that limit is.
[158] Note that this is often caused by one of the entries having Postcode information.

The programme distinguishes between Callsigns not found in the archive and those already present. For the latter, possible matches (where the Callsigns appear to be very similar, or Locators the same) are displayed.

Ticking the **"Show Similar Locators"** box will additionally display contacts where the Locator is similar rather than exact match.

Remember people do change QTH from time to time and change callsign, particularly from Foundation to Intermediate to Full etc. Good locations are also used by multiple stations, sometimes even at the same time!

Whether a problem exists is entirely up to your judgement.

- **Merge Archives**
  The content of the first selected .CSL is added to the second selected CSL.

  The main programme window then shows all the Entries from the first file that has been added to the second.

All of these utilities have **"Close the Archive Maker Application"** buttons (or **'X'** can be used) and (extensive) **'Help'**.

6.2 CSL Manager (G4APJ)

CSLmanager is a simple portable application[159] to add contacts from one or more EDI (Minos) or LOG (old G0GJV Logger) files to the designated CSL file.

It handles (pun intended) the names of remote operators, and merges Postcode information into the archive.

'CSLmanager (V2.2.1a).zip' can also be downloaded from https://sourceforge.net/projects/minos/files/

The Zip file includes release notes, and usage information.

[159] It may be necessary to register an OCX file if your system does not already use it; instructions are given.

7 Appendix

7.1 Hints, tips, and 'tricks'

Reader contributions are welcomed here!

As with all software it is a good idea to take regular backups, but it is particularly important to save the configuration information before and after reconfiguring Applications – especially if you are doing an in-place upgrade and/or using a beta release. The internal format of the configuration files has changed several times, and there is no guarantee that it will not do so again!

It is also a good idea to periodically purge the TraceLog folder since multiple logs accumulate in there.

- If Rigcontrol panel disappears it means there is a problem connecting; could be target not present OR duplicate servername!If it appears to hang, check that there is not an error message displayed in a window that is hidden behind everything else on the screen.
- If you are getting weird bearings (and distances) check that you are using 4 char locators (if you are).
- If 'error' is displayed in the RigControl box when using remote access go check the rig – it has probably disconnected.
- When working NAC stations that don't give serials it is a fair few hoops to jump through to get Minos to accept the QSO with a serial less than 1, i.e. 000 as suggested by RSGB. When you click log the Dialog box comes up. Make sure the insert manual score now is ticked then click force. After the contest it is useful to add a comment in the comment box for each such QSO saying "No serial number received".
- The TS2000X supports 23cms, but Hamlib *does not contain a suitable definition for that band*. This results in a message 'No 1296 MHZ Band found for this radio'. To work round this, add a transverter for 1296 in the TS2000X radio config, setting both Radio and Transmit frequencies to 1296. It may be necessary to restart the Logger.
- If you accidentally lose the contents of the cluster client frame then you can look on the cluster server DX spots tab which retains them indefinitely.
- Using the K3 with (multiple) transverts can present problems – but does seem to be possible. Get in touch!

7.2 Suggestions

Improvements may be suggested by anyone, and if they make sense to G0GJV and, possibly after some rethinking of the request, fit in with the general philosophy for the program they may be included. Several improvements have been made at the request of users.

However, since the project is now Open Source you also make the change you want and submit it for possible inclusion in the main project.

Current possibilities include:

- There are around 50 feature requests logged which may, or may not, get implemented!

7.3 Future directions

- The remainder of the Applications defined in section 4.3.

7.4 Known problems (as of 11 Apr 22)

This section should be considered the equivalent of ‘release notes’.

- There are a few *known* ‘glitches’ in more complex configurations, but nothing likely to affect simple single band contest operation; please report any problems found.
- Similarly there are a couple of issues with the current release of Hamlib V4.5 The FT817 presents particular problems if using RigCtld. These can be avoided by defining the rig as an FT857 for the time being.
- A number of minor cosmetic changes are outstanding, along with checking a few of the French translations.
- If you switch between languages then Log and CSL preloads may be inhibited.
- There are a number of yellow markers in this text which indicate work in progress or questions outstanding.
- If rigctld is started with the rig not connected/turned off then it will cause problems.

A detailed list of known problems and feature requests (of which there are many) is held at **https://minos.groups.io/g/users/files/OutstandingIssues.pdf** although it is currently out of date and needs to be updated with the latest situation.

See https://sourceforge.net/p/minos/minos/ci/master/tree/MINOSChangeLog.txt for information about issues fixed in this release (this also needs to be updated).

<<Problem with Locator Map marking for old contests?>>

7.5 Minos Transverter Switch

Minos Rigcontrol can be configured to send a switching message when a transverter is selected from a radio. The ASCII string message is sent out of the rigcontrol computer comport at the time of selection. A ‘clear switches’ message is sent when no transverter is selected.

A micro-controller such as an Arduino Uno can be utilised with a suitable relay board. The Arduino USB port is connected the computer USB.

Minos can also use a com port defined in the radio configuration tab in Minos (see Figure 102). The serial port is hard coded as 9600 Baud, 8 data, 1 stop, no parity.

Messages are sent of the form <‘:’><Message Body><‘\n’>.

<Message Body> can be any ascii sequence of characters when a transverter is selected. The values are entered in to the Switch Number box (see Figure 101 – Bandswitch Messages), available for each transverter. If this is left blank, no message will be sent.

An example interface that will probably suits most users’ requirements is shown below. This software will switch relays 1 to 8. If ‘1’ is entered in to the transverter switch number box, relay 1 will be turned on when that transverter is selected. If more than one number is entered – for example ‘158’, then relay 1, 5 and 8 will be selected. So, it is possible to turn on more than one relay at a transverter selection, if that is required. Use ‘0’ to clear all relays.

The 5 volt relay board is an example that is available on the internet for a reasonable price, boards with less relays can also be found if 8 aren’t required. Arduino Unos are also available for a reasonable price. This document does not take you through loading the software, for which there are many tutorials on the internet to guide you through the process.

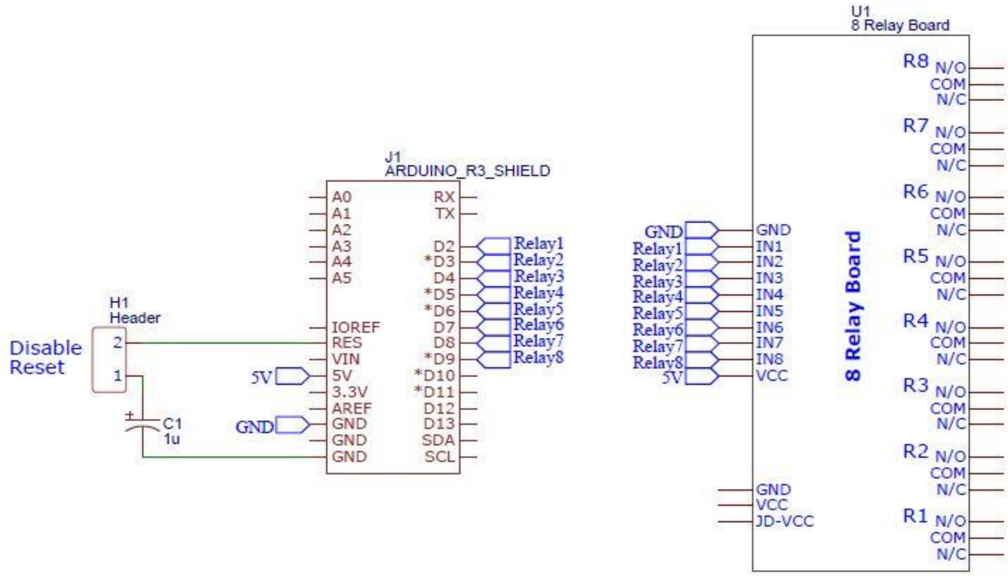

Figure 174 – Minos Transverter Switch Circuit

Figure 174 shows how to connect the IO ports to the relay board. Power for the relay board and Arduino can be taken from the computer USB port. The capacitor C1 is important, as it prevents the board performing a reset when the comport is initialised on the computer. **C1 needs to be out of circuit when uploading the switch control software to the UNO, and in circuit when the software is running.**

Inter-connections between the two boards can be achieved with the male-female Dupont jumper leads that can be purchased on the internet.

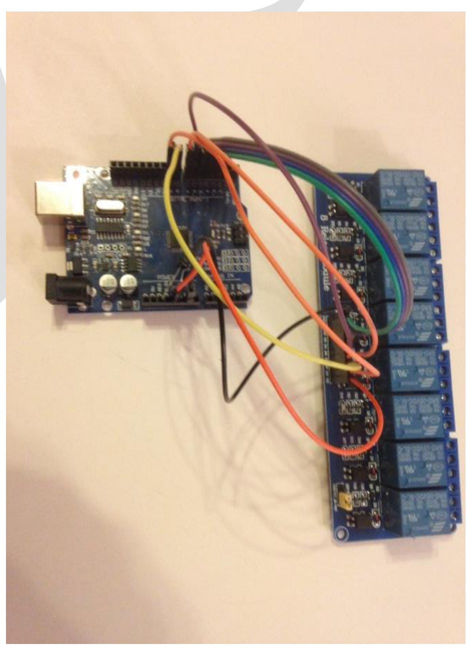

Figure 175 – Arduino and Relay Board

The Arduino Sketch can be downloaded from https://gitlab.com/dgbbal/Minos_Arduino_Transverter_Switch

7.6 Shortcut Keys

There are numerous shortcut key combinations available. They are all case-insensitive.

Rotator within Logger			
Function	**Shortcut**	**Function**	**Shortcut**
Rotate Left (CCW)	Ctrl-L	Rotate Right (CW)	Ctrl-R
Rotate to Bearing - Turn	Ctrl-T, Enter	Stop	Ctrl-S
Preset 1	Ctrl-alt-1	Preset 6	Ctrl-alt-6
Preset 2	Ctrl-alt-2	Preset 7	Ctrl-alt-7
Preset 3	Ctrl-alt-3	Preset 8	Ctrl-alt-8
Preset 4	Ctrl-alt-4	Preset 9	Ctrl-alt-9
Preset 5	Ctrl-alt-5	Preset 10	Ctrl-alt-0

Figure 176 – Rotator Shortcuts

RigControl within Logger			
Function	**Shortcut**	**Function**	**Shortcut**
Select KHz in Frequency Box	Ctrl-f		
Select Run Frequency 1	Ctrl- shift- [	Select Run Frequency 2	Ctrl- shift-]
Turn RIT On/Off	Ctrl-o	Select RIT 100Hz digit (if on)	Ctrl-i
Clear RIT frequency	Ctrl-k		

Figure 177 – Rigcontrol Shortcuts

Bandmap within Logger			
Function	**Shortcut**	**Function**	**Shortcut**
Zoom out	Ctrl-<	Zoom in	Ctrl->
Step up through markers	Ctrl-↑	Step down through markers	Ctrl-↓
Ditto, only unworked Locs	Ctrl-alt-↑	Ditto, only unworked Locs	Ctrl-alt-↓

Figure 178 – Bandmap Shortcuts

7.7 Beginners Guide to Contest Logging

This text is based on input from Ian, G0FCT, chair of the Contest Support Committee.

Regardless of the medium used, a contest QSO must be logged:

- RSGB VHF Contest Rules state: “The contest exchange consists of at least both callsigns, RS(T) signal reports followed by a serial number, and the IARU locator. Particular contests may require additional information to be exchanged as described in the individual contest rules.”

- The equivalent HF Contests is slightly less demanding: "A contest contact requires establishment of communication and a two-way contest exchange. The information contained in a contest exchange often consists of an incrementing serial number starting from 001, acknowledgement of the call sign received and of the contest exchange received. Individual contest rules may vary this exchange."

Surprisingly, neither of them mentions the date/time, but that too is required to allow validation that the QSO took place.

If keeping a paper log (which could later be transferred to Minos for checking/scoring) it would normally contain columns to fit the above requirements:

- Date/time
- Callsign of station being worked
- RS(T) and Serial number Sent to the other station
- RS(T) and Serial number Received from the other station
- Locator and/or Comments/Remarks for noting the contest exchange(s) sent by the other station

Plus any others the operator wants such as Frequency, Mode, Power, Operator (if multi-operator station).

An example from such a log might read:

Date	Time	Callsign	Sent	Received	Locator	Comments
26/11/21	10:41	G9XYZ	57 001	56 002	IO81DD	12km S Okehampton, Devon

When filling in such a log there are two methods of operation:

- **Search & Pounce**, which is initiating a call to other stations (quite possibly calling CQ) as they are heard. When the station responds, the required information will typically be sent immediately and should be entered in the Received, Locator, and Comments columns.The 'Sent' information (plus own Locator etc) is then passed to the other station and entered into the log.

- **Call mode**, which is calling CQ and responding to callers, is the inverse of S&P. When a caller is identified the Callsign is entered in the Callsign field, the station is acknowledged (by callsign) and the 'Sent' information (plus own Locator etc) is then transmitted (and logged). That station then responds with their required information which should be entered in the Received, Locator, and Comments columns.

The MINOS logging screen mimics the above behaviour (see Figure 179) and automates several of the steps - the Date and Time information and the serial number to be sent.

The Callsign of the station being worked is entered in the **"Callsign (F1)"** box.

The **"RS(T)Tx(F2)"** field contains the report being sent to the other station. If a report of 58 is sent, enter 58 into the field.

"RS(T)Rx(F3)" and **"Serial Rx (F4)"** contain the report and serial number received from the other station, e.g. if 53007 is received **"RS(T)Rx(F3)"** is set to 53 and **"Serial Rx (F4)"** is set to (00)7[160]

The received Locator is entered into **"Loc (F5)"**. The **"Exchange (F6)"** field is used for contest specified information, e.g. PostCodes. Any other received information is entered in the **"Comments"** field.

[160] Leading zeros are of no significance and can be inserted or omitted as desired.

The requisite box can be selected by clicking in it with the mouse, using the indicated function key (F1/F2/F3/F4/F5), or by automated sequencing using the <TAB> or <CR>[161] keys. The automated sequence order is controlled by the setting of the **"S&P" / "Run"** buttons. If operating **"S&P"** then <TAB>/<CR> from **"Callsign"** moves the cursor to **"RS(T)Rx(F3)"** ready for the remote station's Report (and Serial, in the next box). If **"Run"** is set (which requires a**"Run Frequency"** to be set) then <TAB>/<CR> from **"Callsign"** moves the cursor moves to **"RS(T)Tx(F2)"** ready to enter the Report being sent. **Note: Fields already containing data are skipped.**

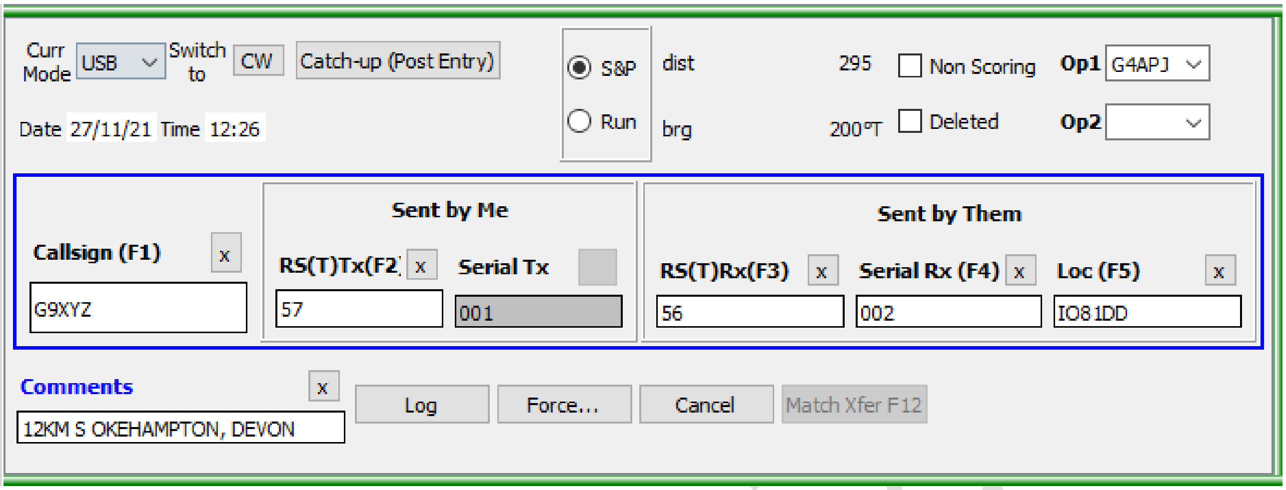

Figure 179 – Example QSO entered into Minos

For further information see Section 2.2.

A common mistake made by contest newcomers is to mix up where the received (Rx) and send (Tx) reports are logged. The report sent is always recorded in the **"RS(T)Tx(F2)"** field and the report received is always recorded in the **"RS(T)Rx(F3)"** with the serial number in the **"Serial Rx (F4)"** field.

[End of Document]

[161] Carriage Return, or Enter key.